

1 The health-relevant architecture of the 2 everyday light exposome

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34 Abstract

35 Personal light exposure has been associated with sleep and non-communicable diseases,
36 yet how ocular exposure varies across the day, among people and between everyday
37 settings remains poorly understood. We analysed 191 adults at nine sites in seven countries;
38 184 contributed 1,478 participant-days (near eye: 141 adults, 816 days; complementary
39 chest: 154 adults, 902 days). The shared 24-hour near-eye pattern accounted for 78.9% of
40 full-model R^2 , participant patterns 12.9%, participant-day shifts 6.2% and site patterns 2.0%.
41 Reported light source and immediate setting produced large contrasts, accounting for 24.2%
42 and 30.7%, respectively, of variation in fitted hourly patterns, the next-largest shares after
43 time of day. Associations with site, day length and personal characteristics differed across
44 exposure measures. Only 24.0% of daytime minutes met the recommendation. By resolving
45 variation across sites, people, days and immediate settings, this study lays an empirical
46 foundation for targeted, harmonised exposure monitoring aligned with health-based light
47 recommendations.

48 Introduction

49 Light reaching the eye supports vision and signals circadian timing, sleep, alertness and
50 neuroendocrine responses^{1–5}. Melanopic equivalent daylight illuminance (melanopic EDI)
51 weights illuminance for melanopsin-related sensitivity and is most relevant to these effects^{6,7}.
52 Expert consensus recommends at least 250 lx melanopic EDI at the eye during daytime, no
53 more than 10 lx during the three hours before sleep and no more than 1 lx in the sleep
54 environment for healthy adults with regular daytime schedules⁸. These physiology-based
55 ranges make daily light a health-relevant exposure, but are not individual risk functions.

56 Population studies and reviews associate personal light patterns with sleep and major non-
57 communicable diseases⁹. Wrist-sensor studies link light timing and day-night distribution with
58 sleep timing and sleepiness, psychiatric disorders, incident type 2 diabetes, cardiovascular
59 disease and mortality^{10–15}. Other cohorts link evening, nocturnal or seasonal light with sleep
60 in bipolar disorder and with obesity, diabetes, hypertension and metabolic health^{16–19}. The
61 exposome framework characterises non-genetic exposures across the life course^{20,21}, yet
62 rarely includes time-resolved ocular melanopic light, the everyday component examined
63 here. These studies establish population relevance but mostly measured broad-spectrum
64 wrist or bedside light, providing only proxy evidence for melanopic ocular exposure.

65 An everyday exposure baseline against these recommendations is missing. Personal
66 exposure to natural and electric light follows an expected 24-hour rhythm, including in large
67 wrist-sensor cohorts²², but previous studies focus on particular populations, occupations or
68 settings^{23–25}. Its amplitude and timing across sites and countries, and the contributions of
69 time, site, individuals, day-to-day variation and immediate context, remain uncertain.
70 Daylight availability, satellite estimates and fixed-site measurements cannot resolve these
71 layers as people move through buildings, transport, outdoor spaces and sleep
72 environments²⁶. Sensor position also matters: near-eye sensing is closer to the incident
73 ocular field but is not a retinal measurement; wrist and chest sensors capture related fields
74 shaped by posture, wear and occlusion^{27–33}.

75 Using the harmonised MeLiDos protocol, we measured personal light across nine sites in
76 seven countries^{34,35}. The near-eye sensor characterised ocular light exposure during wear,
77 alongside diaries, light-source and immediate-setting reports, sleep information and
78 questionnaires. The complementary chest sensor provided non-ocular environmental
79 evidence and enabled participation among people reluctant to wear the glasses-mounted
80 sensor. We sought a multisite recommendation-based baseline, assessing temporal,
81 environmental and routine contributions alongside selected person-level correlates. We use

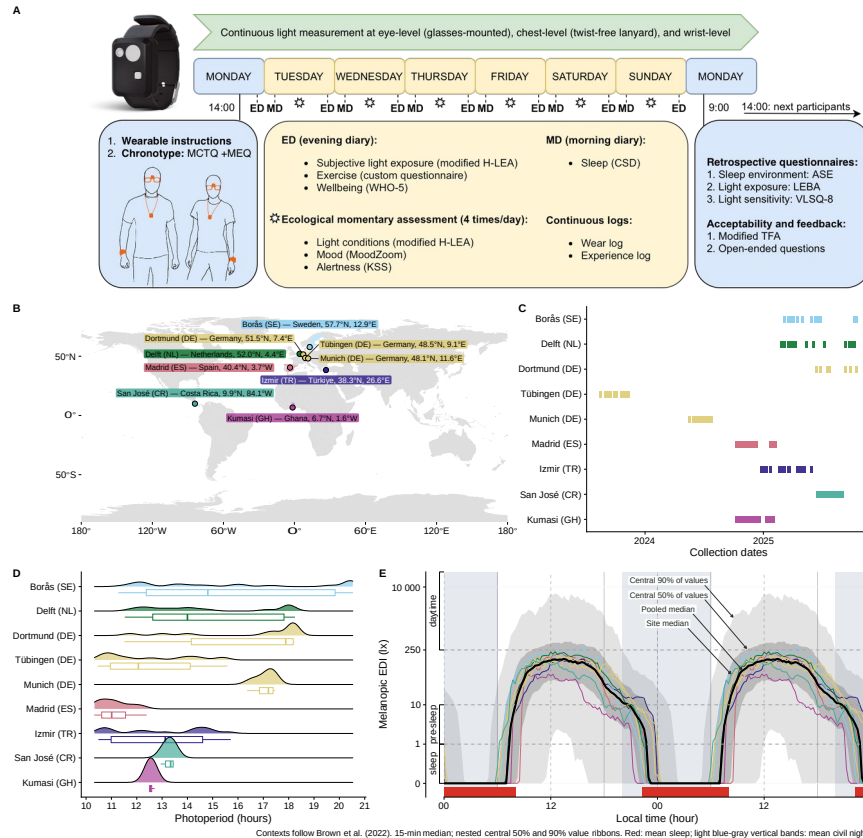
82 architecture to describe how everyday light exposure is organised across time, sites, people,
83 days and immediate settings, and benchmark this component of the exposome against
84 health-based light recommendations. Although not global, the protocol enables harmonised
85 measurement elsewhere, progressively closing gaps across locations, civil photoperiods,
86 latitudes and climates and building towards a global account of ocular light exposure.

87 Results

88 A multisite baseline and recommendation adherence

89 The normalised roster comprised 191 participants across Borås (SE), Delft (NL), Dortmund
90 (DE), Tübingen (DE), Munich (DE), Madrid (ES), Izmir (TR), San José (CR) and Kumasi
91 (GH). Of these, 184 contributed 1,478 participant-days with recorded, non-all-zero light data
92 at either position before quality screening. After screening, the near-eye dataset contained
93 141 participants and 816 participant-days, the complementary chest dataset 154 and 902,
94 and the common sample 112 participants and 643 participant-days. Near-eye
95 measurements provide ocular-exposure evidence during wear, whereas chest
96 measurements are non-ocular; both describe the bedside sleep environment during reported
97 sleep. [Supplementary Table S1](#) gives the sample flow.

98 The pooled near-eye record followed the expected 24-hour rhythm: low overnight exposure,
99 a morning rise, higher daytime values and an evening decline. Seventeen complementary
100 metrics captured thematic streams rather than one score. Level metrics described daily
101 geometric mean melanopic EDI and the brightest and darkest supported 10-hour means;
102 duration and timing metrics quantified exposure ranges, sustained bright periods, bright or
103 dark windows and threshold crossings. Dose integrated intensity, regularity captured stability
104 and fragmentation, and melanopic daylight efficacy (MDER) described spectral composition
105 relative to visual illuminance. These features are not interchangeable health-response
106 estimates. [Supplementary Table S2](#) and [Supplementary Figure S1](#) give the metric dictionary
107 and site distributions; [Supplementary Figure S2](#) shows a worked derivation. [Figure 1](#) and
108 [Table 1](#) summarise the design and data structure.



109

110 **Figure 1: Five-panel overview of the eight-day protocol, sensor positions, nine**
 111 **country-coded sites, collection windows, civil photoperiod, and the pooled near-eye**
 112 **daily profile. *A*, Eight-day field protocol and sensor positions. The wrist logger is**
 113 ***shown as a protocol device but is not an exposure channel analysed in this***
 114 ***manuscript. *B*, Nine sites in seven countries. *C*, Site collection periods. *D*, Observed***
 115 ***civil photoperiod. *E*, Site and pooled 15-minute median near-eye melanopic***
 116 ***equivalent daylight illuminance profiles from 816 participant-days contributed by 141***
 117 ***participants, repeated over 48 hours for display only. Grey ribbons show the central***
 118 ***50% and 90% of measured values, not confidence intervals; coloured lines are site***
 119 ***medians, the black line is the pooled median, and red and blue shading show mean***
 120 ***reported sleep and civil night (sun elevation < -6°). During reported sleep, the***
 121 ***measurement represents the bedside sleep environment.***

122

123 Table 1: Participant and measurement characteristics overall and by country-coded study site. The table distinguishes the 191-person roster, 124 participants and participant-days with recorded light data, and the screened near-eye and complementary chest samples. Model-specific samples 125 are reported with each analysis. Of the 191 roster participants, 184 contributed the 1,478 recorded participant-days. Continuous participant 126 characteristics are median with the stated percentile range.

	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)
Site specifics										
Institution	Overall	RISE	THUAS	BAuA	MPI	TUM	FUSPCEU	IZTECH	UCR	KNUST
Country	—	Sweden	Netherlands	Germany	Germany	Germany	Spain	Türkiye	Costa Rica	Ghana
City	—	Borås	Delft	Dortmund	Tübingen	Munich	Madrid	Izmir	San José	Kumasi
Coordinates	—	57.7°N, 12.9°E	52.0°N, 4.4°E	51.5°N, 7.4°E	48.5°N, 9.1°E	48.1°N, 11.6°E	40.4°N, 3.7°W	38.3°N, 26.6°E	9.9°N, 84.1°W	6.7°N, 1.6°W
Data collection										
Participants ¹	191 roster (g:143; c:157; p:116)	17 roster (g:14; c:17; p:14)	20 roster (g:13; c:15; p:13)	24 roster (g:19; c:22; p:19)	26 roster (g:26; c:0; p:0)	10 roster (g:10; c:10; p:10)	23 roster (g:23; c:22; p:22)	17 roster (g:17; c:17; p:17)	39 roster (g:6; c:39; p:6)	15 roster (g:15; c:15; p:15)
Participant-days ¹	1478 roster (g:1134 c:1246; p:902)	137 roster (g:107 c:137; p:107)	125 roster (g:107 c:124; p:106)	176 roster (g:145 c:163; p:132)	208 roster (g:208 c:0; p:0)	80 roster (g:80 c:80; p:80)	182 roster (g:182 c:174; p:174)	138 roster (g:138 c:138; p:138)	312 roster (g:48 c:312; p:48)	120 roster (g:119 c:118; p:117)
Participant time ²	116w 4d	11w 23h	11w 23h	15w 2d	21w 3d	8w 4d	18w 3d	14w 3d	4w 4d	11w 4d
Declared non-wear ²	g:4w 1d (3.66%)	g:2d 14h (3.34%)	g:3d 1h (3.92%)	g:3d 15h (3.42%)	g:6d 10h (4.30%)	g:2d 17h (4.55%)	g:2d 20h (2.20%)	g:3d 3h (3.13%)	g:1d 11h (4.66%)	g:3d 21h (4.80%)
Screened days ^{1,3}	g:816 c:902	g:78 c:96	g:78 c:93	g:107 c:114	g:150 c:0	g:60 c:60	g:129 c:123	g:101 c:102	g:32 c:230	g:81 c:84
Civil photoperiod ^{4,2}	13.25 h (10.62–18.20) (n=1478 d)	14.81 h (11.80–20.49) (n=137 d)	14.00 h (11.88–18.09) (n=125 d)	17.88 h (11.76–18.22) (n=176 d)	12.07 h (10.55–15.41) (n=208 d)	17.20 h (16.45–17.45) (n=80 d)	11.01 h (10.35–12.22) (n=182 d)	13.13 h (10.52–15.57) (n=138 d)	13.34 h (13.01–13.47) (n=312 d)	12.54 h (12.49–12.68) (n=120 d)

Table continued

	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)
Participant information										
Age ⁴	28 y (21 y–52 y) (n=191 N)	38 y (22.6 y–64.8 y) (n=17 N)	30.5 y (19.9 y–57.1 y) (n=20 N)	35.5 y (20.1 y–58.5 y) (n=24 N)	27 y (22 y–37 y) (n=26 N)	27.5 y (22.2 y–29 y) (n=10 N)	31 y (20.2 y–54.3 y) (n=23 N)	24 y (21 y–29.6 y) (n=17 N)	34 y (22 y–48.1 y) (n=39 N)	23 y (20.1 y–25 y) (n=15 N)
Sex ⁵	F:105 / M:86 (n=191)	F:6 / M:11 (n=17)	F:8 / M:12 (n=20)	F:13 / M:11 (n=24)	F:14 / M:12 (n=26)	F:6 / M:4 (n=10)	F:15 / M:8 (n=23)	F:11 / M:6 (n=17)	F:24 / M:15 (n=39)	F:8 / M:7 (n=15)
Employment status ⁶	F:163 / P:26 / N:2 (n=191)	F:16 / P:0 / N:1 (n=17)	F:11 / P:9 / N:0 (n=20)	F:21 / P:3 / N:0 (n=24)	F:23 / P:3 / N:0 (n=26)	F:9 / P:1 / N:0 (n=10)	F:21 / P:2 / N:0 (n=23)	F:15 / P:2 / N:0 (n=17)	F:34 / P:5 / N:0 (n=39)	F:13 / P:1 / N:1 (n=15)
Chronotype group ⁷	m:62 / i:94 / e:30 (n=186)	m:8 / i:4 / e:5 (n=17)	m:3 / i:9 / e:3 (n=15)	m:8 / i:13 / e:3 (n=24)	m:9 / i:9 / e:8 (n=26)	m:0 / i:8 / e:2 (n=10)	m:5 / i:17 / e:1 (n=23)	m:2 / i:10 / e:5 (n=17)	m:16 / i:20 / e:3 (n=39)	m:11 / i:4 / e:0 (n=15)
Sleep-corrected midsleep on free days ^{4,8}	04:03 (02:08–06:15) (n=185 N)	02:55 (02:04–05:09) (n=17 N)	04:31 (03:05–05:37) (n=15 N)	03:47 (02:17–04:49) (n=24 N)	04:33 (02:41–06:04) (n=26 N)	05:08 (04:12–06:04) (n=10 N)	04:48 (03:10–06:52) (n=22 N)	05:29 (03:35–06:49) (n=17 N)	03:42 (02:17–05:31) (n=39 N)	02:19 (01:22–03:25) (n=15 N)

¹ g, near-eye glasses position; c, complementary chest position; p, paired common sample in which the same participant-day is available at both positions.

² w: weeks; d: days; h: hours; min: minutes. Participant time and declared non-wear use eligible real minutes on screened days; both are shown for near eye only.

³ Screened days are the final participant-days after the 80% completeness screen and exclusion of exact-all-zero melEDI days.

⁴ Median (5th percentile, 95th percentile).

⁵ Sex categories: Female and Male.

⁶ Employment categories: Full/studying, Part/marginal, and Not employed.

⁷ Chronotype groups follow the Morningness–Eveningness Questionnaire score.

⁸ MCTQ: Munich Chronotype Questionnaire; MEQ: Morningness–Eveningness Questionnaire. Clock summaries are circular.

129 Exposure was within the applicable recommendation during 24.0% of daytime minutes,
130 63.3% of pre-sleep minutes and 87.7% of sleep minutes. This represented 137,792 of
131 573,712 waking daytime minutes, 81,894 of 129,390 pre-sleep minutes and 336,052 of
132 383,366 minutes during reported sleep. The sleep quantity describes the bedside sleep
133 environment because the device was not worn at the eye; the Discussion addresses low-
134 light measurement accuracy. [Supplementary Table S3](#) gives exact numerators,
135 denominators and complementary ranges.

136 These pooled-minute percentages describe how often exposure lay within range, not
137 participant adherence. Model-based comparisons retained each window's adherent and
138 valid-minute counts.

139 The adherence model included 2,298 periods from 140 participants and 794 cycles
140 (1,043,192 valid minutes), linking preceding sleep, daytime and following pre-sleep to the
141 wake-start date's day type; windows qualified independently. Free-versus-work adherence
142 was 5.1 percentage points lower during daytime (95% confidence interval [95% CI] 2.4 to 7.8
143 lower), 5.9 higher pre-sleep (1.4 to 10.5) and 6.5 lower during sleep (3.8 to 9.1). At least
144 80% coverage preserved daytime and sleep interval conclusions, but pre-sleep was
145 inconclusive (4.1 points, 95% CI -0.9 to 9.1). Temporal dependence remained unresolved.

146 The window-by-site-by-day-type interaction was retained (FDR-adjusted $p = 0.008$), with site
147 variation during daytime and sleep (both $p = 0.002$), not detected pre-sleep ($p = 0.818$).
148 Relative to the site-average free-minus-work contrast, Dortmund (DE) was 14.8 points higher
149 during daytime (95% CI 6.7 to 22.9; FDR-adjusted $p = 0.004$), Madrid (ES) 9.2 lower (-15.6
150 to -2.8; $p = 0.044$), and Kumasi (GH) 6.3 higher during sleep (3.5 to 9.1; $p < 0.001$). These
151 departures met a separate 27-test FDR criterion and retained directions and interval
152 conclusions at 80% coverage.

153 On the design-standardized response scale, marginal R^2 was 58.0% and conditional R^2
154 60.2%; the participant increment was 2.3 percentage points and observation/distribution
155 variation 39.8%. Shapley allocated 94.2% of marginal R^2 to recommendation window, 4.8%
156 to site and 1.0% to day type. Site's 2.8-point allocation was about 1.2 times the participant-
157 intercept increment and about five times the 0.6-point day-type allocation. Allocations are
158 point-only; the window share partly reflects different thresholds. [Table 2](#) and [Supplementary](#)
159 [Figures S4](#) and [S5](#) give adherence estimates.

160 Table 2: Recommendation adherence by Brown et al. recommendation window and day type. Observed valid-minute fractions are descriptive;
 161 work-day and free-day estimates and their difference are model-based, with 95% confidence intervals and false-discovery-rate-adjusted p-values.
 162 Each of nine sites receives equal weight in the model estimates. Daytime, Pre-sleep and Sleep identify the recommendation windows: daytime
 163 excludes the three hours before reported sleep, pre-sleep comprises those three hours, and sleep describes the bedside sleep environment. The
 164 wake-start date assigns the day type to the preceding sleep, daytime and following pre-sleep windows; each window qualifies independently. Only
 165 the primary any-valid sample is shown: 2,298 periods, 140 participants, 794 cycles and 1,043,192 valid minutes. The 80% coverage sensitivity
 166 remains reported in the Results and Methods; its pre-sleep interval includes zero, and temporal dependence remains unresolved. Confidence-
 167 interval exclusion and FDR decisions are separate summaries. Descriptive pooled-minute fractions are detailed in Supplementary Table S3.
 168 Adherence does not classify participants.

Recommendation adherence by recommendation window and day type

Recommendation window	Recommendation	Observed pooled-minute adherence	Site-average recommendation-window model			
		Valid minutes meeting recommendation, n/N (%)	Work-day adherence % (95% CI)	Free-day adherence % (95% CI)	Free minus Work percentage points (95% CI)	FDR-adjusted p
Daytime	At least 250 lx melanopic EDI during wake, excluding the three hours before sleep	137,792/573,712 (24.0%)	26.6 (24.5 to 28.7)	21.5 (19.2 to 23.8)	-5.1 (-7.8 to -2.4)	<0.001
Pre-sleep	No more than 10 lx melanopic EDI during the three hours before reported sleep	81,894/129,390 (63.3%)	62.0 (58.8 to 65.2)	67.9 (64.2 to 71.6)	+5.9 (+1.4 to +10.5)	0.011
Sleep	No more than 1 lx melanopic EDI during reported sleep; bedside sleep environment	336,052/383,366 (87.7%)	90.4 (88.9 to 91.9)	83.9 (81.4 to 86.5)	-6.5 (-9.1 to -3.8)	<0.001

Observed pooled-minute fractions and fitted recommendation-window estimates are different quantities. Each of the nine sites contributes equally to the fitted estimates; the displayed model results use only the primary any-valid sample.

The recommendation boundaries are inclusive: daytime ≥250 lx, pre-sleep ≤10 lx and sleep ≤1 lx melanopic EDI. Near-eye measurements describe ocular exposure during wear; reported sleep describes the bedside sleep environment, not worn ocular exposure. Chest measurements are not pooled here.

Day type is assigned from the wake-start date to the preceding sleep, daytime and following pre-sleep windows; windows qualify independently. Temporal dependence remained unresolved. The 80% coverage sensitivity, including the inconclusive pre-sleep interval, is reported in the Results and Methods, not as additional rows here.

169

170 Exploratory cross-window within-participant contrasts per 10-percentage-point difference in
171 daytime adherence were -0.26 points for sleep (95% CI -0.95 to 0.43) and -1.09 for pre-
172 sleep (-2.75 to 0.56). Unresolved temporal dependence precluded a within-participant claim;
173 intervals including zero do not establish absence. Between participants, 10 points higher
174 average daytime adherence was associated with 2.52 points lower sleep adherence (1.48 to
175 3.55 lower; FDR-adjusted $p < 0.001$) and 3.59 lower pre-sleep adherence (1.25 to 5.94
176 lower; $p = 0.005$), consistent with higher evening and nighttime exposure during monitoring.
177 Both met the four-effect FDR criterion; directions and interval conclusions persisted at 80%
178 coverage (maximum change 0.60 points). [Supplementary Table S4](#) gives estimates and
179 [Supplementary Figure S6](#) participant profiles.

180 **Personal and day-to-day differences exceed site** 181 **differences**

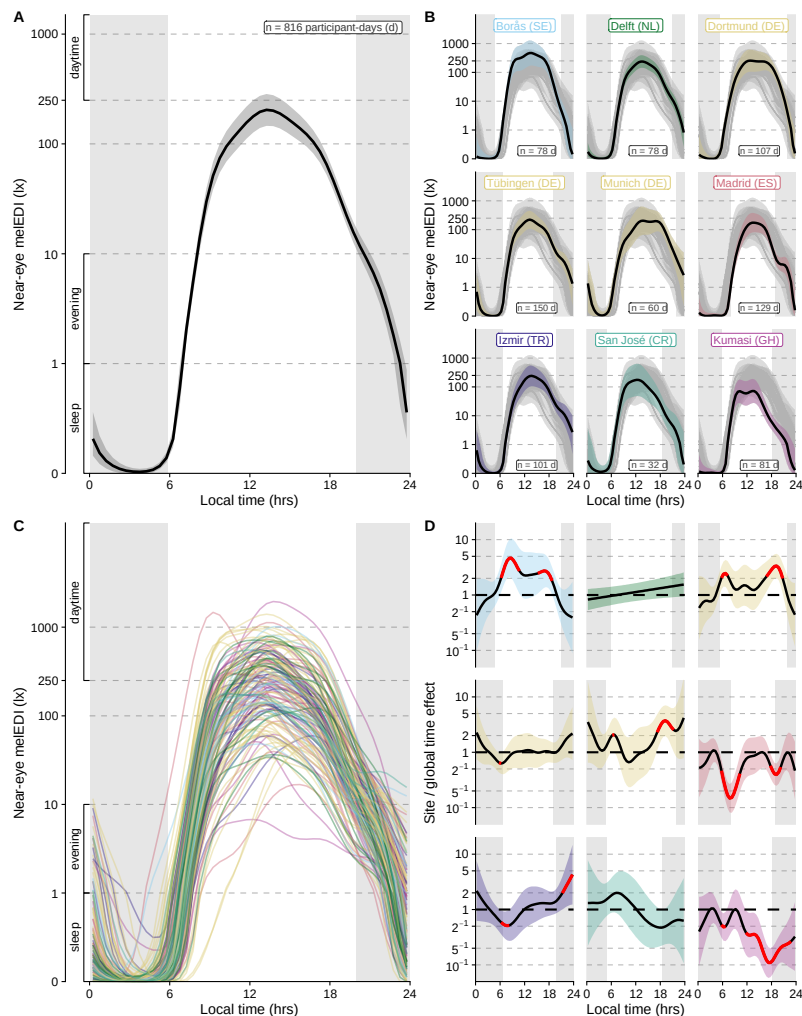
182 A nonlinear model decomposed personal light exposure into common local-clock, site and
183 participant curves, and participant-day shifts. Near-eye in-sample R^2 was 0.775. Participant
184 curves were 1.80 times as dispersed as site curves (95% CI 1.16 to 4.33); participant curves
185 plus day shifts had a dispersion ratio of 1.99 (1.29 to 4.77). After accounting for time of day,
186 fitted variation among people and their days within sites exceeded variation among sites.

187 Shares of full-model R^2 showed the same hierarchy: 78.9% for the shared local-clock
188 pattern, 12.9% for participant patterns, 6.2% for participant-day shifts and 2.0% for site
189 patterns. The participant-pattern share was 6.41 times the site-pattern share (3.85 to 15.14),
190 and the combined participant-pattern and participant-day-shift share was 9.49 times the site-
191 pattern share (5.77 to 21.81).

192 The shared rhythm did not imply uniform amplitude. Relative to the fitted site-average daily
193 pattern, Borås (SE) ratios were 2.04 to 4.64 from 06:00 to 11:00 and 1.80 to 2.73 from 15:00
194 to 19:00. Madrid (ES) ratios were 0.15 to 0.47 from 05:30 to 10:30 and 0.40 to 0.57 from
195 17:30 to 20:30. Izmir (TR) shifted from 0.51 to 0.59 at 06:00 to 08:30 to 1.98 to 4.28 at 21:00
196 to 24:00, while Kumasi (GH) was 0.11 to 0.38 from 11:30 to 23:00. These are clock-specific
197 departures from the average curve.

198 The chest model had an in-sample R^2 of 0.737. Of this full-model R^2 , common time
199 accounted for 79.1%, participant patterns 11.9%, participant-day shifts 7.0% and site
200 patterns 2.1%. Common-sample comparisons supported the combined participant-plus-day
201 ordering, but the participant-only contrast was less stable. Cross-position similarity is

202 complementary context, not replication, equivalence or interchangeability. [Figure 2](#) shows
203 the fitted layers; [Supplementary Tables S5](#) and [S6](#) give both decompositions.



204

205 **Figure 2: Multiscale daily pattern of near-eye melanopic equivalent daylight**
 206 **illuminance. A, Site-average daily curve. B, Fitted site curves. C, Fitted participant**
 207 **curves after the shared local-clock and site patterns were considered. D, Ratio of**
 208 **each fitted site curve to the site-average curve on the melanopic equivalent daylight**
 209 **illuminance plus 0.1 lx scale. Grey bands show mean civil night. Shaded bands are**
 210 **clock-specific 95% confidence intervals; red segments identify displayed half-hours**
 211 **whose interval did not include the site-average curve and are descriptive clock-**
 212 **specific contrasts. The analysis included 141 participants and 816 participant-days**
 213 **across nine sites. The label “global time effect” denotes the site-average local-clock**
 214 **curve (panel A).**

215 Geography and photoperiod provide partial context

216 Site associations were stream-specific. Site differences met the FDR criterion for 8 of 17
 217 near-eye metrics: all three level metrics, waking time above 250 lx, the brightest- and
 218 darkest-10-hour midpoints, and the mean and last timing above 250 lx. They were not FDR-
 219 retained for dose, spectral composition, stability, fragmentation, sustained bright exposure,
 220 the other threshold-duration metrics or first timing above 250 lx. Across metric models,
 221 conditional R^2 averaged 36.4%; participant-associated R^2 averaged 25.6% among

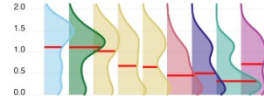
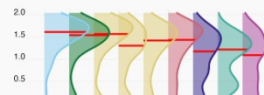
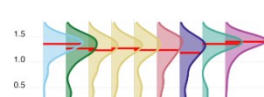
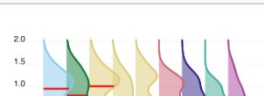
222 participant-day metrics. Among the eight metrics with site support, site-associated part R^2
223 averaged 9.0%. These non-additive components reinforce the stronger individual-level
224 contribution in the daily-pattern analysis.

225 Civil-photoperiod associations were selective. Per additional hour, the fitted daily geometric
226 mean was 1.18 times higher (95% CI 1.11 to 1.26), the brightest-10-hour mean 1.24 times
227 higher (1.13 to 1.36), waking time above 250 lx 1.14 times higher (1.07 to 1.20) and daily
228 dose 1.28 times higher (1.17 to 1.39). Each hour was also associated with 0.134 h less time
229 below 10 lx before sleep (0.066 to 0.201 h less) and the last exposure above 250 lx
230 occurring 0.323 h later (0.186 to 0.460 h). Civil photoperiod met the FDR criterion for 12 of
231 17 metrics but not stability, fragmentation, sleep time below 1 lx, the brightest-window
232 midpoint or first exposure above 250 lx. Among supported models, photoperiod-associated
233 part R^2 averaged 5.2%; the corresponding mean for the six supported latitude models was
234 4.0%, meaning that photoperiod had a stronger effect than latitude, when compared across
235 models.

236 Across the recorded 10.33 to 20.52 h of civil photoperiod, six of nine level, exposure-history,
237 and duration-based metrics changed from increasing to a sustained near-flat tail between
238 14.24 and 16.20 hours of photoperiod: the daily mean, both 10-hour window means, time
239 above 1,000 lx, waking time above 250 lx and dose. Pre-sleep time below 10 lx, sleep-
240 environment time below 1 lx and the longest period above 250 lx did not. This transition
241 means that the effect of photoperiod becomes uncertain around 14 to 16 hours. The fitted
242 slope became compatible with zero and remained so through the observed maximum. Some
243 classifications were sensitive to site, sample or model form. Several limitations apply as
244 discussed below: latitude was confounded with site, collection periods and photoperiod
245 overlap differed, and weather, built environment, culture and routine could not be separated.
246 The metric-specific results are summarised in [Table 3](#), [Supplementary Figure S7](#) and
247 [Supplementary Tables S7](#) and [S8](#).

248 Table 3: Near-eye personal light-exposure metrics and their geographic and photoperiod context. Descriptive exposure values are medians with
249 interquartile ranges (25th to 75th percentiles). Density plots show site distributions and are descriptive. Metrics are grouped into thematic streams
250 and describe distinct features of the daily exposure pattern rather than interchangeable health-response estimates. Model cells report overall-site
251 evidence, civil-photoperiod associations with 95% confidence intervals, FDR decisions, participant-associated model information and model-
252 specific R^2 summaries where applicable. Site and photoperiod part- R^2 values may contain overlapping fitted information and must not be summed.
253 Associations are observational. Absolute latitude is fixed within and confounded with study site, so separate models containing either site or
254 latitude are reported; complete latitude and model-adequacy results are provided in the Supplementary Information.

Near-eye personal light-exposure metrics and their geographic and photoperiod context

		Descriptive summary		Association evidence		Modelled variation
Definition/relevance		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Duration						
Time above 1,000 lx meEDI	Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	Unit h Median 0.683 IQR 0.183 to 1.558 Participants 141 Days 816		FDR not supported; adjusted p 0.071 Part R-squared 4.7% (2.6% to 12.1%)	Estimate Ratio per 1 h: 1.206 (1.128 to 1.288) FDR supported; adjusted p <0.001 Part R-squared 9.0% (3.7% to 15.7%)	Marginal 21.5% (16.3% to 32.3%) Conditional 50.1% (40.2% to 57.6%) Participant-associated 28.6% (17.7% to 34.2%)
Time above 250 lx meEDI during wake	Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	Unit h Median 2.417 IQR 0.850 to 4.617 Participants 141 Days 737		FDR supported; adjusted p 0.007 Part R-squared 8.6% (5.2% to 18.7%)	Estimate Ratio per 1 h: 1.136 (1.073 to 1.202) FDR supported; adjusted p <0.001 Part R-squared 6.4% (2.0% to 12.7%)	Marginal 18.0% (12.9% to 29.1%) Conditional 50.1% (41.6% to 58.0%) Participant-associated 32.1% (20.7% to 36.7%)
Time below 10 lx meEDI before sleep	Low-light time before bed; limits evening melatonin suppression and circadian delay.	Unit h Median 1.883 IQR 1.025 to 2.583 Participants 139 Days 655		FDR supported; adjusted p 0.049 Part R-squared 5.4% (3.2% to 13.4%)	Estimate Difference per 1 h: -0.134 (-0.201 to -0.066) FDR supported; adjusted p <0.001 Part R-squared 4.9% (1.2% to 10.5%)	Marginal 7.6% (4.6% to 17.2%) Conditional 38.3% (31.8% to 48.5%) Participant-associated 30.7% (21.7% to 38.1%)
Time below 1 lx meEDI during sleep	Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	Unit h Median 7.133 IQR 5.967 to 8.250 Participants 141 Days 778		FDR not supported; adjusted p 0.058 Part R-squared 5.4% (3.2% to 14.2%)	Estimate Ratio per 1 h: 0.987 (0.968 to 1.006) FDR not supported; adjusted p 0.227 Part R-squared 0.6% (0.0% to 3.8%)	Marginal 8.0% (5.1% to 18.2%) Conditional 40.9% (32.8% to 48.5%) Participant-associated 32.9% (22.0% to 37.8%)
Longest period above 250 lx meEDI	Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	Unit h Median 0.633 IQR 0.283 to 1.204 Participants 141 Days 816		FDR not supported; adjusted p 0.398 Part R-squared 2.2% (1.4% to 8.1%)	Estimate Ratio per 1 h: 1.125 (1.065 to 1.187) FDR supported; adjusted p <0.001 Part R-squared 4.9% (1.3% to 9.6%)	Marginal 10.0% (6.1% to 17.9%) Conditional 36.3% (29.9% to 45.3%) Participant-associated 26.3% (18.7% to 33.0%)

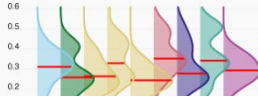
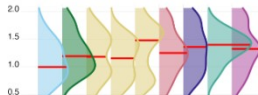
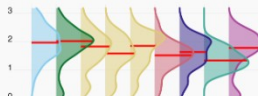
Near-eye personal light-exposure metrics and their geographic and photoperiod context						
Definition/relevance		Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Dynamics						
Interdaily stability	Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	Unit dimensionless Median 0.308 IQR 0.248 to 0.380 Participants 141 Days 816		FDR not supported; adjusted p 0.228 Part R-squared 6.9% (4.1% to 20.4%)	Estimate Odds ratio per 1 h: 0.983 (0.944 to 1.024) FDR not supported; adjusted p 0.401 Part R-squared 0.4% (0.0% to 4.6%)	Marginal 15.4% (10.4% to 30.8%) Conditional 15.4% (10.4% to 30.8%) Participant-associated Not available
Intradaily variability	Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	Unit dimensionless Median 1.253 IQR 0.930 to 1.502 Participants 141 Days 816		FDR not supported; adjusted p 0.483 Part R-squared 5.1% (3.8% to 18.7%)	Estimate Difference per 1 h: –0.019 (–0.056 to 0.019) FDR not supported; adjusted p 0.327 Part R-squared 0.7% (0.0% to 5.7%)	Marginal 5.9% (4.6% to 21.2%) Conditional 5.9% (4.6% to 21.2%) Participant-associated Not available
Exposure history						
meLEDI dose	Intensity–duration-weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	Unit klx·h Median 4.960 IQR 1.936 to 12.313 Participants 141 Days 761		FDR not supported; adjusted p 0.209 Part R-squared 2.8% (1.7% to 8.6%)	Estimate Ratio per 1 h: 1.275 (1.169 to 1.391) FDR supported; adjusted p <0.001 Part R-squared 7.5% (3.1% to 12.9%)	Marginal 11.8% (7.7% to 19.7%) Conditional 33.8% (26.7% to 42.8%) Participant-associated 22.0% (14.5% to 28.7%)

Table continued

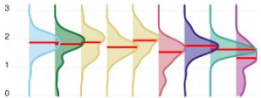
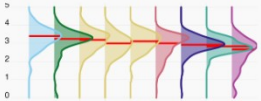
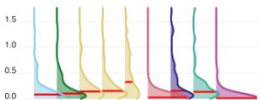
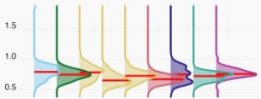
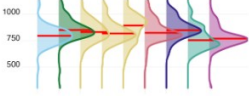
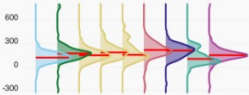
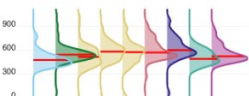
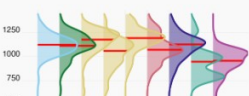
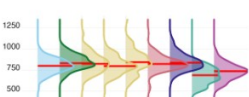
Near-eye personal light-exposure metrics and their geographic and photoperiod context						
		Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Level						
Mean melEDI	Geometric average of daily melEDI values, including zeros; summarizes overall exposure while reducing peak influence.	Unit lx Median 5.154 IQR 2.831 to 9.225 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 8.4% (4.8% to 16.6%)	Estimate Ratio per 1 h: 1.181 (1.109 to 1.257) FDR supported; adjusted p <0.001 Part R-squared 7.7% (2.9% to 13.9%)	Marginal 25.4% (18.6% to 34.4%) Conditional 57.7% (51.9% to 65.1%) Participant-associated 32.3% (24.5% to 39.4%)
Brightest 10 h geometric mean	Offset geometric mean melEDI in the brightest supported 10-hour window; describes the strength of the main sustained daily exposure.	Unit lx Median 110.566 IQR 41.513 to 243.212 Participants 141 Days 816		FDR supported; adjusted p 0.011 Part R-squared 5.7% (3.3% to 12.7%)	Estimate Ratio per 1 h: 1.243 (1.133 to 1.363) FDR supported; adjusted p <0.001 Part R-squared 5.8% (1.9% to 11.1%)	Marginal 17.6% (12.1% to 26.0%) Conditional 46.0% (39.6% to 54.3%) Participant-associated 28.4% (20.9% to 35.4%)
Darkest 10 h geometric mean	Offset geometric mean melEDI in the darkest supported 10-hour window; describes the sustained low-light period.	Unit lx Median 0.103 IQR 0.020 to 0.253 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 13.7% (9.2% to 23.9%)	Estimate Ratio per 1 h: 1.088 (1.034 to 1.144) FDR supported; adjusted p 0.002 Part R-squared 3.4% (0.5% to 8.3%)	Marginal 22.0% (15.8% to 31.8%) Conditional 61.1% (55.3% to 68.1%) Participant-associated 39.1% (30.4% to 46.2%)
Spectrum						
Melanopic daylight efficacy ratio	Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	Unit dimensionless Median 0.724 IQR 0.643 to 0.795 Participants 137 Days 687		FDR supported; adjusted p <0.001 Part R-squared 9.6% (6.0% to 17.8%)	Estimate Difference per 1 h: 0.024 (0.017 to 0.031) FDR supported; adjusted p <0.001 Part R-squared 13.0% (6.7% to 20.6%)	Marginal 27.8% (21.3% to 38.1%) Conditional 60.5% (54.6% to 68.2%) Participant-associated 32.7% (24.1% to 40.0%)

Table continued

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
		Descriptive summary		Association evidence		Modelled variation
	Definition/relevance	Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Timing						
Midpoint of the brightest 10 hours	Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	Unit clock time Median 13:44 IQR 12:48 to 15:00 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 6.5% (4.4% to 12.6%)	Estimate Difference per 1 h: 0.054 (-0.045 to 0.153) FDR not supported; adjusted p 0.305 Part R-squared 0.2% (0.0% to 1.8%)	Marginal 6.7% (4.6% to 13.0%) Conditional 22.9% (16.9% to 31.6%) Participant-associated 16.2% (9.6% to 22.4%)
Midpoint of the darkest 10 hours	Centre time of the darkest 10 hours; indexes the main daily darkness cue.	Unit clock time Median 02:54 IQR 02:01 to 03:50 Participants 141 Days 816		FDR supported; adjusted p 0.020 Part R-squared 4.6% (3.0% to 10.5%)	Estimate Difference per 1 h: -0.148 (-0.245 to -0.050) FDR supported; adjusted p 0.004 Part R-squared 2.1% (0.3% to 5.6%)	Marginal 8.6% (5.9% to 15.8%) Conditional 28.2% (22.1% to 36.7%) Participant-associated 19.6% (12.6% to 25.8%)
First light timing above 250 lx meEDI	First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	Unit clock time Median 09:08 IQR 08:02 to 10:40 Participants 140 Days 727		FDR not supported; adjusted p 0.112 Part R-squared 3.8% (2.3% to 10.2%)	Estimate Difference per 1 h: -0.097 (-0.247 to 0.053) FDR not supported; adjusted p 0.227 Part R-squared 0.5% (0.0% to 2.8%)	Marginal 6.8% (4.4% to 14.4%) Conditional 31.0% (24.6% to 40.5%) Participant-associated 24.2% (16.2% to 31.5%)
Last light timing above 250 lx meEDI	Last bright-light exposure; later timing may delay circadian phase and sleep onset.	Unit clock time Median 18:08 IQR 16:27 to 19:42 Participants 141 Days 687		FDR supported; adjusted p <0.001 Part R-squared 11.6% (8.3% to 18.6%)	Estimate Difference per 1 h: 0.323 (0.186 to 0.460) FDR supported; adjusted p <0.001 Part R-squared 4.5% (1.5% to 8.5%)	Marginal 22.2% (17.0% to 30.2%) Conditional 37.9% (31.6% to 46.3%) Participant-associated 15.8% (9.7% to 22.3%)
Mean timing of exposure above 250 lx meEDI	Average bright-light timing; summarizes the phase of daily circadian stimulation.	Unit clock time Median 13:29 IQR 12:29 to 14:39 Participants 141 Days 742		FDR supported; adjusted p <0.001 Part R-squared 12.7% (9.4% to 19.2%)	Estimate Difference per 1 h: 0.121 (0.028 to 0.213) FDR supported; adjusted p 0.013 Part R-squared 1.2% (0.1% to 3.6%)	Marginal 14.7% (10.7% to 21.9%) Conditional 25.7% (19.8% to 34.2%) Participant-associated 11.0% (5.3% to 17.2%)
Overall distributions are median (interquartile range). Participants is the participant count and Days is the participant-day count. Site and photoperiod part-R ² values can contain overlapping fitted information and must not be summed. Participant-associated R ² is the conditional-minus-marginal contribution associated with the participant random intercept and is undefined for participant-level outcomes. The MDER uses the mean of viable minute-level ratios in both the descriptive summary and geographic-association model cells.						

259 Immediate environments and routines reorganise exposure

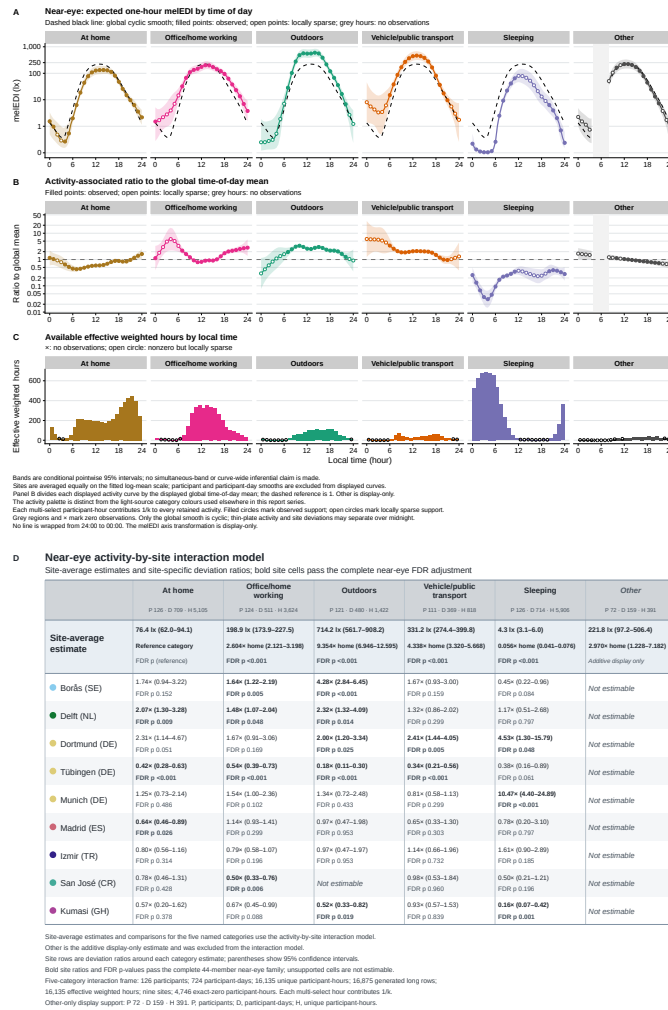
260 Self-reported light source was strongly associated with one-hour near-eye exposure, with an
261 FDR-supported light-source-by-site interaction. Site-average interaction-model estimates
262 were 959.6 lx melanopic EDI for outdoor daylight (95% CI 786.8 to 1,170.2), 198.8 lx for
263 indoor daylight (169.3 to 233.4), 85.1 lx for indoor electric light (71.6 to 101.1) and 23.8 lx for
264 a display screen (17.4 to 32.5). Outdoor daylight was 11.27 times the indoor-electric
265 estimate (8.71 to 14.60) and exceeded the 250-lx daytime recommendation, whereas the
266 indoor-daylight and indoor-electric intervals were below it. During reported sleep, darkness
267 and external-light estimates were 2.24 lx (1.54 to 3.26) and 11.18 lx (8.06 to 15.51), both
268 above the 1-lx bedside upper limit. Outdoor electric light remained imprecise relative to
269 indoor electric light near eye (ratio 1.019, 0.336 to 3.090; FDR-adjusted $p = 0.973$). These
270 contextual averages do not classify people or hours as adherent.

271 External light during sleep in Munich (DE) was estimated at 130.2 lx, 11.65 times the
272 corresponding site average; sleep darkness in Kumasi (GH) was 0.371 lx, 0.166 times its
273 site average. Sparse site-category cells limited some estimates.

274 The primary analysis used a population-mean quasi-Tweedie model with participant-cluster-
275 robust covariance for uncertainty; lag-one residual correlation remained 0.288. Fitted versus
276 observed exact-zero proportions were 39.8% and 27.8%, respectively. An exploratory
277 mixed-effects model quantified variance explained after adding a participant intercept to site-
278 by-light-source fixed terms. Marginal R^2 was 0.796 and conditional R^2 0.876, an 8.0-
279 percentage-point increase in explained variance from the participant intercept. Of the
280 marginal R^2 , light source accounted for 89.3%, site 7.1% and their interaction 3.7%.
281 Separately, an exploratory time-of-day model attributed 60.4% of variation in fitted hourly
282 patterns to global time of day and 24.2% to light-source deviations. Among these two
283 components, light source accounted for 28.6%. The remaining shares were 5.2% for site,
284 7.6% for participant patterns and 2.5% for day-to-day shifts.

285 A sensitivity adjusted the population-average light-source comparisons for each participant's
286 overall category distribution, distinguishing hour-to-hour differences within participants from
287 differences in category composition between participants. The overall category pattern was
288 retained. For external light during sleep, the near-eye ratio relative to indoor electric light
289 changed from 0.449 in the primary additive model to 0.333 (95% CI 0.190 to 0.585), a 25.8%
290 decrease. Thus, between-participant category composition influenced the magnitude, but not
291 the direction, of this contrast. [Supplementary Figure S8](#) shows the temporal, support and
292 site-specific results.

293 Immediate setting combined behaviour and environment, with an FDR-supported activity-by-
294 site interaction. Site-average near-eye estimates were 714.2 lx outdoors (95% CI 561.7 to
295 908.2), 331.2 lx on the road or in a vehicle (274.4 to 399.8), 198.9 lx during office or home
296 work (173.9 to 227.5), 76.4 lx while awake at home (62.0 to 94.1) and 4.28 lx during sleep
297 (3.06 to 5.99). The sleep category included many low or zero observations and occasional
298 high stray-light readings, so its fitted mean need not represent a typical observation. Relative
299 to awake time at home, the ratios were 9.35 outdoors, 4.34 for travel, 2.60 for work and
300 0.056 during sleep. The outdoor estimate exceeded the 250-lx daytime recommendation, the
301 work estimate remained below it, and the sleep estimate exceeded the 1-lx bedside-
302 environment upper limit. Borås (SE) had a fitted outdoor estimate of 3,054 lx, 4.28 times the
303 site average, whereas Tübingen (DE) had 127 lx, 0.18 times the site average. In an
304 exploratory mixed model of five setting categories, marginal R^2 was 0.766 near eye and
305 0.768 at the chest. Adding the participant intercept raised conditional R^2 to 0.858 and 0.859,
306 an explained-variance increment of 9.2 percentage points at both positions. Of the marginal
307 R^2 , immediate setting accounted for 80.7% near eye and 86.1% at the chest, site 12.5% and
308 5.9%, and their interaction 6.9% and 8.0%, respectively. [Figure 3](#) shows the immediate-
309 setting patterns and site-specific estimates. Separately, an exploratory near-eye time-of-day
310 model attributed 52.5% of variation in fitted hourly patterns to time of day and 30.7% to
311 immediate setting. Among these two components, immediate setting accounted for 36.9%.
312 The remaining shares were 5.6% for site, 8.4% for participant patterns and 2.8% for day-to-
313 day shifts.



314

315 *Figure 3: Personal light exposure by immediate setting across time of day and study*
316 *sites. The diary variable labelled activity combines behaviour and environmental*
317 *setting. A–C, Exploratory time-of-day analysis. Panel A shows site-average*
318 *expected one-hour melanopic EDI with clock-specific 95% confidence intervals; the*
319 *dashed black curve is the site-average local-clock smooth. Panel B shows each*
320 *setting curve relative to that smooth, with 1 as the reference. Panel C shows*
321 *effective weighted hours; open circles indicate locally sparse support and grey*
322 *regions indicate times without observations. D, Site-average category estimates and*
323 *site-specific deviation ratios from the activity-by-site interaction model, with*
324 *participant-cluster-robust 95% confidence intervals and FDR-adjusted p values.*
325 *Each multi-select participant-hour contributes weight 1/k to every retained setting*
326 *category. The Other category is sparse and display-only. The fitted frame contains*
327 *126 participants, 724 participant-days, 16,135 unique participant-hours, 16,135*
328 *effective weighted hours and 4,746 exact-zero participant-hours. Estimates are*
329 *observational.*

330 Across models of free versus work day, active versus sedentary daily activity status and
331 previous-night sleep duration, the site-average reference was a work, sedentary day
332 following an average 7.8 h of sleep; expected near-eye exposure was 89 lx melanopic EDI
333 (95% CI 67 to 119). Active versus sedentary status was associated with 2.06 times the

334 expected exposure (1.54 to 2.76; FDR-adjusted $p < 0.001$). A model constraining the free-
335 versus-work association across sites estimated 1.45 times higher exposure on free days
336 (1.14 to 1.85; FDR-adjusted $p = 0.004$). Allowing the association to differ by site gave an
337 inconclusive site-average ratio of 1.15 (0.93 to 1.44; FDR-adjusted $p = 0.298$), while some
338 sites differed (interaction $p = 0.017$). Exploratory ratios were FDR-retained in Borås (SE;
339 1.95, 1.26 to 3.01) and Dortmund (DE; 2.44, 1.42 to 4.22), with FDR-adjusted $p = 0.013$ for
340 each. The free-versus-work result was also inconclusive under the alternative working
341 variance and sensitive to omitting either site. Each additional hour of previous-night sleep
342 duration, centred at 8 h, had an inconclusive ratio of 0.98 (0.85 to 1.12; FDR-adjusted $p =$
343 0.745). Excluding six participants outside the employment requirement left 15,871 supported
344 hours from 684 participant-days and 131 participants across all nine sites. The free-versus-
345 work, active-versus-sedentary and previous-sleep associations remained stable within model
346 uncertainty; the predictor-by-site interaction for day type also remained FDR-supported.
347 [Supplementary Table S9](#) and Supplementary Figures [S9](#), [S10](#), [S11](#) and [S12](#) report common,
348 temporal, position-specific and site-specific results.

349 **Person-level correlates are selective**

350 Self-report analyses did not support general exposure profiling. None of 68 prespecified
351 near-eye associations between four light-exposure-related behaviour (LEBA) factors and 17
352 metrics retained FDR support. Unsatisfactory model checks made the sleep-environment
353 metric unfit for inference, so its estimates were suppressed. Neither average associations
354 nor site interactions across nine visual-light-sensitivity (VLSQ-8) outcomes retained FDR
355 support at either position. Supplementary Tables [S10](#), [S11](#) and [S12](#), and Supplementary
356 Figures [S13](#) and [S14](#), give the full results.

357 Each hour later in corrected midsleep on free days (MCTQ questionnaire) was associated
358 with first exposure above 250 lx occurring 0.381 h later (95% CI 0.146 to 0.616; FDR-
359 adjusted $p = 0.003$). Each 10-point shift towards greater morning preference (MEQ
360 questionnaire) was separately associated with first exposure above 250 lx occurring 0.444 h
361 earlier (0.189 to 0.699 h earlier; FDR-adjusted $p = 0.001$). The brightest- and darkest-10-
362 hour midpoints followed the same directions; last exposure and the longest-bright-period
363 midpoint did not retain support. The distinct questionnaires were not combined.
364 [Supplementary Table S13](#) and [Supplementary Figure S15](#) show adjusted estimates and
365 observed timing distributions.

366 Each additional decade of age was associated with higher near-eye brightest-10-hour mean,
367 time above 1,000 lx and daily dose, with ratios from 1.27 to 1.31. These cross-sectional

368 associations may reflect cohort, occupation, behaviour or other confounding, not ageing
369 itself. Neither near-eye metric associations with biological sex nor near-eye age-by-site or
370 sex-by-site interactions met the FDR criterion. In a separate daily-pattern analysis, near-eye
371 and chest profiles differed by biological sex (FDR-adjusted $p = 0.028$ and 0.029), but the
372 difference did not meet the FDR criterion after restriction to records with complete
373 immediate-setting data and adjustment. This attenuation is compatible with immediate-
374 setting patterning, but restriction and adjustment cannot be disentangled as mechanisms.
375 Gender was recorded separately but not analysed; its category matched biological sex in all
376 but one case. Supplementary Tables [S10](#), [S14](#) and [S15](#), and Supplementary Figures [S16](#)
377 and [S17](#), give the complete results.

378 **Measurement position and robustness constrain** 379 **interpretation**

380 The paired sample of 112 participants and 643 participant-days allowed separate near-eye
381 and chest models to be compared within the same participants and participant-days. The full
382 chest dataset of 154 participants and 902 participant-days broadened coverage beyond the
383 141 participants and 816 participant-days near eye. At several sites, especially San José
384 (CR), offering a chest sensor enabled recruitment among people reluctant to wear the
385 glasses-mounted sensor. This improved feasibility, not the ocular-exposure sample.

386 A parallel analysis of overlapping MeLiDos data reported structured position differences
387 across scale, metric, context, site, day and participant³³. Our separate and common-sample
388 analyses likewise show directional agreement but differences in magnitude or support. Eight-
389 site chest adherence retained daytime and pre-sleep directions and interval conclusions,
390 with calibration and temporal qualifications, but provided no sleep ocular claim or paired test.
391 We neither pooled positions nor inferred equivalence or a universal correction.

392 Non-estimable planned quantities were four sleep-environment variants (main plus three
393 sensitivity variants) at each position in the light-behaviour analysis (LEBA), and one longest-
394 period sensitivity comparison incompatible with the metric representation. We retained these
395 outcomes rather than replacing them with zero or null findings. Qualifications for zeros,
396 residuals, dependence, nonlinear-term overlap, sparse support and influence remain with
397 the affected analyses and in Methods.

398 Discussion

399 The expected 24-hour rhythm was evident across nine sites in seven countries. The
400 advance was a multiscale account of this everyday exposome component: amplitude across
401 sites, people and days, and its reorganisation across immediate settings. The shared local-
402 clock pattern accounted for the largest share of full-model R^2 in the daily-pattern analysis.
403 Separate hourly decompositions showed large light-source and immediate-setting contrasts,
404 with the next-largest shares of variation in fitted hourly patterns after time of day. Participant
405 patterns and participant-day shifts together exceeded site patterns in the daily-pattern
406 model, while site, photoperiod and measured person-level associations were metric-specific.
407 Different outcomes and decompositions make this a qualitative synthesis rather than a single
408 quantitative ranking.

409 The consensus recommendations provide the clearest health-relevant benchmark. Only
410 24.0% of waking minutes reached the daytime range, while 63.3% of pre-sleep and 87.7% of
411 bedside sleep-environment minutes met their upper limits. Modelled free-day adherence was
412 lower during daytime and preceding sleep, varying by site. Pre-sleep adherence was higher
413 but inconclusive under stricter coverage; temporal dependence remained unresolved.
414 Outdoor exposure exceeded the daytime recommendation, but all other conditions remained
415 below it on average. The fitted sleep-category mean exceeded the bedside limit, while
416 87.7% of pooled bedside sleep-environment minutes met it. These quantities describe
417 different aspects of the exposure distribution and should be interpreted together. These
418 pooled fractions, recommendation-window proportions and hourly means are distinct
419 estimands. Together, they identify intervention opportunities and shortfalls. The
420 recommendations principally address healthy adults aged 18 to 55 with regular daytime
421 schedules⁸.

422 This baseline connects mechanistic recommendations with population evidence. Large wrist-
423 sensor cohorts have associated day and night light patterns with psychiatric disorders,
424 incident type 2 diabetes, cardiovascular disease and mortality, and light timing with sleep
425 timing and sleepiness¹⁰⁻¹⁵. Other cohorts have linked evening, nocturnal or strongly seasonal
426 light with sleep in bipolar disorder, obesity, diabetes, hypertension and metabolic health¹⁶⁻¹⁹.
427 These outcomes span sleep and major non-communicable diseases. The exposome frames
428 environmental contributions across the life course^{20,21} but rarely includes time-resolved
429 ocular light. This evidence supports personal-light measurement, not clinical risk
430 classification of our exposure categories. Most cohorts used wrist or bedside illuminance

431 without resolving melanopic exposure at the eye, immediate light source or its multiscale
432 architecture.

433 Exploratory inverse associations of average daytime with sleep and pre-sleep adherence
434 show why the three recommendation windows should not be collapsed into one score. They
435 may reflect schedules, environments or other between-participant differences, not a within-
436 person trade-off or stable phenotype. In exploratory nonlinear models, light source and
437 immediate setting accounted for substantially more variation in fitted hourly patterns than
438 participant-specific patterns: 24.2% versus 7.6%, and 30.7% versus 8.4%, respectively. This
439 finding is consistent with schedules and immediate environments being prominent correlates
440 of hourly exposure, but it is not a causal explanation for the cross-window adherence
441 association. Longitudinal models of dependence and changing routines are needed to test
442 whether improving one part of the pattern displaces or supports another.

443 Location mattered, but not as one north-to-south ranking. Site differences concentrated in
444 daily level, waking bright-light duration and several timing metrics, while dose, spectrum and
445 regularity did not show the same site pattern. Civil photoperiod was associated with a
446 broader set of metrics, with larger day-length associations for bright-light level and dose than
447 for the darkest daily window. Between 14.2 and 16.2 h of civil photoperiod, six of nine
448 metrics showed signs of uncertainty about a photoperiod effect beyond that point. Sites were
449 sampled in distinct calendar periods and covered unequal photoperiod ranges, leaving
450 latitude, season, weather, temperature, culture, mobility and the built environment entangled.
451 Repeated seasonal sampling across more sites is needed to separate these influences.

452 Immediate setting patterned exposure, consistent with environmental opportunity, built
453 settings and behaviour³⁶. Outdoor daylight and being outdoors exceeded indoor electric light
454 and awake time at home, respectively; active days exceeded sedentary days. Site
455 interactions placed outdoors-category exposure higher in Borås (SE) than Tübingen (DE),
456 while the free-versus-work association was most evident in Borås (SE) and Dortmund (DE).
457 Participant intercepts captured remaining between-participant variation, not individual
458 responses or predictions. Future monitoring should preserve local clock time, location,
459 immediate setting, daylight access and site; self-selection, occupation, mobility and weather
460 remain potential confounders.

461 Light-behaviour and visual-sensitivity scores did not meet their FDR criteria, whereas
462 chronotype aligned with selected timing outcomes and age with selected bright-light and
463 dose metrics. This accords with evidence linking circadian preference or timing to daylight
464 exposure, sleep and routine, without identifying causal direction^{37–40}. Metric-level biological-
465 sex comparisons were inconclusive near the eye. The complete available-data curve differed

466 by biological sex, but immediate-setting-complete analyses did not retain this separation.
467 Person-level patterns may concern specific exposure dimensions rather than a general high-
468 versus-low phenotype.

469 Near-eye sensing approximates the incident ocular field during wake but is not a retinal
470 measurement; chest sensing samples a related environmental field. Studies show structured
471 differences by body position, morphology, posture and surrounding light²⁹⁻³¹. The matched
472 sample and parallel placement study show metric- and context-specific MeLiDos differences
473 without a universal correction³³. The chest option broadened recruitment among people
474 unwilling to wear glasses-mounted sensors, especially in San José (CR). Study design must
475 balance biological proximity, burden and recruitment while keeping estimands position-
476 specific.

477 The harmonised protocol combined repeated near-eye sensing with local-clock and
478 contextual information across nine sites and seven countries. A generalised additive model
479 separated the shared 24-hour pattern from site, participant and participant-day layers
480 obscured by daily summaries. Collecting these data demanded substantial effort from
481 participants and study teams. Glasses-mounted loggers required fitting, explanation and
482 daily handling; removal, occlusion, charging, discomfort, weather and social acceptability all
483 affected measurement quality⁴¹⁻⁴³. Despite this burden, the protocol measured closer to the
484 incident ocular field at high temporal and contextual resolution.

485 Several limitations qualify these findings. Modest site samples limit site estimates,
486 interactions and subgroup evaluation. Healthy working-age adults may not represent
487 children, adolescents, older or clinical populations, shift workers, people outside
488 employment, or those with restricted mobility or atypical living conditions. Sparse latitude-by-
489 photoperiod coverage leaves latitude, seasonality, culture, climate and built environment
490 entangled and requires broader seasonal sampling. Retrospective hourly setting and light-
491 source reports enabled harmonisation but could not distinguish fine-grained settings or
492 behaviours within home, work or transport. Very low readings require caution. The
493 manufacturer specifies an operating range starting at 1 lx, and independent laboratory
494 testing began at 2 lx⁴⁴. A scene-based field validation found high overall agreement with
495 laboratory-grade spectral references, but underestimation increased at lower intensities and
496 varied with lighting, time, scene complexity and site⁴⁵. Single-digit values therefore
497 approximate low-light exposure rather than provide precise measurements. Finally, no health
498 outcome was measured.

499 This nine-site reference locates everyday ocular light exposure within the exposome,
500 benchmarks it against physiology-based recommendations, highlights within-site variation

501 among people and days, and identifies temporal, contextual and positional information for
502 future cohorts. Everyday ocular light exposure is structured, measurable and strongly
503 shaped by behaviour and micro-environment, but cannot be reduced to site, latitude or
504 photoperiod. Next steps are broader sampling across photoperiods, latitudes and locations
505 and prospective links to sleep, mental, metabolic, cardiovascular and other outcomes.
506 Causal health benefits or policy effects require interventions and representative longitudinal
507 designs. Potential interventions through clinical advice, occupational health, building design,
508 urban planning or public-health guidance must address the contexts in which people receive
509 light. Time-resolved near-eye measurement offers a practical route from circadian biology to
510 precision prevention and context-sensitive environmental health.

511 Methods

512 Study design and sites

513 MeLiDos is a prospective multicentre observational field study of personal light exposure
514 conducted with a harmonised protocol³⁵. Participants completed an eight-day ambulatory
515 protocol comprising continuous light logging, repeated smartphone assessments and daily
516 diaries under free-living conditions. At an in-person registration visit, participants provided
517 consent, completed baseline questionnaires, received study instructions and were fitted with
518 the loggers. Seven consecutive recording days separated registration and return visits; most
519 sites recorded from Monday to Monday, while Dortmund (DE) recorded from Tuesday to
520 Tuesday. If a return visit was delayed, participants continued until formal study termination
521 and device retrieval. Data were collected between August 2023 and October 2025 at Borås
522 (SE), Delft (NL), Dortmund (DE), Tübingen (DE), Munich (DE), Madrid (ES), Izmir (TR), San
523 José (CR) and Kumasi (GH). The Kumasi (GH) and Izmir (TR) collections are described in
524 site-specific data notes^{46,47}. Sites were an opportunity sample of consortium partners and
525 were not selected to represent national populations. Questionnaire materials were
526 administered in the local language except in Tübingen (DE) and Kumasi (GH), where
527 English was used, and translated materials followed a harmonised translation, review,
528 adjudication, pretesting and documentation process⁴⁸.

529 Participants, recruitment and ethics

530 The registered target population comprised adults aged 18 to 65 years in full-time
531 employment or part-time employment above 80%. Exclusion criteria included diagnosed
532 psychiatric or sleep disorders, shift work in the preceding two months, regular tobacco or
533 recreational drug use, photosensitising medication, visual impairment incompatible with the
534 devices and residence outside the local study area during recording. The retained near-eye
535 sample included nine participants outside the registered age or employment criteria: one
536 participant older than 65 years, two recorded as not employed and seven recorded as
537 marginally employed; one participant met both the age and employment criteria for
538 exclusion. As a near-eye-only eligibility sensitivity, we refitted the additive and predictor-by-
539 site models after excluding participants recorded as not employed or marginally employed.
540 Students and trainees were retained, and age above 65 years was not an exclusion criterion
541 for this sensitivity. The target was at least 15 participants per site, based on prior power
542 calculations and allowing for up to one-third dropout or incomplete recording⁴⁹. Participants
543 were recruited through local advertisements and institutional mailing lists. At some sites,

544 especially San José (CR), reluctance to wear the glasses-mounted near-eye sensor limited
545 recruitment, and the chest option broadened recruitment possibilities.

546 All participants provided written informed consent. Approvals were obtained from the Ethics
547 Committee of the Technical University of Munich for Tübingen (DE; 2023-115-S-KK) and for
548 Munich (DE) and Dortmund (DE; 2024-118-S-SB); the Dortmund (DE) site at the Federal
549 Institute for Occupational Safety and Health (BAuA) operated under this TUM multicentre
550 approval. The other approvals were from the Swedish Ethical Review Authority for Borås
551 (SE; 2024-06792-01), the THUAS Ethics Committee for Delft (NL; positive review dated 29
552 May 2024), the Research Ethics Committee of Universidad San Pablo-CEU for Madrid (ES;
553 834/24/106), the Scientific Research and Publication Ethics Committee of the Faculty of
554 Science and Engineering at İzmir Institute of Technology for Izmir (TR; decision 5/1), the
555 Comité Ético Científico of Universidad de Costa Rica for San José (CR; CEC-279-2025),
556 and the Committee on Human Research, Publications and Ethics of Kwame Nkrumah
557 University of Science and Technology for Kumasi (GH; CHRPE/AP/644/24). The protocol
558 was first trialled with participants at the Tübingen (DE) site, whose feedback informed
559 procedural refinements before implementation at the remaining sites; further details are
560 reported in the published protocol³⁵.

561 **Personal light measurements**

562 Personal light was recorded every 10 s with spectrally sensitive ActLumus loggers (Condor
563 Instruments, São Paulo, Brazil). Eight visible-band channels were processed by the device
564 to generate photopic and alpha-opic quantities, including melanopic EDI⁶. The manufacturer
565 specified a factory-calibrated operating range of 1 to 100,000 lx and 10% precision at 1,000
566 lx. Independent testing against a criterion spectrometer across indoor LED and outdoor
567 daylight conditions covered 2 to 100,000 lx and reported mean photopic error of +3.3% (SD
568 13.4%) for ActLumus⁴⁴. A subsequent scene-based field validation found high overall
569 agreement with laboratory-grade spectral reference instruments, together with systematic
570 underestimation that varied with intensity and setting⁴⁵. Participants wore the near-eye
571 device centrally on non-prescription spectacle frames and, where available, a
572 complementary pendant-mounted device at the chest. The near-eye position measures light
573 closer to the eye but is not a direct retinal measurement. The chest position provided a
574 separate measurement of the local environmental light field. Paired common-sample
575 analyses evaluated how closely it tracked the near-eye measurement as a potential ocular
576 proxy. During reported sleep, devices were placed facing upward on a bedside surface near

577 head level, so both positions characterise the bedside sleep environment. Participants
578 recorded removals and placed devices in an opaque bag during waking non-wear^{43,50}.

579 The placement analysis from the same project used overlapping MeLiDos data and is cited
580 as related methodological evidence, not independent replication³³. No analysis pooled
581 sensor positions or estimated a universal conversion. Common-sample analyses retained
582 the same participants and participant-days or participant-hours at both positions and fitted
583 each position separately.

584 **Contextual and person-level measures**

585 Participants completed the core Consensus Sleep Diary after awakening⁵¹, together with
586 evening reports and paper diaries recording light source and an activity-environment
587 category for each hour. Smartphone assessments of sleepiness and mood were scheduled
588 at 11:00, 14:00, 17:00 and 20:00, and evening questionnaires assessed wellbeing and
589 physical activity⁵²⁻⁵⁴. Smartphone measures and baseline or end-of-study questionnaires
590 used MyCap integrated with REDCap^{55,56}. Participants photographed and uploaded the
591 paper hourly diaries each day.

592 The modified Harvard Light Exposure Assessment categories were indoor electric light,
593 outdoor electric light, indoor daylight, outdoor daylight including shade, an emissive display,
594 darkness during sleep and external light during sleep⁵⁷. The primary light-source analysis
595 used one retained category per participant-hour. The diary variable labelled activity
596 combined behaviour and environmental setting. Its categories distinguished sleep, awake
597 time at home, motorised travel, active travel, office or home work, outdoor work, outdoor
598 leisure and other activities. We therefore refer to this variable as immediate setting in the
599 narrative. For analysis, outdoor travel, outdoor work and outdoor leisure were combined as
600 Outdoors. Reports could contain concurrent categories; duplicates were removed and an
601 hour with k retained labels contributed weight $1/k$ to each, preserving total weight one. Day
602 type was coded as work or free day, daily activity status as sedentary or active, and previous
603 sleep duration was linked from the preceding sleep period.

604 Baseline measures recorded age, biological sex, gender, employment and chronotype; end-
605 of-study measures covered visual light sensitivity, the sleep environment and habitual light-
606 exposure behaviour. Chronotype was represented separately by corrected midsleep on free
607 days from the Munich Chronotype Questionnaire and the Morningness-Eveningness
608 Questionnaire, whose higher scores indicate greater morning preference^{58,59}. Visual
609 sensitivity used the VLSQ-8⁶⁰, sleep environment used the Assessment of Sleep
610 Environment⁶¹, and four Light Exposure Behaviour Assessment (LEBA) factors represented

611 outdoor time, device use in bed, ambient light before bed and morning or daytime light use⁶².
612 Biological sex and gender were recorded as separate variables. Models analysed biological
613 sex, coded Female or Male, but did not analyse gender; its recorded category matched
614 biological sex in all but one case.

615 **Time, sleep/wake status and coverage preparation**

616 Processing preserved both actual time in Coordinated Universal Time and local wall-clock
617 time. Actual time ordered observations and measured elapsed gaps; local time aligned daily
618 patterns across sites. Repeated local minutes during autumn clock changes remained
619 distinct in actual time, and structurally absent spring minutes remained missing. Melanopic
620 EDI values at or above the validated 100,000-lx operating boundary were unavailable
621 analytically but retained in the raw channel with a flag.

622 Diary-defined sleep began at reported sleep preparation and ended at reported wake. Sleep
623 values were retained as the intended bedside record. Declared waking non-wear made the
624 analytical light value unavailable; missing wear labels were not treated as non-wear.
625 Complete 1,440-minute local participant-day grids preserved unsupported periods as
626 missing rather than deleting rows or replacing them with zero. A day was eligible when at
627 least 80% of this combined waking and bedside-sleep record had valid melanopic EDI.
628 Hourly summaries required at least 50% valid minutes in that hour, but an unsupported hour
629 did not erase valid minutes from the daily decision or minute-based metrics. Two otherwise
630 eligible near-eye days and three chest days with finite but entirely zero melanopic EDI were
631 excluded by the defined exact-all-zero screen.

632 **Exposure metrics**

633 Metrics were derived in R⁶³ with LightLogR^{64,65}. The 17 outcomes represented
634 complementary properties of the daily pattern. Three level metrics described the daily
635 geometric mean melanopic EDI and the mean melanopic EDI in the brightest and darkest
636 10-hour windows. Five duration metrics quantified time above 1,000 lx, waking time above
637 250 lx, pre-sleep time below 10 lx, sleep-environment time below 1 lx and the longest
638 continuous period above 250 lx. Five timing metrics located the brightest and darkest
639 windows and the first, last and mean timing above 250 lx. Melanopic EDI dose integrated
640 exposure over time, melanopic daylight efficacy ratio characterised spectral composition
641 relative to visual illuminance, and interdaily stability and intradaily variability characterised
642 regularity and fragmentation. Geometric means were calculated from $\log_{10}(\text{melanopic EDI} +$
643 $0.1 \text{ lx})$, followed by inverse transformation and removal of the same offset⁶⁶. All-observed-

644 zero windows were retained as exact zero within verified numerical tolerance. Metric-specific
645 support rules made only the affected metric unavailable, not the participant-day. A longest-
646 period duration remained a lower bound when missing data could conceal a longer period;
647 exactly identified periods were examined separately.

648 The melanopic daylight efficacy ratio was the arithmetic mean of viable one-minute
649 melanopic EDI to photopic-illuminance ratios on a complete 1,440-position local-clock grid.
650 Both channels had to be finite and strictly positive, and at least 720 positions were required.
651 Repeated autumn minutes were averaged within each channel before division, and absent
652 spring minutes remained missing.

653 **Recommendation-window and adherence analyses**

654 We compared valid one-minute values with the Brown et al. recommendation ranges⁸: at
655 least 250 lx melanopic EDI during daytime, no more than 10 lx during pre-sleep and no more
656 than 1 lx during sleep. Daytime comprised diary-defined wake excluding the three hours
657 before reported sleep, pre-sleep comprised those three hours and sleep comprised the
658 reported sleep interval. Each descriptive numerator was divided by all valid near-eye-record
659 minutes in the corresponding window and pooled across the analysed sample. Daytime and
660 pre-sleep used the worn near-eye record; sleep used the bedside sleep-environment record.

661 The main adherence analysis retained one row per window period and defined adherence as
662 adherent valid minutes divided by all valid minutes in that period. Each cycle linked the sleep
663 interval ending at wake, the ensuing daytime interval excluding pre-sleep, and the following
664 three hours before the next reported sleep. All three inherited the work or free label of the
665 wake-start calendar date, even when a window crossed midnight. Actual diary dates and
666 UTC boundaries established chronology; gaps between diary dates were not bridged.
667 Windows qualified independently, so a cycle need not contain three eligible windows. The
668 any-valid primary sample retained at least one valid comparison; the coverage sensitivity
669 required valid minutes for at least 80% of each expected window interval. Expected duration
670 defined coverage, not the likelihood denominator. Missing minutes were not imputed. Exact
671 0% and 100% periods remained in an endpoint-inflated beta-binomial model. Its mean
672 included the full window-by-site-by-day-type interaction and a participant intercept; extra
673 wholly non-adherent and wholly adherent periods and window-specific dispersion were
674 modelled separately. Population-average adherence integrated the participant distribution
675 and gave each of nine sites equal weight. Separate FDR families covered three window-
676 specific free-minus-work contrasts, three window-specific site interactions, the global
677 interaction, 27 site-specific contrasts against zero, 54 site-minus-average levels and 27 site-

678 effect-minus-average contrasts. The last family's coverage repetition assessed stability
679 without another FDR adjustment. Variance decomposition used balanced 54-cell global and
680 18-cell within-window references; Shapley allocations and R^2 were point-only. Chest models
681 used a separate two-window, eight-site sample and did not test paired placement
682 differences.

683 The exploratory cross-window extension used daytime adherence as the predictor and the
684 preceding sleep or following pre-sleep adherence as outcomes under the same wake-start-
685 date alignment. It retained pairwise-complete observations, not necessarily complete three-
686 window cycles: 1,376 outcome rows comprised 758 daytime-sleep and 618 daytime-pre-
687 sleep pairs from 140 participants and 761 cycles. Its frozen eligibility rules did not include
688 every potential main-model pair. The model separated each participant's cycle-specific
689 daytime deviation from their observed average, adjusting for window, site, day type, their
690 corresponding interactions and the participant's day-type composition. Response-scale
691 contrasts used equal-site and 50:50 work/free weighting. The four within- and between-
692 participant associations formed a separate FDR family. Positive serial dependence remained
693 unresolved and some sensitivity fits were non-estimable, so no within-participant day-level
694 claim was stated. Descriptive profiles used unweighted equal-cycle participant means, with
695 daytime deduplicated within participant and cycle, for 139 complete profiles and 417
696 participant-window points. These are anonymous monitoring-period summaries, not ranks or
697 stable participant types.

698 The main model passed all five applicable endpoint-calibration checks in both samples, but
699 temporal dependence still qualified inference. The 80% coverage restriction retained the
700 daytime and sleep directions and interval conclusions; pre-sleep retained direction but not
701 exclusion of zero. Diary-indexed grouping was a separate sensitivity. Pre-sleep interval
702 conclusions also changed under 70% and 90% coverage and restriction to participants
703 observed on both day types. Actual-date temporal endpoint models failed structural checks.
704 Their beta-binomial fallbacks changed both the response family and random-effects
705 structure, retained boundary cautions and failed all five endpoint-calibration checks in both
706 samples. They did not replace the primary model or resolve its temporal limitation. All nine
707 site-deletion checks and four participant-deletion checks were estimable; one participant
708 deletion and the ordinary-binomial diagnostic failed and supplied no derived inference. The
709 separate chest model retained calibration and temporal qualifications. In the exploratory
710 extension, the 80% restriction preserved all four directions and interval conclusions, but
711 incomplete influence checks and unresolved dependence limited robustness claims.

712 Daily temporal architecture

713 The daily-pattern analysis used supported 30-minute arithmetic means, retaining a bin when
714 at least 15 one-minute values were valid. The outcome was $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$. A
715 generalised additive model fitted with `mgcv::bam()` and fREML combined a cyclic common
716 local-clock smooth, sum-to-zero site curves, participant curves and participant-day random
717 intercepts^{67–70}. A boundary-aware first-order autoregressive working structure prevented
718 residual sequences from crossing participant-day, clock-repeat or gap boundaries. Fitted-
719 curve dispersion was compared on a common 48-bin grid. A Shapley decomposition
720 averaged each component's addition to R^2 across all component orders, thereby distributing
721 the model's explained variance^{71,72}. The decomposition partitioned the full model's row-
722 weighted in-sample R^2 for $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$, defined as the reduction in squared
723 prediction error relative to predicting the fitted-sample mean. The percentages therefore
724 partition full-model R^2 on that transformed response scale. They are not percentages of
725 variance explained in raw melanopic EDI and do not indicate causal or out-of-sample
726 predictive importance. Confidence intervals for dispersion and R^2 -share ratios used 2,000
727 hierarchical cluster resamples while holding the fitted model structure fixed.

728 Geographic and photoperiod analyses

729 We analysed 17 participant-level or participant-day metrics with metric-specific Gaussian or
730 Tweedie exponential-dispersion models^{73,74}. We treated overall site differences, civil
731 photoperiod (day length including civil twilight), absolute latitude and the adequacy of a
732 linear latitude term relative to site as four separate model questions, each adjusted as a
733 complete 17-test FDR family. Participant-day models included participant random intercepts.
734 Mixed-effects models were fitted with `lme4` or `glmmTMB`, and component R^2 values were
735 obtained with `performance` where reported^{75–77}. Site and latitude were not entered as
736 independent fixed predictors in one model because each site has one latitude; site and
737 latitude models instead used the same observations and photoperiod adjustment. Site
738 follow-up contrasts were calculated with `emmeans` only after a supported overall site test and
739 compared each site with a site-average estimate, obtained by giving each included site
740 equal weight⁷⁸.

741 Nonlinear photoperiod analyses fitted each of nine planned metrics separately with a
742 generalised additive model estimated by restricted maximum likelihood, a thin-plate
743 photoperiod smooth with basis dimension six and participant random effects nested within
744 site. The derivative, meaning the slope of the fitted curve, was evaluated with `gratia` at 100

745 equally spaced photoperiod values with 95% intervals⁷⁹. A qualifying descriptive transition
746 required the preceding slope interval to be wholly above zero, the next interval to include
747 zero and every later interval through the recorded maximum to include zero as well. This
748 procedure located a change in the fitted slope but did not test equivalence, a mechanism or
749 a physiological ceiling. Expanded-basis, fixed-site, common-sample and leave-one-site-out
750 analyses assessed stability. The gap-timing-unaware sensitivity dataset still passed the
751 50%-per-hour and 80%-per-day rules, but did not use the timing of remaining missing
752 observations for an additional metric-specific adjustment.

753 **Immediate-context and routine analyses**

754 Light-source and activity-environment analyses modelled the zero-aware one-hour
755 geometric mean melanopic EDI with fixed-power quasi-Tweedie log-mean models and
756 participant-cluster-robust inference^{73,74}. Additive fixed-site models supplied primary category
757 tests. Category-by-site interaction models supplied site-average category estimates, ratios
758 and site-specific context when the interaction was supported. The activity-environment
759 analysis retained fractional $1/k$ weights for concurrent labels. FDR adjustments were applied
760 to the declared category and site-contrast families. Exact-zero calibration, residual
761 dependence, sparse cells and deletion refits were retained as model checks. A secondary
762 population-mean sensitivity added six participant-level category-proportion terms to the light-
763 source model while retaining participant-clustered covariance. This separated hour-to-hour
764 category differences within participants from between-participant differences in category
765 composition; it did not estimate participant random slopes. Exploratory `glmmTMB` Tweedie
766 log-link models fitted by maximum likelihood then added one common participant intercept to
767 site-by-category fixed effects at the near-eye position for light source and separately at both
768 positions for the activity-environment variable, with the power fixed at 1.539919^{76,77}. The
769 activity-environment models retained the five named categories, excluded Other-only hours
770 and used exact $1/k$ weights. Their fixed-predictor variance used the same fractional weights
771 so every participant-hour contributed one unit. Marginal and conditional R^2 used a lognormal
772 distribution-specific variance approximation, and hierarchy-respecting Shapley allocations
773 distributed marginal R^2 across site, category and their interaction. Percentages from the
774 exploratory time-of-day models instead partition variance among fitted linear-predictor
775 values. Because they do not compare this with observed outcome variance, they are not R^2 .
776 Results describe them as variation in fitted hourly patterns, not variance explained in raw
777 exposure. These were model-based point descriptions without bootstrap intervals. No model
778 included random category slopes.

779 The participant-hour analysis modelled each supported participant-hour with a zero-inclusive
780 quasi-Poisson log-link model, fixed site associations and participant-cluster HC3 uncertainty.
781 Free versus work day, active versus sedentary daily activity status and previous-night sleep
782 duration centred at 8 h were entered together. Three-test FDR families were applied
783 separately to pooled associations, predictor-by-site interactions and practical contrasts.

784 **Person-level analyses**

785 Light-related behaviour, visual sensitivity, chronotype, age and measured biological sex
786 were evaluated with site-adjusted metric-specific models. Participant-day outcomes
787 generally used Gaussian or Tweedie mixed models with participant random intercepts;
788 participant-level dynamics outcomes used fixed-effect models. Each analysis retained its
789 complete prespecified false-discovery-rate families, including non-estimable outcomes.
790 Chronotype instruments were modelled separately across five timing outcomes. Age
791 associations were scaled per decade, and female-minus-male contrasts used male as the
792 reference. Average associations and predictor-by-site interactions were distinct questions.

793 The complete biological-sex curve analysis used supported 30-minute arithmetic means
794 transformed as $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$. The generalised additive model included
795 cyclic common and female-specific curves, site curves, participant curves, participant-day
796 random effects and boundary-aware first-order autoregression. The primary test jointly
797 assessed the time-constant and time-varying female-minus-male contributions across the
798 complete 24-hour curve. Clock-specific uncertainty was participant-cluster robust. Level-
799 versus-shape attribution and adjustment for the activity-environment variable were
800 secondary or exploratory analyses.

801 **Multiplicity, uncertainty, model checks and sensitivities**

802 Tests were two-sided. We controlled the false discovery rate with the Benjamini-Hochberg
803 procedure separately within each declared complete family⁸⁰. We report 95% confidence
804 intervals and FDR-adjusted p values for confirmatory families. A result that did not meet an
805 FDR-adjusted criterion was treated as inconclusive rather than as evidence of absence or
806 equivalence. Unless stated otherwise, site-average estimates gave each included site equal
807 weight on the fitted scale before back-transformation. Exploratory analyses retained
808 separate multiplicity families and are labelled as such.

809 Model checks addressed convergence, rank, singularity, residual distribution, zero mass,
810 fitted support, heteroscedasticity, serial dependence, nonlinear-term overlap and participant
811 or site influence. Qualifications were attached to the affected results. Prespecified or

812 structured sensitivities included the separately prepared gap-timing-unaware dataset,
813 common sensor-position samples, alternative support or metric definitions, model-form
814 changes, participant deletion and leave-one-site-out checks. Near-eye and chest models
815 remained separate in every comparison. Analyses were conducted in R, with tidyverse
816 packages supporting data handling and reporting^{63,81}. Exact package versions, formulas,
817 inputs and output files are documented in the Supplementary Information and reproducibility
818 records.

819 Preregistration and deviations

820 The study was preregistered with AsPredicted (#273407)⁸². The final analyses preserve the
821 registered scientific questions but include documented changes to placement priority, state
822 and coverage rules, metric definitions, outcome construction, model families, site structure,
823 multiplicity and sensitivity handling. The main manuscript reports the resulting estimands
824 rather than the historical hypothesis sequence. A complete mapping in the Supplementary
825 Information distinguishes scientific deviations and qualifications from analytical history and
826 technical provenance.

827 Artificial-intelligence assistance

828 During the original manuscript preparation, generative artificial intelligence tools (ChatGPT
829 5.5) were used to support language editing, structural refinement and text condensation. AI
830 assistance was used to improve clarity, coherence, concision and journal fit, but **not** to
831 generate, analyse or interpret primary data.

832 During primary data analysis, AI assistance was used to debug code sections, support figure
833 and table styling, and resolve modelling convergence and singularity issues. Participant
834 comments in their native languages were translated into English with AI assistance. All
835 translations were reviewed and approved or corrected by the respective site authors. AI
836 assistance was **not** used to develop the primary analysis strategy, write analysis scripts or
837 interpret results.

838 During review, an agentic AI assistant (ChatGPT 5.6 Sol) was used to review the analyses
839 for potential issues. This review prompted additional sensitivity analyses to evaluate data-
840 preparation and model-selection choices, expanded model diagnostics, and improved effect-
841 size reporting through Shapley decompositions and more precise language. AI assistance
842 also supported expanded analysis documentation and reproducibility checks.

843 All AI-assisted text was critically reviewed, edited, and approved by the authors, who take
844 full responsibility for the content, accuracy, integrity, and conclusions of the manuscript.

845 Data availability

846 All data used in this study are available through the [MeLiDos Project GitHub organisation](#)
847 and are archived in the [MeLiDos Zenodo community](#). Public site datasets have persistent
848 records for Kumasi (GH), Madrid (ES), Dortmund (DE), Izmir (TR), Tübingen (DE), San José
849 (CR), Munich (DE), Borås (SE) and Delft (NL)^{83–91}. These records and other project
850 resources are indexed through the MeLiDos Project Data Hub⁹² and accessed reproducibly
851 through the documented `melidosData` interface⁹³.

852 Code availability

853 All analysis code is available in the [project repository](#), with rendered reports on [GitHub](#)
854 [Pages](#) and an archive on [Zenodo](#). The archived version will be reconciled with the final
855 manuscript snapshot before submission.

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- 1120 • **Software:** J.Z.
- 1121 • **Supervision:** J.Z., A.D., S.A., J.B. and M.S.
- 1122 • **Validation:** J.Z., A.D. and M.S.
- 1123 • **Visualization:** J.Z.
- 1124 • **Writing – original draft:** J.Z.
- 1125 • **Writing – review & editing:** all authors.

1126 Competing interests

1127 M.S. declares the following potential competing interests between 2021 and 2025: academic
1128 roles as a member of the Board of Directors of the Society of Light, Rhythms, and Circadian
1129 Health, Chair of Joint Technical Committee 20 of the International Commission on
1130 Illumination, member of the Daylight Academy, and Chair of the Research Data Alliance
1131 Working Group Optical Radiation and Visual Experience Data; remunerated roles as
1132 Speaker of the Steering Committee of the Daylight Academy, ad hoc reviewer for the Health
1133 and Digital Executive Agency of the European Commission and the Swedish Research
1134 Council, Associate Editor for *LEUKOS*, examiner for the University of Manchester, Flinders
1135 University and the University of Southern Norway, and consultant for LyS Technologies and
1136 RoX Health; research funding and support from the Max Planck Society, Max Planck
1137 Foundation, Max Planck Innovation, Technical University of Munich, Wellcome Trust,
1138 National Research Foundation Singapore, European Partnership on Metrology, VELUX
1139 Foundation, Bayerisch-Tschechische Hochschulagentur, BayFrance/Bayerisch-
1140 Französisches Hochschulzentrum, BayFOR/Bayerische Forschungsallianz and Reality Labs
1141 Research; honoraria for talks from ISGlobal, the Research Foundation of the City University
1142 of New York and Stadt Ebersberg, Museum Wald und Umwelt; travel reimbursements from
1143 the Daimler und Benz Stiftung; and being named on European Patent Application
1144 EP23159999.4A, "System and method for corneal-plane physiologically-relevant light
1145 logging with an application to personalized light interventions related to health and well-
1146 being." M.S. declares that these roles and relationships had no influence on the work
1147 presented here.

1148 J.Z. declares the following potential competing interests between 2021 and 2025: academic
1149 roles as a member of Joint Technical Committee 20 of the International Commission on
1150 Illumination, member of the Research Data Alliance Working Group Optical Radiation and
1151 Visual Experience Data, and Speaker of group 2, melanopic effects of light, of the Technical
1152 Scientific Committee of the German Society of Lighting Technology and Design;
1153 remunerated roles as examiner for the Swiss Lighting Society; teacher for the German
1154 Society of Lighting Technology and Design, Munich University of Applied Sciences and the
1155 Technical University of Applied Sciences Rosenheim; associated partner at 3lpi lighting
1156 design + engineering, Munich; tool and three-dimensional-model designer for Zumtobel
1157 Lighting GmbH; and course designer for Munich University of Applied Sciences and Virtual
1158 University Bavaria; honoraria for talks from the German Society of Lighting Technology and
1159 Design, Lamilux/Heinrich Strunz GmbH, Robert Bosch Hospital Stuttgart, Ergotopia GmbH,

1160 the German Social Accident Insurance Institution for the Administrative Sector, BRIXEN
1161 CULTUR, KITEO GmbH & Co. KG and Augsburg University of Applied Sciences; travel
1162 reimbursements from the Daimler und Benz Stiftung; and, together with 3lpi, holding
1163 European Union Intellectual Property Office design patent 008194021-0001 through -0006
1164 for a non-visually optimized luminaire.

1165 A.D. declares academic roles as a member of Joint Technical Committee 20 of the
1166 International Commission on Illumination and Division Reporter DR6-50 for the “4th
1167 Manchester Workshop on Light Metrics for Biology: Light Pollution.” A.D. declares that these
1168 roles had no influence on the work presented here. D.B. is also owner of Minus 15 B.V., a
1169 company active in the lighting sector. All other authors declare no competing interests.

1170 **Supplementary Information**

1171 **Study support and exposure definitions**

1172 **Supplementary Table S1. Descriptive sample flow**

Descriptive sample flow and analytical support

Stage	Quantity	Count
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Participant roster

Available normalized participant metadata	Participants	191
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Primary near-eye sensor position

At least 80% complete before all-zero screen	Participant-days	818
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Exact all-zero days excluded	Participant-days	2
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Final descriptive dataset	Participants	141
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Final descriptive dataset	Participant-days	816
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Final descriptive dataset	One-minute real observations	1,175,160
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Complementary chest sensor position

At least 80% complete before all-zero screen	Participant-days	905
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Exact all-zero days excluded	Participant-days	3
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Final descriptive dataset	Participants	154
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Final descriptive dataset	Participant-days	902
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Final descriptive dataset	One-minute real observations	1,298,880
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Paired main subset

Paired main subset	Participants	112
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Paired main subset	Paired participant-days	643
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Counts use their stated denominator. An excluded participant-day is an all-zero day, not necessarily an excluded participant. The paired row reports paired participant-days, not minute-level observation pairs.

1174 **Supplementary Table S1.** Participant, participant-day, and one-minute observation counts across the roster, sensor-specific completeness and all-zero
1175 screens, final descriptive datasets, and paired main subset. Each row names its denominator explicitly. An excluded participant-day is not necessarily an
1176 excluded participant.

1177 **Supplementary Table S2. Full near-eye metric dictionary and distributions**

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Duration													
Time above 1,000 lx meEDI Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	HH:MM	00:41 (00:11, 01:33) <i>01:10 ± 01:28</i> N=141; d=816	01:18 (00:31, 03:24) <i>02:00 ± 01:54</i> N=13; d=78	01:18 (00:29, 02:12) <i>01:33 ± 01:28</i> N=13; d=78	01:03 (00:20, 02:32) <i>01:48 ± 02:14</i> N=18; d=107	00:40 (00:07, 01:25) <i>01:06 ± 01:22</i> N=26; d=150	00:39 (00:10, 01:46) <i>01:09 ± 01:18</i> N=10; d=60	00:27 (00:06, 00:53) <i>00:38 ± 00:47</i> N=23; d=129	00:30 (00:06, 01:04) <i>00:44 ± 00:51</i> N=17; d=101	00:19 (00:09, 00:43) <i>00:32 ± 00:33</i> N=6; d=32	00:43 (00:17, 01:14) <i>00:58 ± 00:57</i> N=15; d=81	Symlog (base 10; threshold 1)	
Time above 250 lx meEDI during wake Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	HH:MM	02:25 (00:51, 04:37) <i>03:00 ± 02:29</i> N=141; d=737	04:06 (02:30, 06:15) <i>04:21 ± 02:24</i> N=13; d=73	03:29 (02:03, 05:16) <i>03:39 ± 02:25</i> N=13; d=64	03:42 (02:05, 05:58) <i>04:05 ± 02:51</i> N=18; d=90	02:01 (00:47, 04:20) <i>02:52 ± 02:39</i> N=26; d=141	02:38 (00:55, 04:19) <i>02:57 ± 02:19</i> N=10; d=54	02:43 (00:54, 04:43) <i>03:02 ± 02:19</i> N=23; d=113	01:30 (00:41, 03:26) <i>02:09 ± 01:57</i> N=17; d=94	01:39 (00:33, 02:40) <i>01:48 ± 01:21</i> N=6; d=30	01:14 (00:25, 02:37) <i>01:45 ± 01:46</i> N=15; d=78	Symlog (base 10; threshold 1)	
Time below 10 lx meEDI before sleep Low-light time before bed; limits evening melatonin suppression and circadian delay.	HH:MM	01:53 (01:01, 02:35) <i>01:51 ± 01:02</i> N=139; d=655	02:16 (01:30, 02:39) <i>02:05 ± 00:53</i> N=13; d=67	01:50 (01:21, 02:52) <i>01:59 ± 00:57</i> N=12; d=50	01:42 (00:53, 02:17) <i>01:41 ± 00:58</i> N=18; d=84	01:52 (01:04, 02:25) <i>01:52 ± 01:06</i> N=26; d=132	01:43 (00:42, 02:32) <i>01:45 ± 01:10</i> N=10; d=52	01:43 (00:58, 02:50) <i>01:50 ± 01:05</i> N=23; d=105	01:31 (00:49, 02:15) <i>01:38 ± 01:00</i> N=17; d=82	02:16 (01:15, 02:39) <i>01:56 ± 01:00</i> N=6; d=25	02:30 (01:28, 02:52) <i>02:09 ± 00:57</i> N=14; d=58	Symlog (base 10; threshold 1)	
Time below 1 lx meEDI during sleep Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	HH:MM	07:08 (05:58, 08:15) <i>07:04 ± 01:57</i> N=141; d=778	07:08 (06:03, 08:18) <i>07:12 ± 01:46</i> N=13; d=76	07:14 (05:56, 08:06) <i>07:01 ± 01:50</i> N=13; d=68	07:08 (05:56, 08:16) <i>07:05 ± 01:50</i> N=18; d=95	06:54 (05:51, 07:46) <i>06:47 ± 01:30</i> N=26; d=150	05:52 (04:41, 07:03) <i>05:58 ± 01:58</i> N=10; d=60	08:06 (07:10, 09:10) <i>08:02 ± 01:57</i> N=23; d=118	06:45 (05:08, 08:15) <i>06:40 ± 02:07</i> N=17; d=99	06:56 (06:06, 07:36) <i>06:39 ± 01:43</i> N=6; d=31	07:31 (06:31, 09:00) <i>07:35 ± 02:15</i> N=15; d=81	Symlog (base 10; threshold 1)	
Longest period above 250 lx meEDI Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	HH:MM	00:38 (00:17, 01:12) <i>00:55 ± 01:00</i> N=141; d=816	00:57 (00:28, 01:27) <i>01:12 ± 01:02</i> N=13; d=78	00:48 (00:23, 01:20) <i>01:09 ± 01:15</i> N=13; d=78	01:00 (00:21, 01:40) <i>01:20 ± 01:31</i> N=18; d=107	00:35 (00:13, 01:06) <i>00:50 ± 00:54</i> N=26; d=150	00:38 (00:17, 01:32) <i>01:01 ± 01:02</i> N=10; d=60	00:38 (00:16, 01:04) <i>00:45 ± 00:36</i> N=23; d=129	00:34 (00:14, 00:55) <i>00:43 ± 00:41</i> N=17; d=101	00:28 (00:16, 00:43) <i>00:30 ± 00:20</i> N=6; d=32	00:23 (00:15, 00:49) <i>00:38 ± 00:45</i> N=15; d=81	Symlog (base 10; threshold 1)	
Dynamics													
Interdaily stability Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	—	0.308 (0.248, 0.38) <i>0.318 ± 0.095</i> N=141; d=816	0.311 (0.272, 0.367) <i>0.315 ± 0.078</i> N=13; d=78	0.258 (0.239, 0.389) <i>0.288 ± 0.094</i> N=13; d=78	0.264 (0.219, 0.319) <i>0.274 ± 0.074</i> N=18; d=107	0.329 (0.263, 0.424) <i>0.344 ± 0.107</i> N=26; d=150	0.244 (0.218, 0.277) <i>0.252 ± 0.058</i> N=10; d=60	0.352 (0.321, 0.455) <i>0.368 ± 0.092</i> N=23; d=129	0.278 (0.254, 0.394) <i>0.323 ± 0.104</i> N=17; d=101	0.342 (0.315, 0.473) <i>0.37 ± 0.118</i> N=6; d=32	0.293 (0.255, 0.354) <i>0.299 ± 0.062</i> N=15; d=81	Identity	
Intradaily variability Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	—	1.253 (0.93, 1.502) <i>1.229 ± 0.389</i> N=141; d=816	0.997 (0.818, 1.388) <i>1.088 ± 0.343</i> N=13; d=78	1.193 (0.96, 1.389) <i>1.238 ± 0.345</i> N=13; d=78	1.182 (0.928, 1.49) <i>1.177 ± 0.407</i> N=18; d=107	1.154 (0.801, 1.442) <i>1.125 ± 0.354</i> N=26; d=150	1.48 (0.869, 1.642) <i>1.332 ± 0.477</i> N=10; d=60	1.251 (0.999, 1.497) <i>1.251 ± 0.413</i> N=23; d=129	1.359 (0.95, 1.546) <i>1.307 ± 0.415</i> N=17; d=101	1.399 (1.282, 1.483) <i>1.355 ± 0.256</i> N=6; d=32	1.324 (1.029, 1.595) <i>1.344 ± 0.413</i> N=15; d=81	Identity	

1178

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Exposure history													
melEDI dose Intensity–duration–weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	klx·h	4.96 (1.936, 12.313) <i>10.358 ± 16.216</i> N=141; d=761	9.296 (3.892, 21.895) <i>19.923 ± 27.784</i> N=13; d=71	10.495 (3.855, 17.395) <i>15.881 ± 20.395</i> N=13; d=74	6.751 (2.469, 14.435) <i>13.809 ± 23.711</i> N=18; d=97	3.862 (1.411, 10.217) <i>7.719 ± 9.847</i> N=26; d=137	7.11 (1.897, 15.789) <i>10.055 ± 11.237</i> N=10; d=55	3.317 (1.703, 6.29) <i>6.115 ± 8.79</i> N=23; d=125	4.35 (1.791, 10.625) <i>7.49 ± 8.898</i> N=17; d=96	2.195 (1.339, 4.117) <i>3.281 ± 2.573</i> N=6; d=30	6.045 (2.49, 14.907) <i>10.016 ± 10.799</i> N=15; d=76	Symlog (base 10; threshold 1)	
Level													
Mean melEDI Geometric average of daily melEDI values, including zeros; summarizes overall exposure while reducing peak influence.	lx	5.154 (2.831, 9.225) <i>7.587 ± 9.496</i> N=141; d=816	7.569 (4.55, 11.613) <i>9.171 ± 7.111</i> N=13; d=78	6.659 (4.18, 11.674) <i>8.274 ± 6.558</i> N=13; d=78	7.704 (3.081, 13.924) <i>12.017 ± 18.865</i> N=18; d=107	5.232 (3.224, 8.188) <i>7.221 ± 6.33</i> N=26; d=150	9.013 (4.011, 15.089) <i>12.102 ± 12.013</i> N=10; d=60	3.505 (1.469, 6.211) <i>4.166 ± 3.272</i> N=23; d=129	5.857 (3.726, 7.986) <i>7.778 ± 6.946</i> N=17; d=101	4.325 (3.585, 6.77) <i>5.248 ± 2.85</i> N=6; d=32	2.144 (0.744, 4.042) <i>3.019 ± 2.896</i> N=15; d=81	Symlog (base 10; threshold 1)	
Brightest 10 h mean Mean of the brightest 10 hours; reflects the strength of the main daytime light episode.	lx	110.566 (41.513, 243.212) <i>214.866 ± 508.56</i> N=141; d=816	252.689 (104.514, 461.953) <i>399.302 ± 518.102</i> N=13; d=78	175.974 (78.625, 299.106) <i>262.206 ± 329.201</i> N=13; d=78	156.184 (67.783, 339.731) <i>422.54 ± 1,209.616</i> N=18; d=107	100.057 (33.255, 209.904) <i>179.372 ± 241.53</i> N=26; d=150	130.78 (54.044, 259.726) <i>201.947 ± 194.042</i> N=10; d=60	101.102 (24.355, 218.457) <i>136.312 ± 134.07</i> N=23; d=129	83.24 (53.17, 151.786) <i>120.172 ± 101.545</i> N=17; d=101	68.143 (45.331, 132.458) <i>96.735 ± 72.311</i> N=6; d=32	46.66 (9.242, 103.175) <i>82.489 ± 115.07</i> N=15; d=81	Symlog (base 10; threshold 1)	
Darkest 10 h mean Mean of the darkest 10 hours; lower values during the biological night favour melatonin preservation and sleep.	lx	0.103 (0.02, 0.253) <i>0.243 ± 0.513</i> N=141; d=816	0.08 (0.016, 0.191) <i>0.183 ± 0.299</i> N=13; d=78	0.1 (0.035, 0.23) <i>0.2 ± 0.332</i> N=13; d=78	0.14 (0.036, 0.282) <i>0.223 ± 0.299</i> N=18; d=107	0.149 (0.074, 0.309) <i>0.273 ± 0.465</i> N=26; d=150	0.325 (0.119, 0.71) <i>0.515 ± 0.653</i> N=10; d=60	0.022 (0, 0.08) <i>0.054 ± 0.075</i> N=23; d=129	0.152 (0.052, 0.481) <i>0.475 ± 0.882</i> N=17; d=101	0.133 (0.055, 0.309) <i>0.411 ± 1.044</i> N=6; d=32	0.01 (0, 0.092) <i>0.059 ± 0.104</i> N=15; d=81	Symlog (base 10; threshold 1)	
Spectrum													
Melanopic daylight efficacy ratio Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	—	0.724 (0.643, 0.795) <i>0.724 ± 0.116</i> N=137; d=687	0.788 (0.725, 0.872) <i>0.796 ± 0.162</i> N=13; d=74	0.743 (0.699, 0.796) <i>0.743 ± 0.077</i> N=13; d=65	0.781 (0.675, 0.853) <i>0.768 ± 0.118</i> N=18; d=97	0.65 (0.592, 0.728) <i>0.667 ± 0.098</i> N=26; d=144	0.724 (0.652, 0.773) <i>0.72 ± 0.093</i> N=10; d=55	0.668 (0.629, 0.721) <i>0.673 ± 0.07</i> N=22; d=83	0.747 (0.634, 0.834) <i>0.735 ± 0.122</i> N=17; d=91	0.719 (0.661, 0.761) <i>0.706 ± 0.067</i> N=6; d=30	0.755 (0.729, 0.82) <i>0.758 ± 0.097</i> N=12; d=48	Identity	

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Timing													
Midpoint of the brightest 10 hours Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	HH:MM clock time	13:44 (12:48, 15:00) 13:56 ± 01:51 N=141; d=816	13:16 (12:28, 14:10) 13:20 ± 01:33 N=13; d=78	14:12 (13:23, 15:18) 14:22 ± 01:51 N=13; d=78	13:52 (12:54, 15:04) 14:00 ± 01:42 N=18; d=107	13:38 (12:47, 14:44) 13:52 ± 01:42 N=26; d=150	14:54 (12:50, 16:03) 14:31 ± 02:14 N=10; d=60	13:44 (13:06, 15:20) 14:10 ± 01:51 N=23; d=129	14:10 (13:32, 15:21) 14:25 ± 01:37 N=17; d=101	12:34 (11:49, 14:02) 12:55 ± 01:44 N=6; d=32	12:49 (12:06, 13:39) 13:03 ± 01:58 N=15; d=81	Circular clock	
Midpoint of the darkest 10 hours Centre time of the darkest 10 hours; indexes the main daily darkness cue.	HH:MM clock time	02:54 (02:01, 03:50) 02:58 ± 01:41 N=141; d=816	02:02 (01:25, 02:48) 02:03 ± 01:07 N=13; d=78	02:55 (02:01, 03:27) 02:50 ± 02:05 N=13; d=78	02:32 (01:45, 03:22) 02:37 ± 01:19 N=18; d=107	03:08 (02:11, 04:10) 03:16 ± 01:34 N=26; d=150	02:37 (01:54, 04:17) 03:10 ± 02:00 N=10; d=60	03:39 (02:43, 04:34) 03:34 ± 01:49 N=23; d=129	03:31 (02:33, 04:14) 03:26 ± 01:25 N=17; d=101	01:45 (01:05, 03:01) 02:10 ± 01:27 N=6; d=32	02:33 (02:04, 03:17) 02:36 ± 01:36 N=15; d=81	Circular clock	
First light timing above 250 lx melEDI First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	HH:MM clock time	09:08 (08:02, 10:40) 09:26 ± 02:22 N=140; d=727	07:57 (06:56, 08:33) 08:03 ± 01:32 N=13; d=74	09:04 (08:18, 09:52) 09:17 ± 01:56 N=13; d=68	08:36 (07:10, 10:13) 08:45 ± 02:21 N=18; d=93	09:43 (08:13, 10:53) 09:45 ± 02:21 N=25; d=133	09:36 (07:58, 10:53) 09:39 ± 03:12 N=10; d=56	09:32 (08:40, 11:21) 10:05 ± 02:18 N=23; d=118	10:01 (09:06, 11:52) 10:26 ± 02:28 N=17; d=95	08:11 (07:26, 09:57) 08:36 ± 01:30 N=6; d=29	08:43 (08:01, 10:10) 09:19 ± 01:52 N=15; d=61	Circular clock	
Last light timing above 250 lx melEDI Last bright-light exposure; later timing may delay circadian phase and sleep onset.	HH:MM clock time	18:08 (16:27, 19:42) 18:03 ± 02:36 N=141; d=687	18:41 (17:38, 19:38) 18:39 ± 01:38 N=13; d=73	18:30 (17:37, 19:59) 18:37 ± 02:08 N=13; d=62	19:34 (18:04, 20:42) 19:16 ± 02:28 N=18; d=89	17:37 (15:35, 19:09) 17:18 ± 02:44 N=26; d=121	19:50 (18:37, 20:50) 19:33 ± 02:11 N=10; d=53	17:48 (16:46, 19:19) 18:01 ± 02:21 N=23; d=106	18:45 (17:02, 19:38) 18:23 ± 02:15 N=17; d=89	15:45 (13:20, 16:38) 15:22 ± 02:38 N=6; d=28	15:54 (14:34, 17:25) 15:46 ± 02:13 N=15; d=66	Circular clock	
Mean timing of exposure above 250 lx melEDI Average bright-light timing; summarizes the phase of daily circadian stimulation.	HH:MM clock time	13:29 (12:29, 14:39) 13:31 ± 01:52 N=141; d=742	13:20 (12:29, 14:24) 13:21 ± 01:25 N=13; d=72	13:52 (13:13, 15:06) 14:02 ± 01:41 N=13; d=71	13:44 (12:28, 14:55) 13:51 ± 01:56 N=18; d=93	13:17 (12:17, 14:31) 13:24 ± 01:45 N=26; d=138	14:04 (12:58, 15:25) 14:09 ± 02:22 N=10; d=55	13:43 (12:55, 14:29) 13:47 ± 01:33 N=23; d=117	13:58 (12:54, 14:58) 14:04 ± 01:39 N=17; d=94	11:30 (10:28, 12:38) 11:26 ± 01:36 N=6; d=31	12:17 (11:04, 13:04) 12:11 ± 01:36 N=15; d=71	Circular clock	
Median (25th percentile, 75th percentile), <i>circular or arithmetic mean ± standard deviation</i> , N=participants; d=participant-days with a finite value for that metric.													
¹ The brief meaning and relevance notes describe established physiological constructs; this descriptive table does not estimate individual health effects.													
² Scaling describes the distribution column only; printed values remain on their stated scale. Symlog uses base 10 with a linear region through 1 in the stated unit; Identity is linear; Circular clock unwraps values around the metric-specific clock centre.													
³ Red lines indicate site medians. Symlog and circular-clock rows are transformed only for plotting.													

1180

1181 **Supplementary Table S2.** Definitions, exact support, median (interquartile range) summaries, and density displays for the 17 near-eye personal light-exposure metrics overall and by site. Values use three decimal places where
1182 applicable. Metrics are distinct summaries and should not be interpreted as interchangeable outcomes.

1183 **Supplementary Table S3. Descriptive recommendation-range fractions**

Site ¹	Minutes in the recommended range: ²				Share of all eligible real minutes: ³			
	Daytime	Pre-sleep	Sleep ⁴	Total ⁵	Daytime	Pre-sleep	Sleep	Unclassified ⁶
Overall	24.0% 137,792 / 573,711	63.3% 81,894 / 129,391	87.7% 336,052 / 383,355	51.2% 555,738 / 1,086,106	51.6% 606,215 / 1,175,140	11.9% 140,152 / 1,175,140	32.6% 383,607 / 1,175,140	3.8% 45,186 / 1,175,140
Borås (SE)	33.2% 19,225 / 57,952	69.9% 8,914 / 12,759	93.4% 33,152 / 35,489	57.7% 61,291 / 106,200	54.0% 60,642 / 112,260	12.2% 13,704 / 112,260	31.7% 35,549 / 112,260	2.1% 2,365 / 112,260
Delft (NL)	29.6% 14,760 / 49,852	66.7% 7,181 / 10,774	86.4% 29,915 / 34,621	54.4% 51,856 / 95,247	47.0% 52,755 / 112,260	10.9% 12,283 / 112,260	30.9% 34,682 / 112,260	11.2% 12,540 / 112,260
Dortmund (DE)	32.4% 22,631 / 69,900	59.6% 9,749 / 16,361	87.5% 42,829 / 48,928	55.6% 75,209 / 135,189	48.1% 74,094 / 154,080	11.3% 17,409 / 154,080	31.8% 48,928 / 154,080	8.9% 13,649 / 154,080
Tübingen (DE)	22.5% 25,082 / 111,396	61.8% 15,623 / 25,283	87.1% 61,030 / 70,029	49.2% 101,735 / 206,700	55.1% 119,150 / 216,120	12.4% 26,881 / 216,120	32.4% 70,089 / 216,120	0.0% 0 / 216,120
Munich (DE)	25.0% 10,681 / 42,685	60.6% 5,963 / 9,833	71.7% 21,485 / 29,951	46.2% 38,129 / 82,469	53.0% 45,831 / 86,400	12.3% 10,618 / 86,400	34.7% 29,951 / 86,400	0.0% 0 / 86,400
Madrid (ES)	24.2% 21,067 / 87,030	62.7% 12,766 / 20,366	95.9% 58,374 / 60,891	54.8% 92,207 / 168,287	48.4% 89,948 / 185,880	11.7% 21,684 / 185,880	32.8% 60,891 / 185,880	7.2% 13,357 / 185,880
Izmir (TR)	17.8% 12,708 / 71,562	55.1% 9,221 / 16,736	79.0% 39,936 / 50,556	44.6% 61,865 / 138,854	51.5% 74,926 / 145,440	12.3% 17,883 / 145,440	34.8% 50,556 / 145,440	1.4% 2,075 / 145,440
San José (CR)	13.8% 3,266 / 23,640	64.6% 3,124 / 4,839	87.1% 12,438 / 14,275	44.0% 18,828 / 42,754	54.9% 25,285 / 46,080	11.5% 5,320 / 46,080	31.0% 14,275 / 46,080	2.6% 1,200 / 46,080
Kumasi (GH)	14.0% 8,372 / 59,695	75.2% 9,353 / 12,439	95.5% 36,893 / 38,626	49.3% 54,618 / 110,760	54.5% 63,584 / 116,640	12.3% 14,370 / 116,640	33.2% 38,686 / 116,640	0.0% 0 / 116,640

¹ Near eye participant and participant-day sample sizes are reported in the participant table; every percentage cell gives its exact minute denominator.

² Recommended ranges follow Brown et al. (2022): Daytime ≥250 lx melanopic EDI, Pre-sleep ≤10 lx melanopic EDI, and Sleep ≤1 lx melanopic EDI. Daytime, Pre-sleep, and Sleep identify the recommendation windows. Each grey numerator/denominator gives minutes within the applicable recommendation range over all valid one-minute observations in that window. Sleep describes the bedside sleep environment.

³ Each grey relevant / all count gives minutes in the displayed diary state over all eligible real minutes before classification by diary state or measurement availability.

⁴ During diary-defined sleep the bedside sensor describes the sleep environment rather than direct ocular exposure.

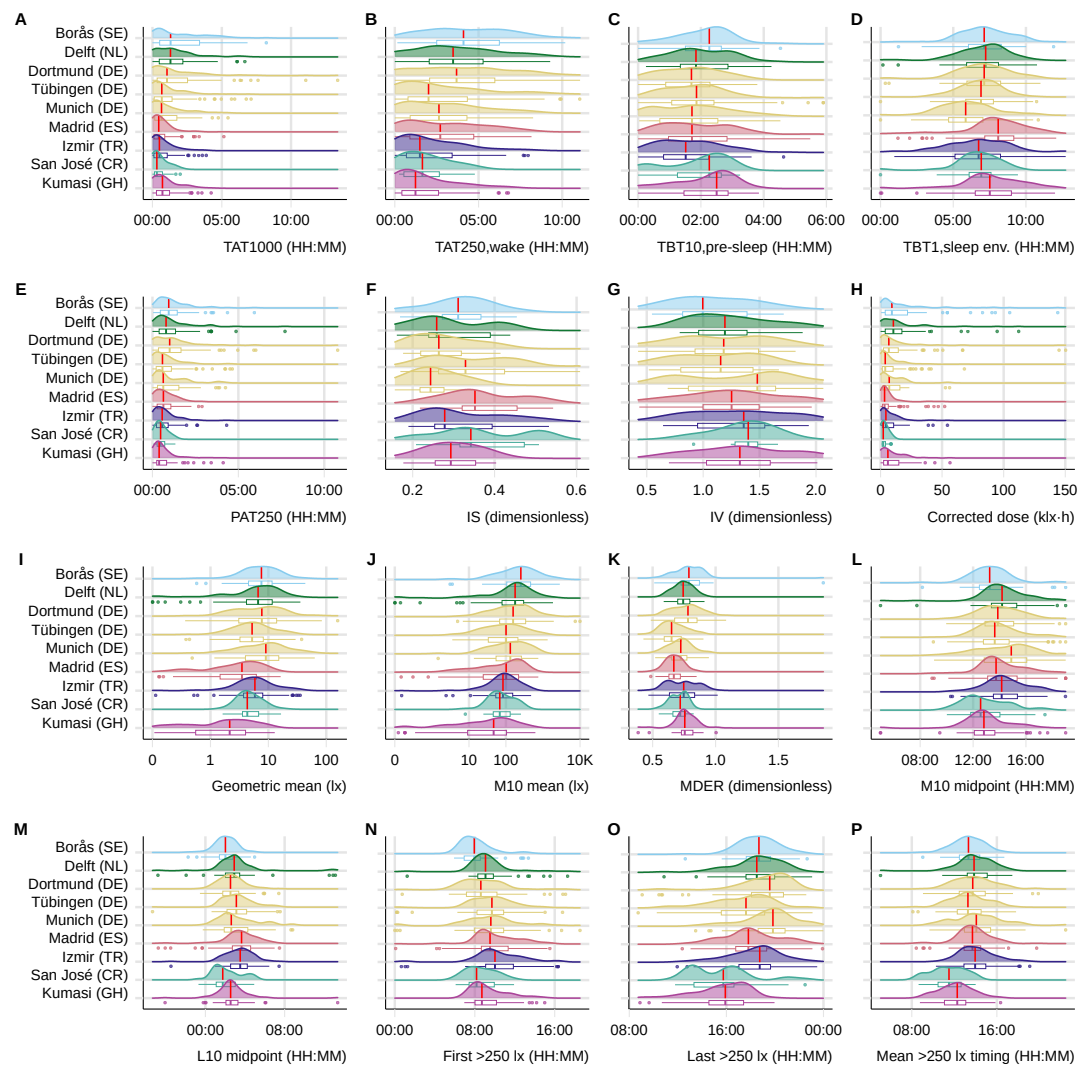
⁵ Total pools all classified valid minutes.

⁶ Unclassified time can result from missing diary state, missing or removed melEDI measurements, or both.

1185 **Supplementary Table S3.** Fractions of valid near-eye minutes meeting each Brown et al. recommendation range, with explicit numerators and denominators.
1186 Daytime, Pre-sleep, and Sleep identify the recommendation windows. The Sleep summaries use near-eye illuminance recorded during sleep and describe the
1187 bedside sleep environment rather than verified ocular exposure. These pooled-minute summaries are descriptive and differ from the modelled
1188 recommendation-window-period estimates in main Table 2.

1189 **Supplementary Figure S1. Near-eye metric distributions**

1190

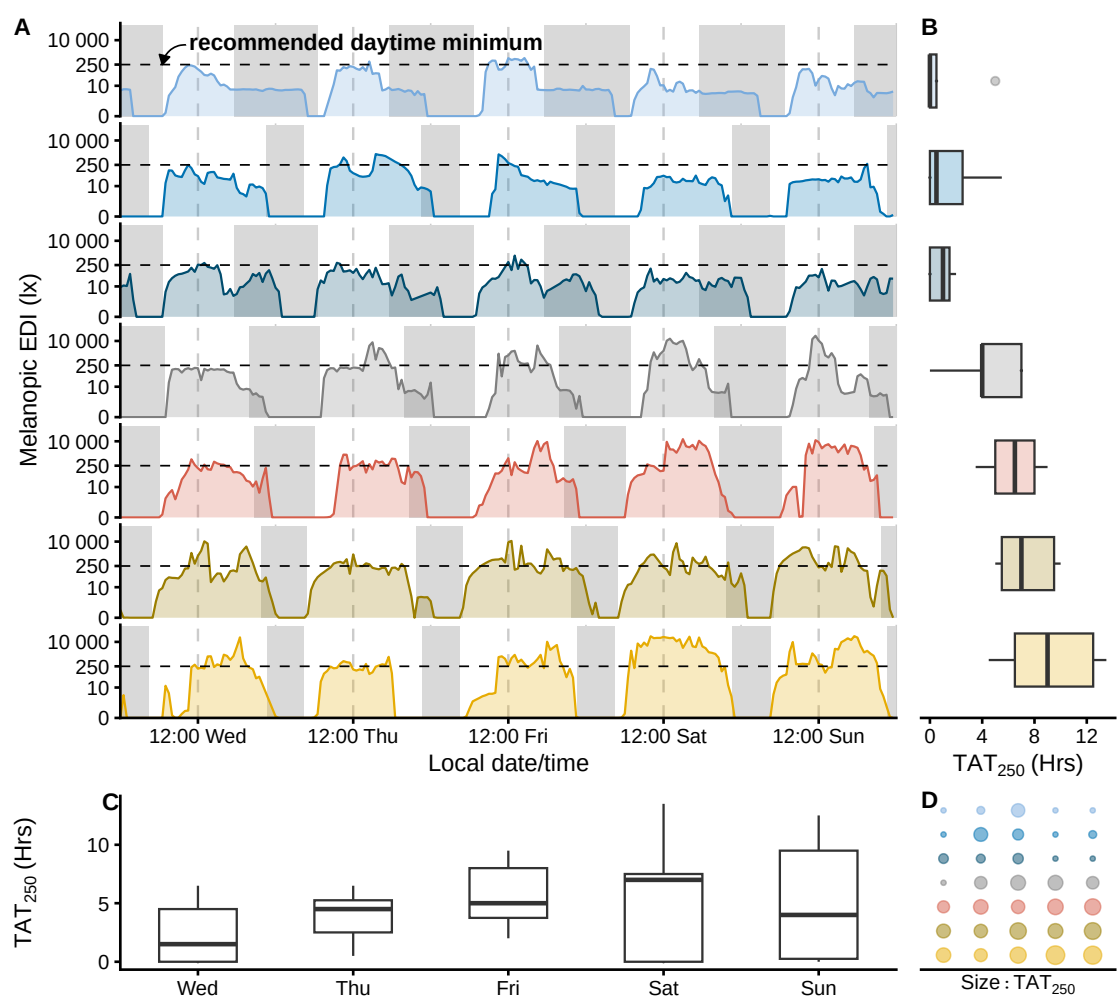


1191

1192 **Supplementary Figure S1. Detailed near-eye metric distributions by country-coded study**
1193 **site.**

1194 **Supplementary Figure S2. Worked metric derivation**

1195



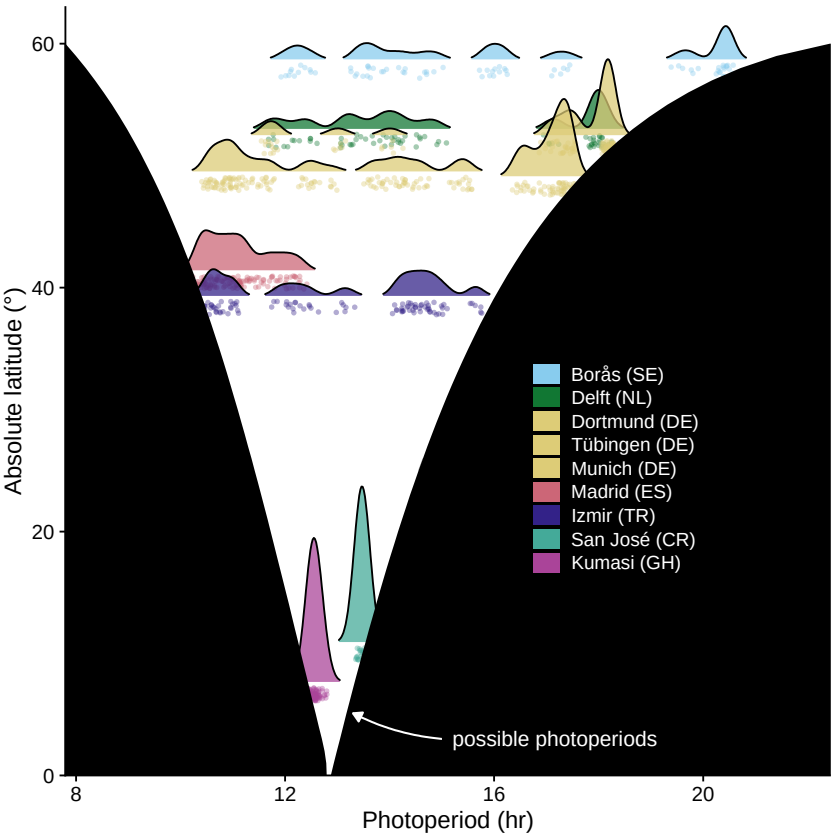
TAT = time above threshold during daytime; subscripts give melEDI thresholds in lx.
Lines and TAT250 values use the same stored 30-minute near-eye samples.

1196

1197 Supplementary Figure S2. Examples linking 30-minute time series to the daily time-
1198 above-250-lx metric.

1199 **Supplementary Figure S3. Latitude and civil-photoperiod**
1200 **measurement range**

1201



1202

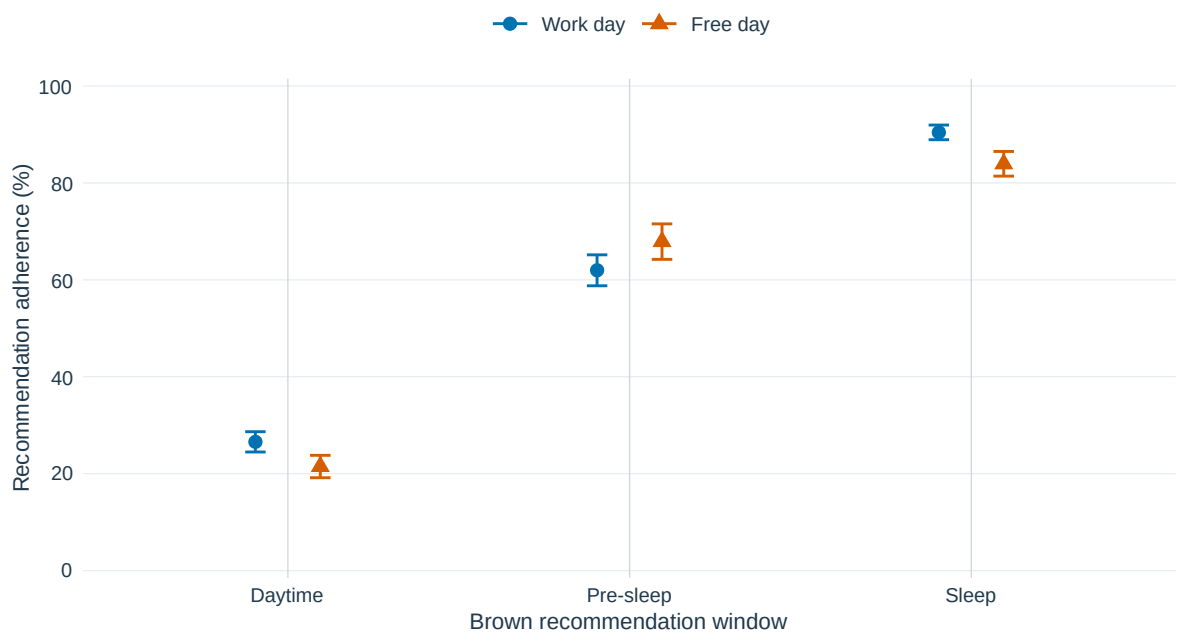
1203 Supplementary Figure S3. Observed civil photoperiod and theoretical bounds by absolute
1204 latitude.

1205 **Recommendation adherence**

1206 Daytime, Pre-sleep, and Sleep identify the Brown et al. recommendation windows. In the
1207 modelled comparisons, the wake-start date assigns the work or free label to the preceding
1208 sleep interval, ensuing daytime interval and following pre-sleep window. Each window
1209 qualifies independently. These modelled comparisons are distinct from the pooled-minute
1210 fractions in Supplementary Table S3.

1211 **Supplementary Figure S4. Site-average recommendation adherence**

1212



1213 Equal weight for nine sites. Bars are 95% confidence intervals.

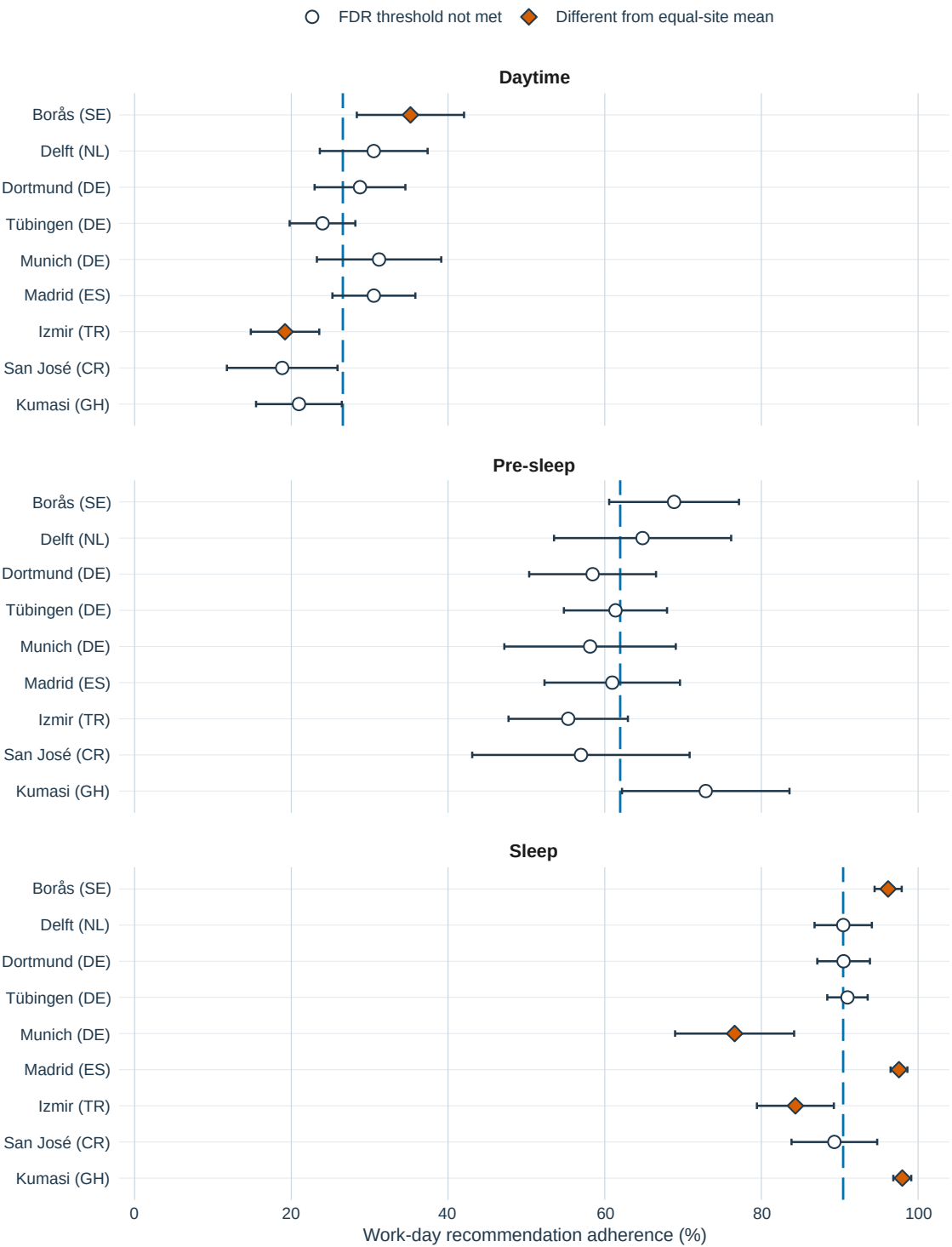
1214 **Supplementary Figure S4. Site-average recommendation adherence on work and free days.**

1215 Daytime, Pre-sleep and Sleep identify the Brown et al. recommendation windows. Each
1216 point gives equal weight to nine sites; bars are 95% confidence intervals. The wake-start
1217 date labels the preceding sleep, daytime and following pre-sleep windows. Sleep describes
1218 the device-recorded bedside environment, not ocular exposure or a whole-bedroom
1219 measurement. The pre-sleep contrast remains positive but its interval includes zero under
1220 the 80% coverage restriction; temporal dependence remains unresolved. The coverage
1221 sensitivity is reported in the Results and Methods.

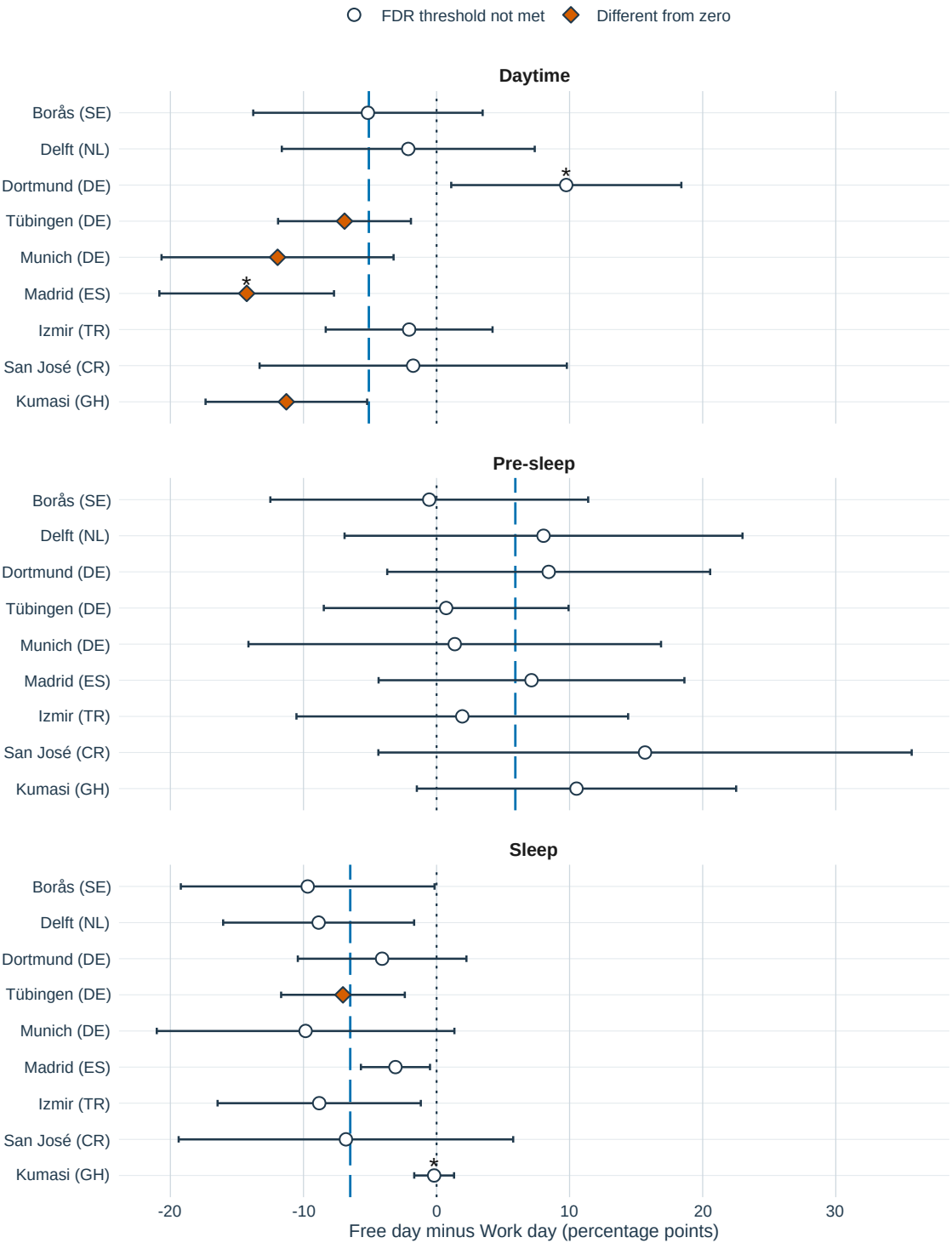
1222 **Supplementary Figure S5. Site-specific adherence and day-type**
1223 **differences**

1224

1225 **A**



1226



* Different from the equal-site Free-minus-Work effect (separate FDR family).
Dotted line: zero. Long-dashed blue line: equal-site effect.

1229 Supplementary Figure S5. Site-specific recommendation adherence, shown as two separate
1230 images. A, Work-day adherence, with filled diamonds identifying site-minus-average
1231 differences that met their FDR criterion. B, Free-minus-work differences, with filled diamonds
1232 identifying differences from zero and asterisks identifying departures from the window-
1233 specific equal-site difference in a separate 27-comparison FDR family. Daytime, Pre-sleep
1234 and Sleep identify the Brown et al. recommendation windows. Points are estimates and bars
1235 are 95% confidence intervals. Each equal-site reference includes the indexed site. Sites are
1236 components of one pooled observational model, not independent replications or causal
1237 effects of location. Day type follows the wake-start date; sleep is the preceding bedside
1238 sleep-environment window. Temporal dependence remains unresolved, and the pre-sleep
1239 coverage qualification applies.

1240 **Supplementary Table S4. Exploratory cross-window recommendation-adherence associations**

Exploratory cross-window recommendation-adherence associations				
Outcome window	Difference per +10 percentage points Daytime adherence, percentage points (95% CI)	FDR-adjusted p	FDR-correction set/family	Interpretation for recommendation adherence
Within participant, linked cycle				
Sleep	-0.26 (-0.95 to 0.43)	0.453	Four prespecified primary cross-window tests	No retained evidence that a cycle with 10 percentage points higher Daytime adherence had different Sleep adherence; the day-level claim is withheld because serial dependence remains unresolved.
Pre-sleep	-1.09 (-2.75 to 0.56)	0.258	Four prespecified primary cross-window tests	No retained evidence that a cycle with 10 percentage points higher Daytime adherence had different Pre-sleep adherence; the day-level claim is withheld because serial dependence remains unresolved.

1241

Table continued

Exploratory cross-window recommendation-adherence associations				
Outcome window	Difference per +10 percentage points Daytime adherence, percentage points (95% CI)	FDR-adjusted p	FDR-correction set/family	Interpretation for recommendation adherence
Between participants, observed monitoring-period average				
Sleep	-2.52 (-3.55 to -1.48)	<0.001	Four prespecified primary cross-window tests	Participants with 10 percentage points higher average Daytime adherence had 2.52 percentage points lower average Sleep adherence over the monitoring period.
Pre-sleep	-3.59 (-5.94 to -1.25)	0.005	Four prespecified primary cross-window tests	Participants with 10 percentage points higher average Daytime adherence had 3.59 percentage points lower average Pre-sleep adherence over the monitoring period.
All four tests form one FDR-correction set. Site-standardized estimates give nine sites equal weight and use 50:50 Work/Free weighting. The primary sample contained 1,376 target-window rows from 140 participants and 761 linked cycles. Within-participant claims are withheld because temporal dependence remains unresolved. Between-participant estimates describe observed monitoring-period averages, not rankings, stable traits, or causal effects. Confidence intervals are not multiplicity adjusted.				

1242

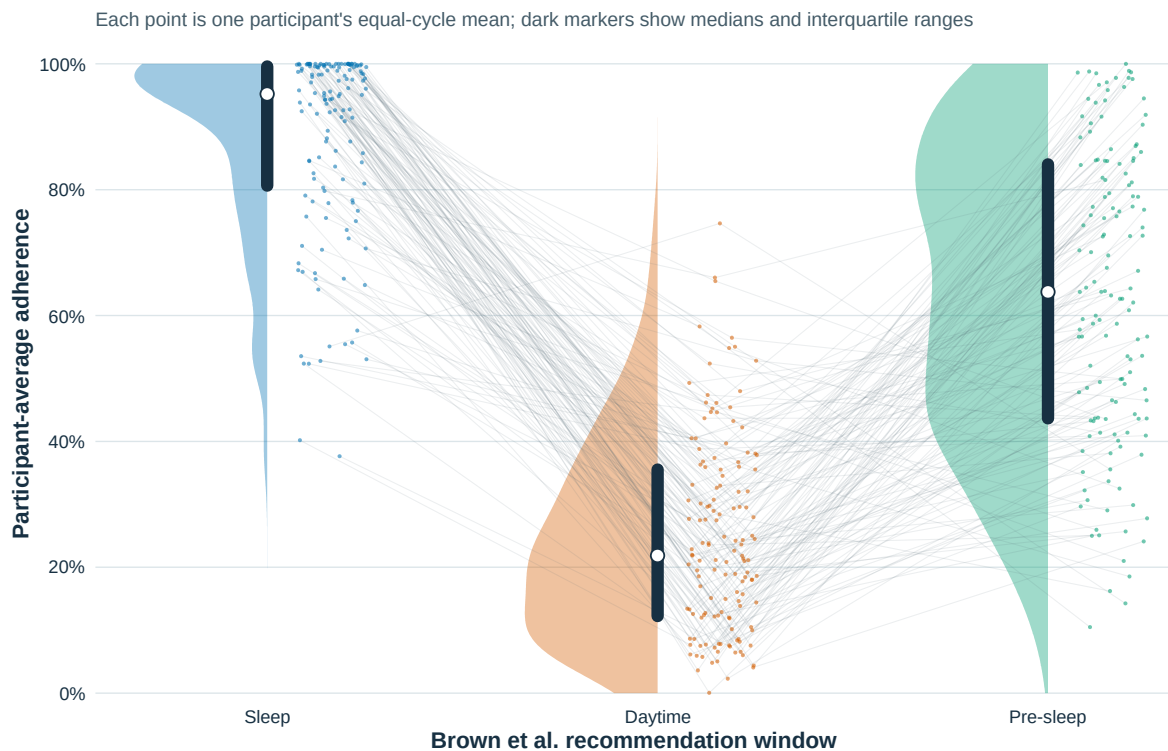
1243 **Supplementary Table S4.** Exploratory within-participant and between-participant associations across Brown et al. recommendation windows. Results are

1244 percentage-point differences per 10 percentage points higher Daytime adherence. The four tests form one FDR family. The within-participant day-level claim is

1245 withheld because serial dependence remains unresolved. Between-participant results describe observed monitoring-period averages and do not establish

1246 rankings, stable traits, or causal effects.

Supplementary Figure S6. Anonymous participant recommendation-adherence profiles



Supplementary Figure S6. Participant-average recommendation adherence across the Brown et al. recommendation windows. Daytime, Pre-sleep, and Sleep are the window labels. Each colored point is one anonymous participant's equal-cycle mean within that window. Faint lines connect the same anonymous participant across the three window summaries; they are not time trajectories or ranks. Half-violin shapes show the window distributions, and dark bars with white points show interquartile ranges and medians. Because each window has its own Brown et al. threshold and direction, absolute percentages should be interpreted within window.

1259 **Daily architecture**

1260 **Supplementary Table S5. Near-eye fitted-curve dispersion and full-model R² allocation**

Near-eye fitted-curve dispersion and conditional Shapley allocation

	Fitted-curve result (95% CI)	Conditional Shapley result (95% CI)	Share (95% CI)
Component summaries			
Shared local-clock curve	Not applicable	0.611 (0.550 to 0.657)	78.9% (73.6% to 82.6%)
Site pattern	0.100 (0.041 to 0.139)	0.016 (0.007 to 0.026)	2.0% (0.9% to 3.5%)
Participant pattern	0.180 (0.126 to 0.214)	0.100 (0.079 to 0.125)	12.9% (10.1% to 16.4%)
Participant-day shift	0.019 (0.010 to 0.024)	0.048 (0.035 to 0.064)	6.2% (4.4% to 8.6%)
Participant pattern + day shift	0.200 (0.140 to 0.230)	Not separately allocated	Not applicable
Participant-to-site comparisons			
Participant / site	1.797 (1.157 to 4.332)	6.41 (3.85 to 15.14)	Not applicable
(Participant + day) / site	1.991 (1.291 to 4.767)	9.49 (5.77 to 21.81)	90.5% (85.2% to 95.6%)

Full-model in-sample R² was 0.775 (0.739 to 0.804). Fitted-curve variation is in squared log₁₀(melEDI + 0.1 lx) prediction units; curve and Shapley ratios are unitless. Conditional Shapley values allocate in-sample R² with the shared local-clock curve as the mandatory baseline. Intervals are 95% percentile hierarchical cluster-bootstrap intervals from 2,000 replicates. Dispersion and Shapley credit are different estimands and must not be added or interpreted causally.

1261

1262 **Supplementary Table S5.** Near-eye fitted-curve dispersion and Shapley allocation of full-model in-sample R². Fitted-curve variation is in squared
1263 log₁₀(melanopic EDI + 0.1 lx) prediction units; dispersion and R²-share ratios are unitless. The allocation includes the shared local-clock curve in every
1264 component model as the baseline. Intervals are 95% hierarchical cluster-bootstrap percentile intervals from 2,000 replicates and are conditional on the fitted
1265 models. Dispersion and R² allocation describe different estimands and should not be added or interpreted causally.

1266 **Supplementary Table S6. Chest fitted-curve dispersion and full-model R^2 allocation**

Complementary chest fitted-curve dispersion and conditional Shapley allocation			
	Fitted-curve result (95% CI)	Conditional Shapley result (95% CI)	Share (95% CI)
Component summaries			
Shared local-clock curve	Not applicable	0.583 (0.523 to 0.629)	79.1% (73.6% to 82.8%)
Site pattern	0.101 (0.049 to 0.116)	0.015 (0.008 to 0.024)	2.1% (1.1% to 3.3%)
Participant pattern	0.148 (0.100 to 0.199)	0.087 (0.071 to 0.110)	11.9% (9.5% to 15.3%)
Participant-day shift	0.035 (0.020 to 0.039)	0.051 (0.040 to 0.067)	7.0% (5.3% to 9.4%)
Participant pattern + day shift	0.183 (0.127 to 0.228)	Not separately allocated	Not applicable
Participant-to-site comparisons			
Participant / site	1.465 (0.973 to 3.179)	5.69 (3.72 to 10.68)	Not applicable
(Participant + day) / site	1.809 (1.219 to 3.736)	9.02 (6.01 to 17.03)	90.0% (85.7% to 94.5%)

Full-model in-sample R^2 was 0.737 (0.703 to 0.766). Fitted-curve variation is in squared $\log_{10}(\text{meIEDI} + 0.1 \text{ lx})$ prediction units; curve and Shapley ratios are unitless. Conditional Shapley values allocate in-sample R^2 with the shared local-clock curve as the mandatory baseline. Intervals are 95% percentile hierarchical cluster-bootstrap intervals from 2,000 replicates. Dispersion and Shapley credit are different estimands and must not be added or interpreted causally.

1267

1268 **Supplementary Table S6.** Complementary chest fitted-curve dispersion and Shapley allocation of full-model in-sample R^2 . The quantities, bootstrap intervals
1269 and interpretation follow Supplementary Table S5; chest and near-eye measurements remain distinct sensor-position estimands.

1270 Geographic and civil-photoperiod context

1271 Supplementary Figure S7. Geographic support and nonlinear 1272 photoperiod classifications

1273 A

	Near eye				Chest			
	Overall site	Photoperiod	Latitude	Site versus linear latitude	Overall site	Photoperiod	Latitude	Site versus linear latitude
Interdaily stability	–	–	–	–	–	–	–	–
Intradaily variability	–	–	–	–	–	–	–	–
Mean melEDI	✓	✓	✓	✓	✓	✓	–	✓
Brightest 10 h mean	✓	✓	✓	–	✓	✓	–	✓
Darkest 10 h mean	✓	✓	–	✓	✓	✓	✓	✓
Time above 1,000 lx melEDI	–	✓	–	–	✓	✓	✓	✓
Time above 250 lx melEDI during wake	✓	✓	✓	–	✓	✓	–	✓
Time below 10 lx melEDI before sleep	✓	✓	–	✓	–	✓	–	–
Time below 1 lx melEDI during sleep	–	–	–	✓	✓	–	✓	✓
Longest continuous period above 250 lx melEDI	–	✓	–	–	–	✓	–	–
Midpoint of the brightest 10 hours	✓	–	✓	✓	✓	–	✓	✓
Midpoint of the darkest 10 hours	✓	✓	–	✓	✓	✓	✓	✓
Mean timing of exposure above 250 lx melEDI	✓	✓	✓	✓	✓	✓	✓	✓
First light timing above 250 lx melEDI	–	–	–	–	✓	–	✓	✓
Last light timing above 250 lx melEDI	✓	✓	✓	✓	✓	✓	✓	–
melEDI dose	–	✓	–	–	✓	✓	–	✓
Melanopic daylight efficacy ratio	✓	✓	✓	✓	✓	–	✓	–

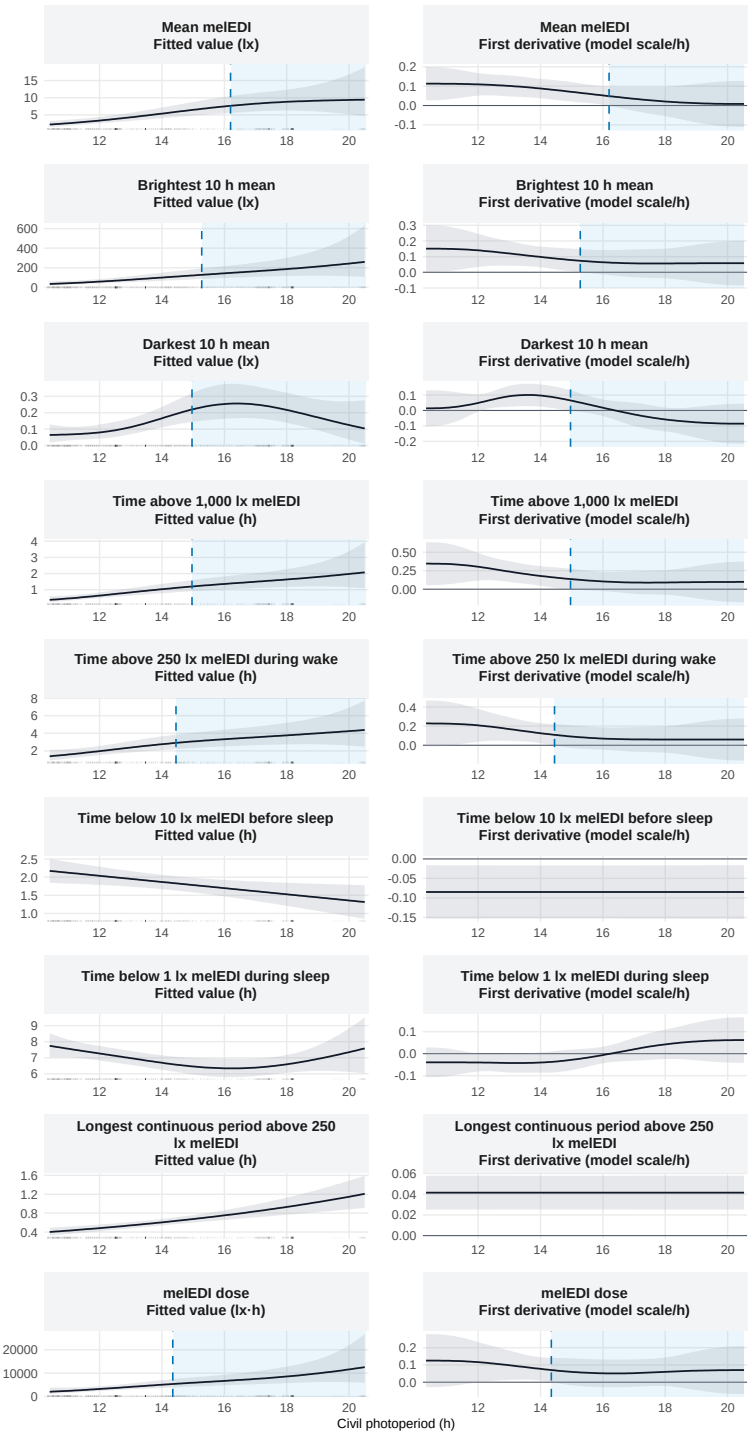
FDR-adjusted result Not supported Supported

1274

1275 Supplementary Figure S7. Geographic and civil-photoperiod associations across near-eye
1276 personal light-exposure metrics. A, Evidence retained after FDR adjustment for overall site,
1277 civil photoperiod, latitude, and site-versus-linear-latitude adequacy. B, Nonlinear civil-
1278 photoperiod associations and their fitted slopes. A qualifying transition denotes an increase
1279 followed by a sustained near-flat tail under the specified slope rule. These descriptive
1280 associations do not identify a mechanism, and no physiological or environmental ceiling was
1281 identified.

Near-eye — primary

Each row pairs the response-scale fitted metric value (left) with the model-scale first derivative used for classification (right). Grey ribbons are unconditional pointwise 95% intervals; dashed lines and blue tails mark the derivative-defined plateau pattern.



1284 **Supplementary Table S7. Geographic and photoperiod associations**

		Photoperiod		Latitude		Site-versus-latitude FDR-adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR-adjusted p	Latitude association per 10° (95% CI)	Latitude FDR-adjusted p		
Dynamics							
Interdaily stability	0.228	×0.98 [0.94–1.02]	0.401	×1.01 [0.96–1.06]	0.903	0.175	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Intradaily variability	0.483	-0.02 [-0.06–0.02]	0.327	-0.04 [-0.08–0.01]	0.139	0.723	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Level							
Mean melEDI	<0.001	×1.18 [1.11–1.26]	<0.001	×1.15 [1.07–1.25]	0.002	0.032	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Brightest 10 h mean	0.011	×1.24 [1.13–1.36]	<0.001	×1.24 [1.11–1.39]	0.001	0.325	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Darkest 10 h mean	<0.001	×1.09 [1.03–1.14]	0.002	×1.04 [0.97–1.11]	0.400	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$

1285

Table continued

	Overall site FDR-adjusted p	Photoperiod		Latitude		Site-versus-latitude FDR-adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR-adjusted p	Latitude association per 10° (95% CI)	Latitude FDR-adjusted p		
Duration							
Time above 1,000 lx melEDI	0.071	×1.21 [1.13–1.29]	<0.001	×1.03 [0.94–1.12]	0.769	0.053	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Time above 250 lx melEDI during wake	0.007	×1.14 [1.07–1.20]	<0.001	×1.12 [1.04–1.20]	0.013	0.053	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 737$
Time below 10 lx melEDI before sleep	0.049	-0.13 [-0.20--0.07] h	<0.001	0.00 [-0.09–0.09] h	0.962	0.040	$n_{\text{participants}} = 139$; $n_{\text{participant-days}} = 655$
Time below 1 lx melEDI during sleep	0.058	×0.99 [0.97–1.01]	0.227	×1.00 [0.97–1.02]	0.903	0.043	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 778$
Longest continuous period above 250 lx melEDI	0.398	×1.12 [1.07–1.19]	<0.001	×1.06 [0.99–1.13]	0.139	0.626	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$

		Photoperiod		Latitude		Site- versus- latitude FDR- adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR- adjusted p	Latitude association per 10° (95% CI)	Latitude FDR- adjusted p		
Timing							
Midpoint of the brightest 10 hours	<0.001	0.05 [-0.05–0.15] h	0.305	0.19 [0.06–0.32] h	0.013	0.010	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Midpoint of the darkest 10 hours	0.020	-0.15 [-0.25--0.05] h	0.004	0.12 [-0.00–0.24] h	0.114	0.043	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Mean timing of exposure above 250 lx meLEDI	<0.001	0.12 [0.03–0.21] h	0.013	0.33 [0.21–0.46] h	<0.001	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 742$
First light timing above 250 lx meLEDI	0.112	-0.10 [-0.25–0.05] h	0.227	0.02 [-0.17–0.21] h	0.903	0.077	$n_{\text{participants}} = 140$; $n_{\text{participant-days}} = 727$
Last light timing above 250 lx meLEDI	<0.001	0.32 [0.19–0.46] h	<0.001	0.46 [0.27–0.64] h	<0.001	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 687$

1287

	Overall site FDR- adjusted p	Photoperiod		Latitude		Site- versus- latitude FDR- adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR- adjusted p	Latitude association per 10° (95% CI)	Latitude FDR- adjusted p		
Exposure history							
melEDI dose	0.209	×1.28 [1.17–1.39]	<0.001	×1.02 [0.91–1.13]	0.903	0.156	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 761
Spectrum							
Melanopic daylight efficacy ratio	<0.001	0.02 [0.02–0.03]	<0.001	-0.01 [-0.02--0.00]	0.013	0.010	<i>n</i> _{participants} = 137; <i>n</i> _{participant-days} = 702
This summary combines the model-level decisions, photoperiod and latitude associations, 95% CIs, and exact fitted samples. Each displayed p-value is FDR-adjusted within its explicitly labelled complete 17-test family; bold values meet the FDR-adjusted p < 0.050 rule. Full raw p-values and test statistics remain in the detailed tables below. Exact fitted samples use italic n with participant and participant-day subscripts. For the 15 participant-day models, the participant-day count equals the number of fitted observations. The two participant-level dynamics models use one fitted observation per participant and retain the participant-day count as contributing repeated-day support. All models include nine sites.							

1288

1289 **Supplementary Table S7.** Overall-site, civil-photoperiod and absolute-latitude associations for the 17 near-eye personal light-exposure metrics. Each displayed
1290 *p* value is FDR-adjusted within its stated 17-test family. Site and latitude were evaluated in separate models because absolute latitude is fixed within and
1291 confounded with study site.

1292 **Supplementary Table S8. Nonlinear civil-photoperiod classifications**

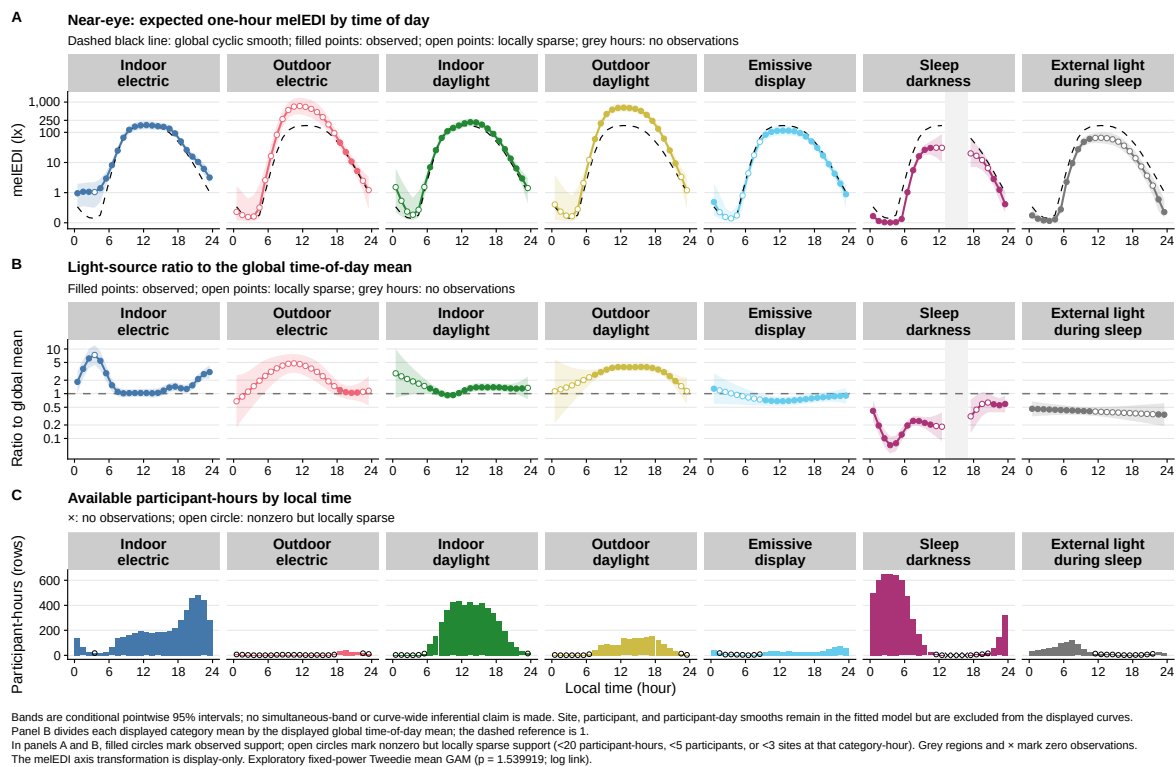
	Qualifying transition	Qualifying transition bracket (h)	Endpoint derivative	Interpretation
Light level				
Mean melEDI	Yes	16.10–16.20	0.008 (95% CI -0.111 to 0.127)	Detected increase followed by a sustained zero-compatible tail
Brightest 10 h mean	Yes	15.17–15.27	0.059 (95% CI -0.083 to 0.201)	Detected increase followed by a sustained zero-compatible tail
Darkest 10 h mean	Yes	14.86–14.96	-0.086 (95% CI -0.216 to 0.044)	Detected increase followed by a sustained zero-compatible tail
Duration and continuous period				
Time above 1,000 lx melEDI	Yes	14.86–14.96	0.098 (95% CI -0.176 to 0.371)	Detected increase followed by a sustained zero-compatible tail
Time above 250 lx melEDI during wake	Yes	14.35–14.45	0.060 (95% CI -0.159 to 0.279)	Detected increase followed by a sustained zero-compatible tail
Time below 10 lx melEDI before sleep	No	—	-0.085 (95% CI -0.154 to -0.016)	No preceding detected increase
Time below 1 lx melEDI during sleep	No	—	0.062 (95% CI -0.042 to 0.165)	No preceding detected increase
Longest continuous period above 250 lx melEDI	No	—	0.041 (95% CI 0.025 to 0.058)	Increase remained detected at the recorded maximum
Exposure history				
melEDI dose	Yes	14.24–14.35	0.070 (95% CI -0.069 to 0.209)	Detected increase followed by a sustained zero-compatible tail
A 'Yes' is the formal derivative-defined classification: the immediately preceding point has an interval wholly above zero, the next contains zero, and every later interval through the recorded maximum remains zero-compatible. Endpoint derivatives and 95% confidence intervals are on each model's linear-predictor scale, in model-scale units per hour of civil photoperiod.				

1294 **Supplementary Table S8.** Derivative-based classifications for nine near-eye personal light-exposure metrics across the observed civil-photoperiod range. A
1295 qualifying transition required the preceding fitted-slope interval to be wholly above zero, the next interval to include zero and every later interval through the
1296 recorded maximum to remain zero-compatible. These classifications are descriptive and do not identify a physiological or environmental ceiling.

1297 Immediate environments and routines

1298 Supplementary Figure S8. Light-source category and local-clock
1299 pattern of near-eye melanopic EDI

1300



1301

1302 **Supplementary Figure S8 (continued)**

D **Near-eye light-source-by-site interaction model**

Site-average estimates and site-specific deviation ratios; bold site cells pass the complete near-eye FDR adjustment

	Indoor electric	Outdoor electric	Indoor daylight	Outdoor daylight	Emissive display	Sleep darkness	External light during sleep
	P 138 · D 695 · H 4,629	P 54 · D 118 · H 203	P 138 · D 692 · H 4,804	P 135 · D 512 · H 1,557	P 88 · D 235 · H 664	P 130 · D 712 · H 5,225	P 66 · D 233 · H 853
Site-average estimate	85.1 lx 95% CI 71.6–101.1 1.000× reference	9.4 lx 95% CI 6.9–12.8 0.110× indoor 95% CI 0.078–0.156	198.8 lx 95% CI 169.2–233.4 2.335× indoor 95% CI 1.877–2.905	959.6 lx 95% CI 786.8–1170.2 11.274× indoor 95% CI 8.709–14.595	23.8 lx 95% CI 17.4–32.5 0.280× indoor 95% CI 0.197–0.398	2.2 lx 95% CI 1.5–3.3 0.026× indoor 95% CI 0.018–0.039	11.2 lx 95% CI 8.1–15.5 0.131× indoor 95% CI 0.091–0.189
Borås (SE)	1.43× (1.02–1.99) FDR p 0.083	Not estimable	1.61× (1.12–2.32) FDR p 0.029	3.07× (1.92–4.90) FDR p <0.001	0.31× (0.16–0.57) FDR p 0.002	1.52× (0.39–5.90) FDR p 0.704	0.70× (0.21–2.36) FDR p 0.707
Delft (NL)	1.83× (1.32–2.54) FDR p 0.002	Not estimable	1.46× (0.93–2.29) FDR p 0.190	1.58× (0.89–2.80) FDR p 0.214	7.17× (4.00–12.86) FDR p <0.001	0.94× (0.49–1.80) FDR p 0.894	Not estimable
Dortmund (DE)	1.26× (0.69–2.29) FDR p 0.595	Not estimable	1.62× (0.83–3.17) FDR p 0.263	1.48× (0.91–2.40) FDR p 0.210	2.86× (1.66–4.93) FDR p 0.002	0.89× (0.45–1.75) FDR p 0.801	9.22× (2.50–34.03) FDR p 0.004
Tübingen (DE)	0.56× (0.38–0.82) FDR p 0.009	20.42× (3.95–105.59) FDR p 0.002	1.04× (0.72–1.50) FDR p 0.866	0.89× (0.51–1.56) FDR p 0.776	0.25× (0.14–0.44) FDR p <0.001	0.78× (0.32–1.92) FDR p 0.710	1.59× (0.71–3.59) FDR p 0.406
Munich (DE)	1.70× (0.63–4.57) FDR p 0.425	Not estimable	1.00× (0.65–1.55) FDR p 0.983	0.94× (0.55–1.61) FDR p 0.866	Not estimable	6.60× (2.11–20.69) FDR p 0.005	11.65× (7.19–18.85) FDR p <0.001
Madrid (ES)	1.48× (1.15–1.90) FDR p 0.007	2.68× (1.56–4.61) FDR p 0.002	0.83× (0.57–1.19) FDR p 0.436	0.65× (0.33–1.26) FDR p 0.332	3.68× (2.24–6.03) FDR p <0.001	1.59× (0.30–8.32) FDR p 0.710	Not estimable
Izmir (TR)	0.69× (0.52–0.92) FDR p 0.029	1.85× (0.84–4.07) FDR p 0.222	0.86× (0.65–1.13) FDR p 0.422	1.12× (0.65–1.94) FDR p 0.776	1.32× (0.70–2.47) FDR p 0.540	0.97× (0.43–2.18) FDR p 0.952	1.59× (0.92–2.76) FDR p 0.190
San José (CR)	0.54× (0.38–0.77) FDR p 0.003	Not estimable	0.72× (0.49–1.05) FDR p 0.189	0.46× (0.21–1.01) FDR p 0.115	1.32× (0.65–2.66) FDR p 0.595	0.60× (0.25–1.42) FDR p 0.390	Not estimable
Kumasi (GH)	0.58× (0.38–0.88) FDR p 0.029	6.24× (2.28–17.06) FDR p 0.002	0.49× (0.26–0.92) FDR p 0.065	0.50× (0.36–0.68) FDR p <0.001	1.27× (0.40–4.07) FDR p 0.776	0.17× (0.06–0.49) FDR p 0.004	Not estimable

Site-average estimates and comparisons use the light-source-by-site interaction model.
Site rows are deviation ratios around each category estimate; parentheses show 95% confidence intervals.
Bold site ratios and FDR p-values pass the complete 53-member near-eye family; unsupported cells are not estimable.
Near-eye frame: 17,935 participant-hours; 140 participants; 801 participant-days; nine sites.
Each accepted participant-hour has one primary light-source category; 1/k multi-label weighting does not apply.
Panels A–C exclude site, participant, and participant-day smooths from displayed curves; panel D uses stored interaction-model estimands.
P, participants; D, participant-days; H, participant-hours. Country-coded labels and row order provide non-colour site cues.

1303

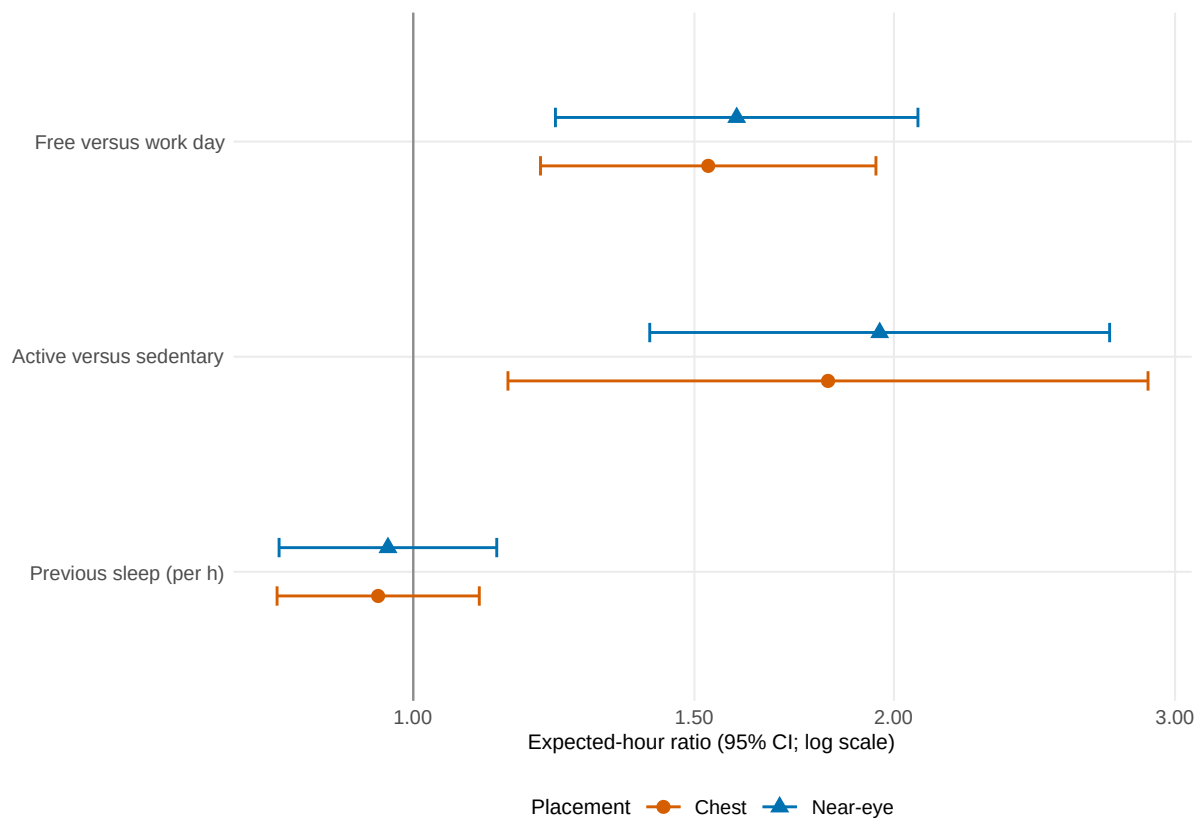
1304 Supplementary Figure S8. Near-eye light exposure by reported light source across time of
1305 day and study site. A–C, Exploratory time-of-day analysis. Panel A shows expected one-
1306 hour melanopic EDI with clock-specific 95% confidence intervals; the dashed line is the site-
1307 average local-clock smooth. Panel B shows each light-source curve relative to that smooth,
1308 with 1 as the reference. Panel C shows available participant-hours; open circles indicate
1309 locally sparse support and grey gaps indicate no observations. D, Site-average category
1310 estimates and site-specific deviation ratios from the light-source-by-site interaction model,
1311 with participant-cluster-robust 95% confidence intervals, FDR-adjusted p values, and exact
1312 support. The analysis includes 17,935 participant-hours from 140 participants, covering 801
1313 participant-days and nine sites. Panels A–C and panel D answer different questions and are
1314 not numerically interchangeable. Estimates are observational. During reported sleep,
1315 measurements describe the bedside sleep environment.

1316 **Hourly routine analyses**

1317 **Supplementary Figure S9. Paired sensor-position hourly**

1318 **associations**

1319

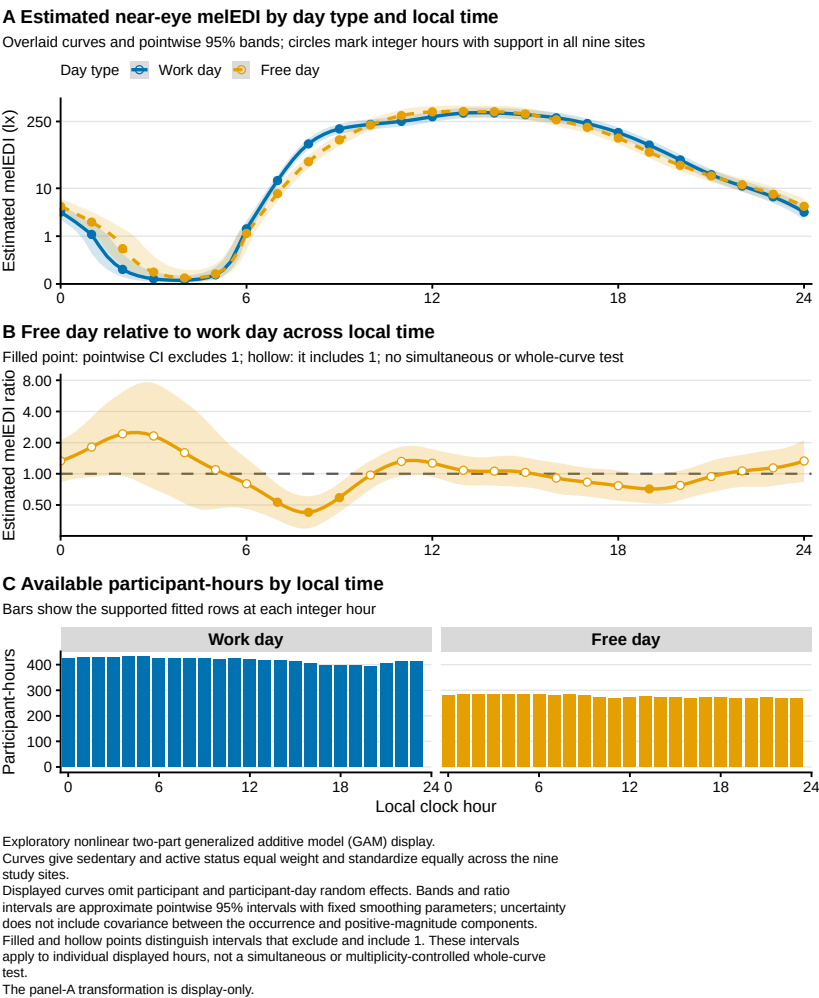


1320

1321 Supplementary Figure S9. Near-eye and complementary chest associations estimated
 1322 separately for the same participants and participant-days. The sets of supported participant-
 1323 hours differ between sensor positions, so the display is not an observation-level identity
 1324 comparison and does not test equivalence.

1325 **Supplementary Figure S10. Free-versus-Work hourly timing**

1326

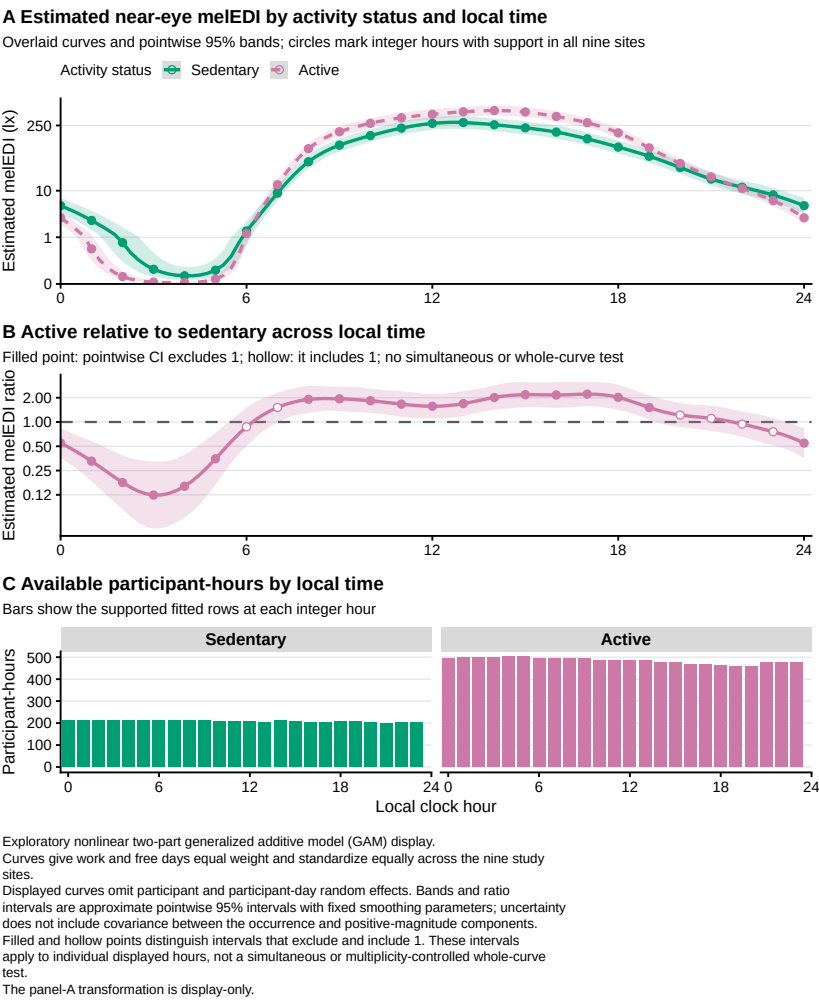


1327

1328 Supplementary Figure S10. Exploratory Free-versus-Work hourly fitted exposure and
1329 response-scale ratio curves with 95% confidence intervals and participant-hour support.

1330 **Supplementary Figure S11. Active-versus-Sedentary hourly timing**

1331



1332

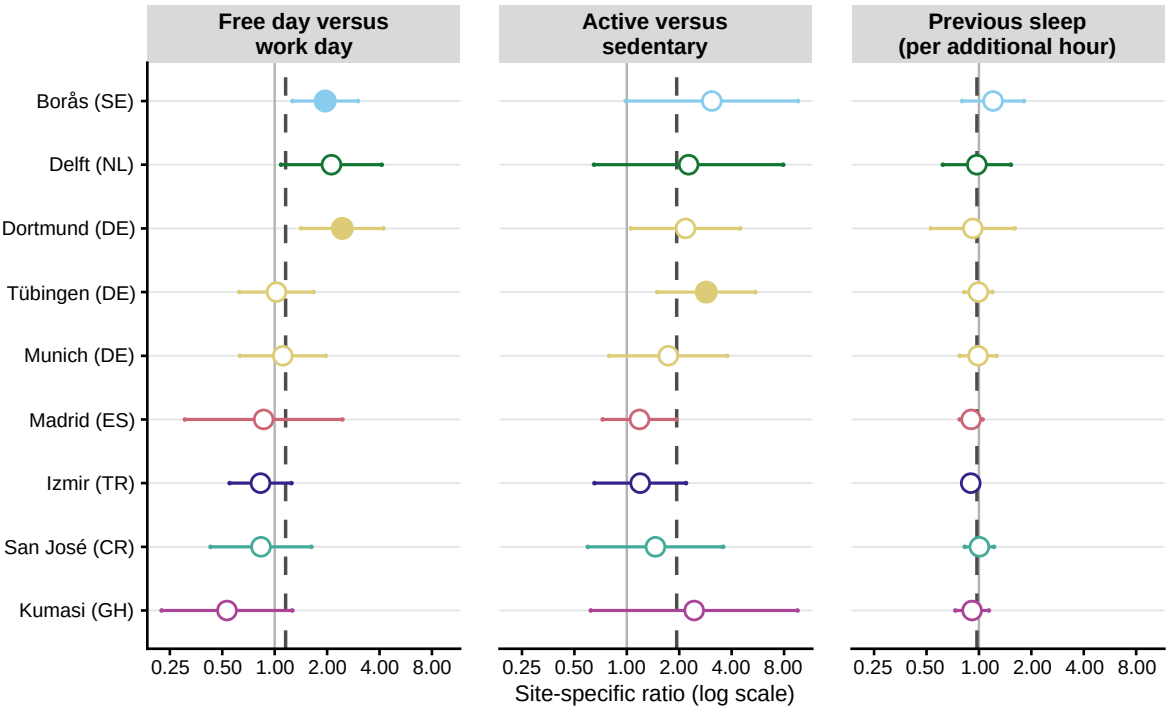
1333 Supplementary Figure S11. Exploratory Active-versus-Sedentary hourly fitted exposure and
1334 response-scale ratio curves with 95% confidence intervals and participant-hour support.

1335 **Supplementary Figure S12. Site-specific hourly routine**
1336 **associations**

1337

Site-specific mean hourly near-eye meEDI ratios

Dashed: site-average estimate from this model; filled: retained after nine-site FDR adjustments; open: not retained



Bars are participant-cluster HC3 pointwise 95% confidence intervals from the current predictor-by-site interaction model. The grey line at 1 is the site-specific association null. The dashed line is the site-average geometric mean of the nine ratios in this same model. Filled points pass a separate nine-site FDR adjustment within that predictor; this does not test deviation from the site-average estimate or the overall predictor-by-site interaction.

1338

1339 Supplementary Figure S12. Site-specific estimates from the predictor-by-site interaction
1340 model. Filled and hollow points distinguish associations that did and did not meet the 0.050
1341 criterion after a separate nine-site FDR adjustment within each predictor. The 95%
1342 confidence intervals are unadjusted; dashed lines show site-average estimates calculated
1343 with equal weight for each of the nine sites. Sites are components of one pooled model, not
1344 independent replications or causal effects of location.

1345 **Supplementary Table S9. Participant-hour routine associations**

Predictor	Ratio (95% CI)	FDR-adjusted p
Free day versus work day	1.45 × (1.14–1.85)	0.004
Active versus sedentary	2.06 × (1.54–2.76)	<0.001
Previous sleep duration (per hour)	0.98 × (0.85–1.12)	0.745
Directions are free versus work day, active versus sedentary day status, and the ratio per one additional hour of previous sleep. Participant-cluster-robust 95% CIs use HC3 with 137 participants. The three p-values form one FDR-adjusted family; the null ratio is 1.		

1346

1347 **Supplementary Table S9.** Common-association estimates from the participant-hour analysis of Free versus Work day, Active versus Sedentary daily activity
1348 status and each additional hour of previous-night sleep duration. Ratios and participant-cluster-robust 95% confidence intervals come from the fixed-site model;
1349 the three *p* values form one FDR-adjusted family. Predictor-by-site differences are shown in Supplementary Figure S12.

1350 **Person-level findings**

1351 **Supplementary Table S10. Person-level evidence synthesis**

Person-level evidence synthesis

	Outcome or analysis scale	Result	FDR-correction set	FDR decision	Exact sample	Interpretation and qualification
Light-exposure behaviour and awareness factors	68 factor-by-metric cells across four participant-level factors	No cell retained FDR support; four sleep-environment cells were non-estimable	68 near-eye factor-by-metric tests	None retained	Metric-specific near-eye samples in Supplementary Table S11	Broad inconclusive result; non-retention is not evidence of no association
Visual light sensitivity	Nine near-eye metrics per one VLSQ-8 SD	Largest displayed ratio shift was melanopic EDI dose, $\times 0.846$ (0.716 to 0.999)	Nine near-eye metric tests	None retained; adjusted $p = 0.160$ for this row	141 participants, 761 participant-days, nine sites for the displayed row	Broad inconclusive result
Corrected midsleep on Free days	Timing outcomes per one-hour later MSFsc	M10 midpoint +0.205 h (0.045 to 0.366); L10 midpoint +0.276 h (0.118 to 0.435); first light above 250 lx +0.381 h (0.146 to 0.616)	Five MCTQ timing outcomes, separate from MEQ	These three rows retained	139 to 140 participants; 722 to 810 participant-days; nine sites	Selective timing associations
Morningness-Eveningness Questionnaire	Timing outcomes per 10 points greater morning preference	M10 midpoint -0.278 h (-0.451 to -0.105); L10 midpoint -0.314 h (-0.487 to -0.142); first light above 250 lx -0.444 h (-0.699 to -0.189)	Five MEQ timing outcomes, separate from MCTQ	These three rows retained	140 to 141 participants; 727 to 816 participant-days; nine sites	Selective timing associations

1352

Table continued

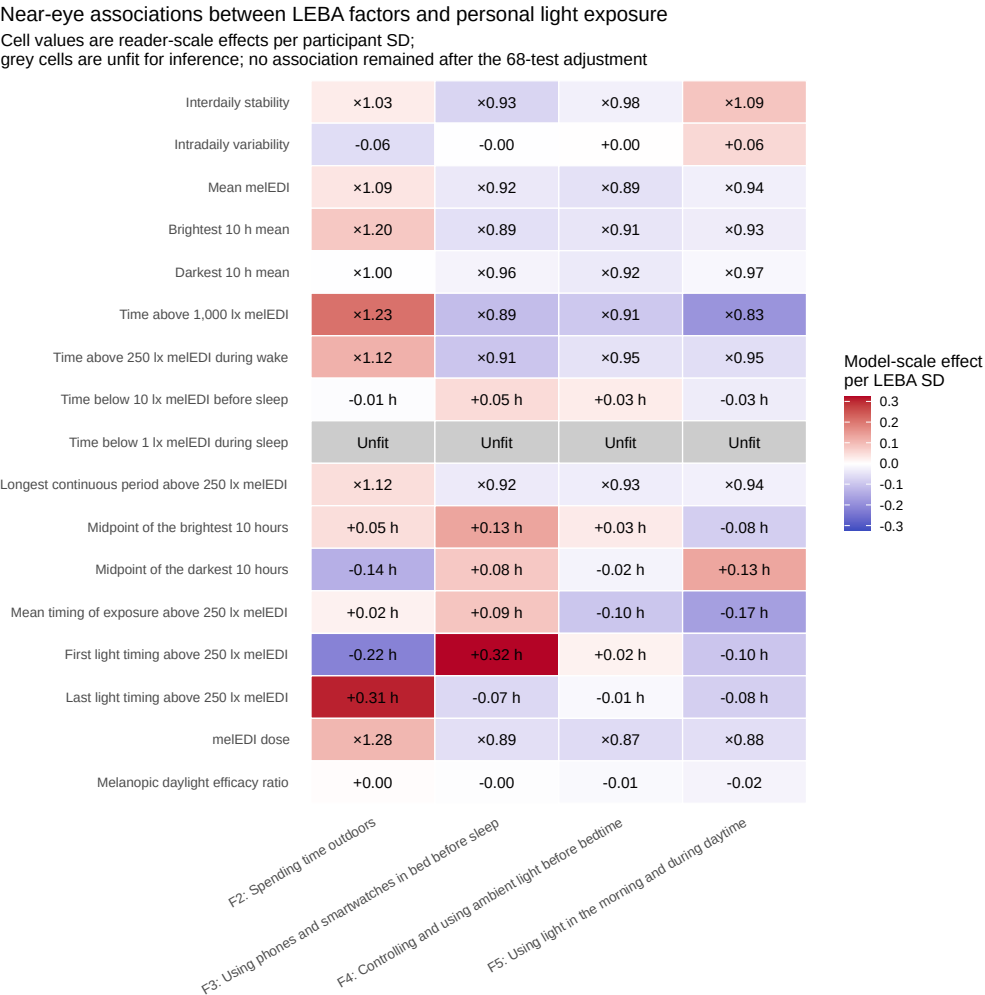
Person-level evidence synthesis						
	Outcome or analysis scale	Result	FDR-correction set	FDR decision	Exact sample	Interpretation and qualification
Age	Near-eye daily metrics per 10 years	Brightest 10 h geometric mean ×1.31 (1.09 to 1.58); time above 1,000 lx ×1.27 (1.11 to 1.45); melanopic EDI dose ×1.31 (1.09 to 1.56)	Near-eye age-metric tests	Three rows retained	141 participants; 761 to 816 participant-days; nine sites	Selective age associations; complementary chest results remain separate
Metric-level biological sex	Near-eye daily metrics, Female minus Male	No near-eye metric-level result retained FDR support	Near-eye biological-sex metric tests	None retained	Metric-specific samples in Supplementary Table S14	Biological sex and gender were recorded separately; gender was not analysed
Biological-sex-specific daily curve	Complete 24-hour curve and activity-complete sensitivity	The all-available near-eye complete-curve test was supported, but separation attenuated and was not retained in the activity-complete near-eye analysis	Global complete-curve tests and separately prespecified sensitivity decisions	Analysis-specific global decisions	All-available near eye: 141 participants, 816 participant-days, 37,756 observations, nine sites	Attenuation-first interpretation; confidence intervals apply at the displayed clock times

Results that did not meet the applicable FDR criterion are inconclusive rather than evidence of no association. The two chronotype instruments were analysed separately. Biological sex and gender were recorded separately; gender was not analysed.

1354 **Supplementary Table S10.** Summary of person-level results in the near-eye analyses. Each row identifies the analysis-specific FDR-correction set or decision
1355 structure applied to that result. Results that did not meet the applicable FDR criterion are inconclusive rather than evidence of no association. Four light-
1356 behaviour sleep-environment cells were non-estimable. The two chronotype instruments were analysed separately. Biological sex and gender were recorded
1357 separately; gender was not analysed. The complete available-data curve difference was attenuated and not retained in the activity-complete near-eye analysis.

1358 **Supplementary Figure S13 and Table S11. Light-exposure**
1359 **behaviour and awareness**

1360



1361

1362 Supplementary Figure S13. Near-eye associations per one participant-level standard
1363 deviation of each Light Exposure Behaviour Assessment factor. None of the 68 tests
1364 retained FDR-adjusted support. Four sleep-environment cells were not estimable and are
1365 labelled Unfit with estimates suppressed. Lack of adjusted support is inconclusive rather
1366 than proof of no association.

	F2	F3	F4	F5
Interdaily stability	OR 1.028 (0.956–1.106) FDR-adjusted p = 0.658	OR 0.931 (0.868–0.999) FDR-adjusted p = 0.260	OR 0.975 (0.906–1.050) FDR-adjusted p = 0.679	OR 1.089 (1.011–1.174) FDR-adjusted p = 0.202
Intradaily variability	-0.057 (-0.124–+0.009) FDR-adjusted p = 0.351	-0.001 (-0.068–+0.065) FDR-adjusted p = 0.978	+0.001 (-0.068–+0.069) FDR-adjusted p = 0.978	+0.057 (-0.013–+0.127) FDR-adjusted p = 0.362
Mean melEDI	×1.092 (0.965–1.235) FDR-adjusted p = 0.379	×0.917 (0.813–1.034) FDR-adjusted p = 0.379	×0.891 (0.787–1.009) FDR-adjusted p = 0.313	×0.941 (0.827–1.070) FDR-adjusted p = 0.585
Brightest 10 h mean	×1.204 (1.008–1.439) FDR-adjusted p = 0.260	×0.888 (0.745–1.058) FDR-adjusted p = 0.384	×0.906 (0.756–1.087) FDR-adjusted p = 0.545	×0.931 (0.772–1.123) FDR-adjusted p = 0.658
Darkest 10 h mean	×0.995 (0.905–1.094) FDR-adjusted p = 0.958	×0.956 (0.871–1.049) FDR-adjusted p = 0.577	×0.921 (0.838–1.013) FDR-adjusted p = 0.351	×0.970 (0.879–1.070) FDR-adjusted p = 0.716
Time above 1,000 lx melEDI	×1.234 (1.084–1.405) FDR-adjusted p = 0.124	×0.895 (0.785–1.020) FDR-adjusted p = 0.362	×0.911 (0.797–1.043) FDR-adjusted p = 0.390	×0.831 (0.725–0.953) FDR-adjusted p = 0.169
Time above 250 lx melEDI during wake	×1.122 (1.006–1.252) FDR-adjusted p = 0.260	×0.907 (0.814–1.010) FDR-adjusted p = 0.351	×0.953 (0.852–1.065) FDR-adjusted p = 0.614	×0.946 (0.843–1.061) FDR-adjusted p = 0.586
Time below 10 lx melEDI before sleep	-0.005 h (-0.135–+0.124 h) FDR-adjusted p = 0.958	+0.052 h (-0.073–+0.177 h) FDR-adjusted p = 0.614	+0.028 h (-0.104–+0.159 h) FDR-adjusted p = 0.781	-0.031 h (-0.167–+0.105 h) FDR-adjusted p = 0.781
Time below 1 lx melEDI during sleep	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display
Longest continuous period above 250 lx melEDI	×1.117 (1.007–1.238) FDR-adjusted p = 0.260	×0.924 (0.835–1.023) FDR-adjusted p = 0.375	×0.933 (0.840–1.037) FDR-adjusted p = 0.393	×0.945 (0.847–1.053) FDR-adjusted p = 0.557

Associations are reported per one participant-level SD of the LEBA score with 95% CIs. FDR-adjusted p is bold only at alpha = 0.050 across all 68 tests. Sleep-environment cells are unfit for inference and suppressed.

1369

1370 *Supplementary Table S11 (continued).*

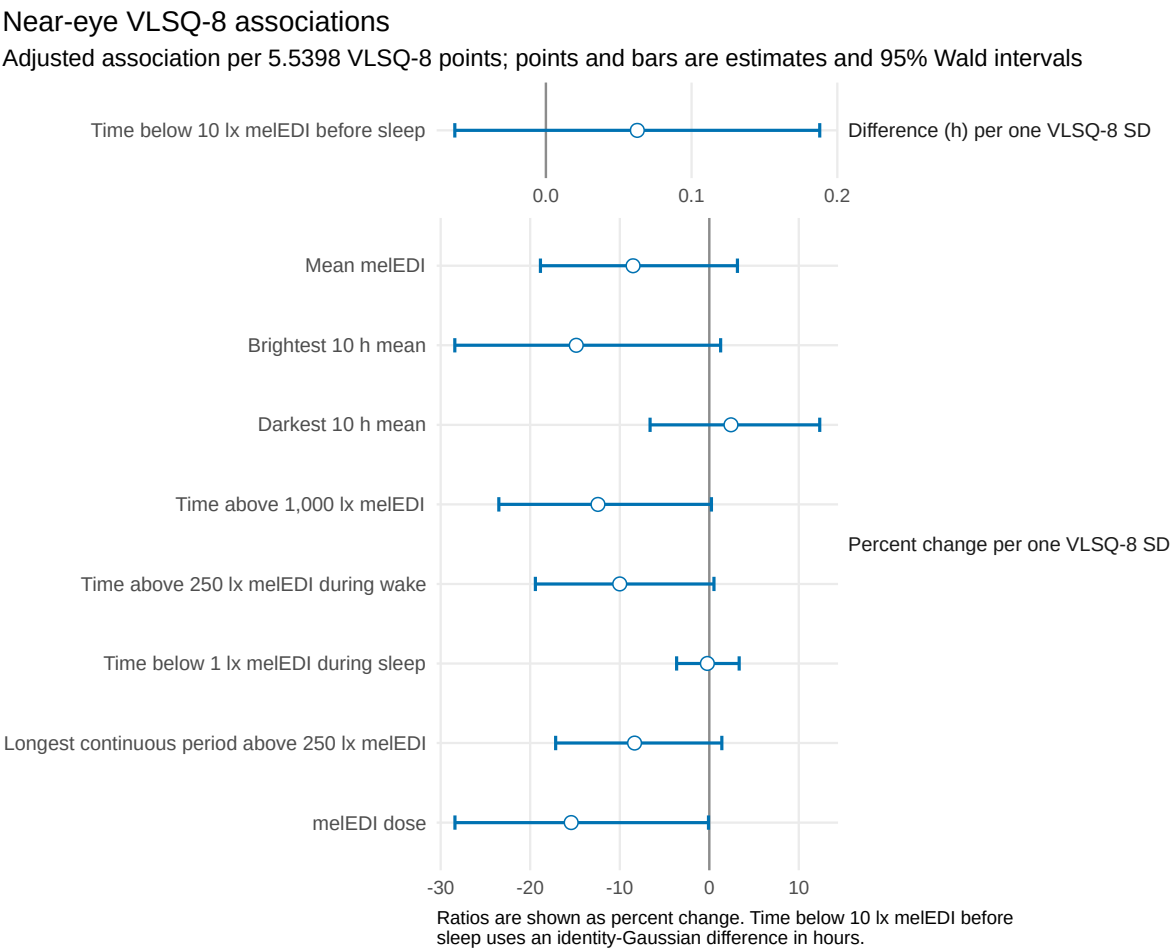
	F2	F3	F4	F5
Midpoint of the brightest 10 hours	+0.049 h (-0.132–+0.229 h) FDR-adjusted p = 0.747	+0.133 h (-0.042–+0.309 h) FDR-adjusted p = 0.375	+0.030 h (-0.153–+0.213 h) FDR-adjusted p = 0.868	-0.083 h (-0.270–+0.105 h) FDR-adjusted p = 0.601
Midpoint of the darkest 10 hours	-0.136 h (-0.317–+0.044 h) FDR-adjusted p = 0.375	+0.080 h (-0.097–+0.258 h) FDR-adjusted p = 0.594	-0.020 h (-0.204–+0.164 h) FDR-adjusted p = 0.935	+0.132 h (-0.056–+0.320 h) FDR-adjusted p = 0.379
Mean timing of exposure above 250 lx melEDI	+0.020 h (-0.152–+0.192 h) FDR-adjusted p = 0.935	+0.086 h (-0.081–+0.253 h) FDR-adjusted p = 0.558	-0.097 h (-0.267–+0.074 h) FDR-adjusted p = 0.510	-0.166 h (-0.339–+0.008 h) FDR-adjusted p = 0.301
First light timing above 250 lx melEDI	-0.219 h (-0.489–+0.052 h) FDR-adjusted p = 0.362	+0.324 h (+0.062–+0.585 h) FDR-adjusted p = 0.179	+0.019 h (-0.252–+0.291 h) FDR-adjusted p = 0.958	-0.097 h (-0.380–+0.185 h) FDR-adjusted p = 0.679
Last light timing above 250 lx melEDI	+0.308 h (+0.043–+0.573 h) FDR-adjusted p = 0.202	-0.066 h (-0.328–+0.196 h) FDR-adjusted p = 0.760	-0.011 h (-0.280–+0.259 h) FDR-adjusted p = 0.958	-0.081 h (-0.361–+0.200 h) FDR-adjusted p = 0.743
melEDI dose	×1.278 (1.079–1.514) FDR-adjusted p = 0.124	×0.885 (0.748–1.048) FDR-adjusted p = 0.379	×0.872 (0.732–1.038) FDR-adjusted p = 0.375	×0.881 (0.736–1.054) FDR-adjusted p = 0.379
Melanopic daylight efficacy ratio	+0.004 (-0.011–+0.019) FDR-adjusted p = 0.747	-0.004 (-0.019–+0.011) FDR-adjusted p = 0.747	-0.010 (-0.026–+0.005) FDR-adjusted p = 0.380	-0.020 (-0.035–+0.004) FDR-adjusted p = 0.169

Associations are reported per one participant-level SD of the LEBA score with 95% CIs. FDR-adjusted p is bold only at alpha = 0.050 across all 68 tests. Sleep-environment cells are unfit for inference and suppressed.

1371

1372 **Supplementary Figure S14 and Table S12. Visual light sensitivity**

1373



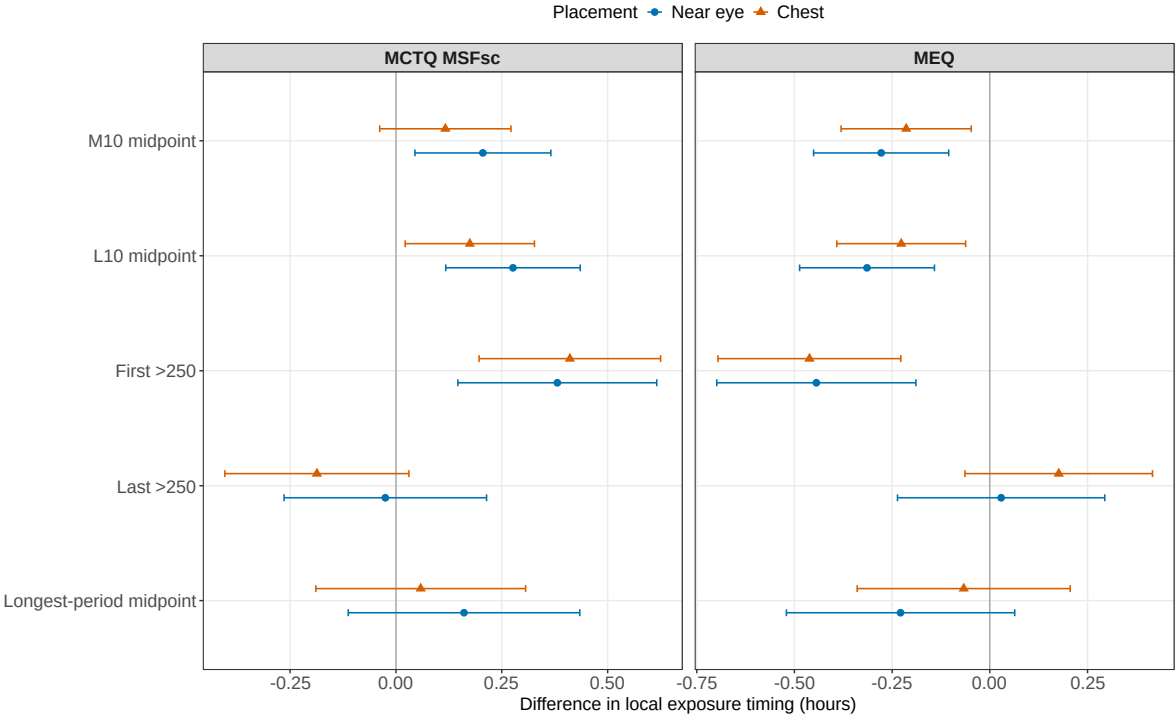
1374

1375 Supplementary Figure S14. Near-eye site-average associations between the Visual Light
1376 Sensitivity Questionnaire-8 and nine personal light-exposure metrics. Associations compare
1377 scores separated by one participant-level standard deviation. Horizontal bars are 95% Wald
1378 confidence intervals. None of the nine associations retained FDR-adjusted support.

Metric	Scale	Association per VLSQ-8 SD (95% CI)	FDR-adjusted p	Participants / participant-days / sites
Mean melEDI	Ratio (percentage change)	×0.915 (0.811–1.031)	0.221	141 / 816 / 9
Brightest 10 h mean	Ratio (percentage change)	×0.851 (0.716–1.013)	0.160	141 / 816 / 9
Darkest 10 h mean	Ratio (percentage change)	×1.024 (0.934–1.123)	0.689	141 / 816 / 9
Time above 1,000 lx melEDI	Ratio (percentage change)	×0.876 (0.765–1.002)	0.160	141 / 816 / 9
Time above 250 lx melEDI during wake	Ratio (percentage change)	×0.900 (0.806–1.005)	0.160	141 / 737 / 9
Time below 10 lx melEDI before sleep	Difference in hours	+0.063 h (-0.062–+0.188 h)	0.422	139 / 655 / 9
Time below 1 lx melEDI during sleep	Ratio (percentage change)	×0.998 (0.964–1.033)	0.903	141 / 778 / 9
Longest continuous period above 250 lx melEDI	Ratio (percentage change)	×0.917 (0.828–1.014)	0.167	141 / 816 / 9
melEDI dose	Ratio (percentage change)	×0.846 (0.716–0.999)	0.160	141 / 761 / 9
Associations compare scores separated by one participant-level SD (5.540 VLSQ-8 points). The hour-scale row is an absolute difference; ratio rows can be read as percentage changes. Intervals are two-sided 95% Wald CIs. FDR adjustment is across all nine metrics; adjusted p-values would be bold at 0.050.				

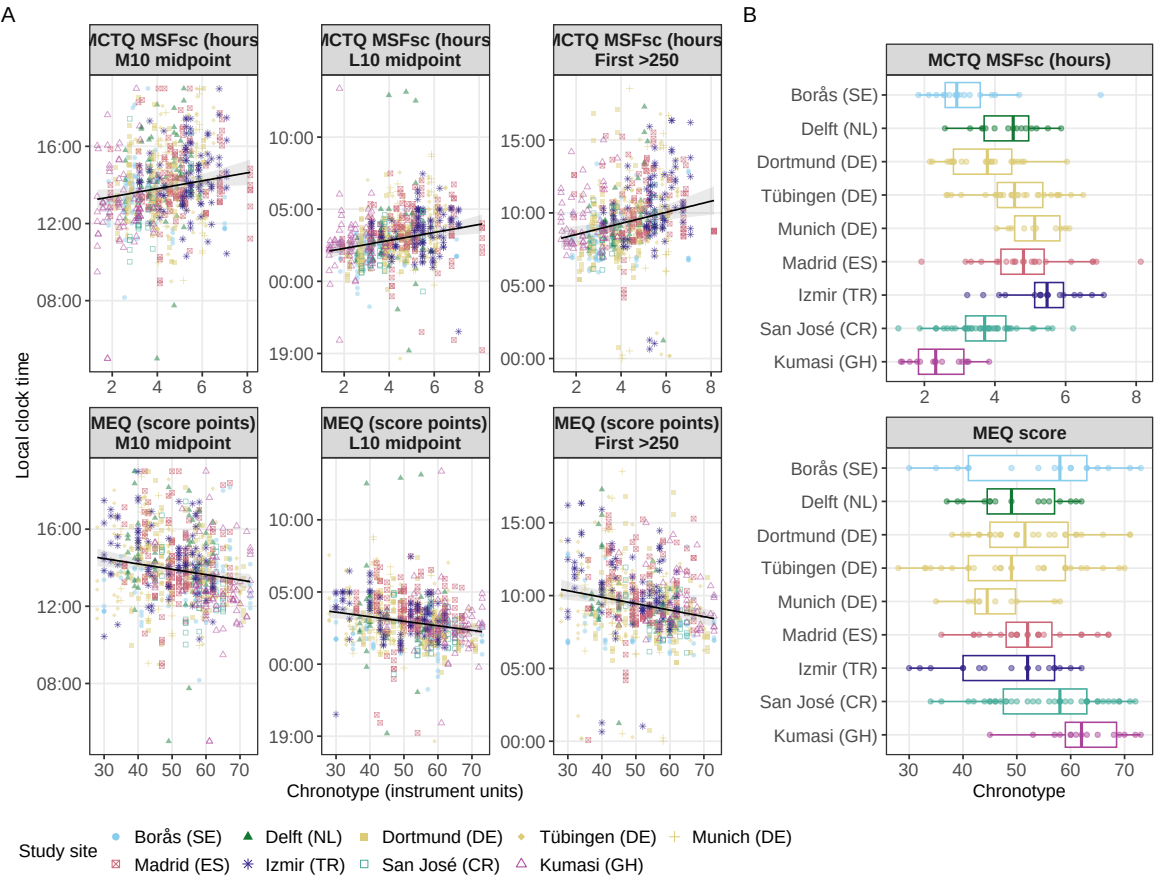
1381 **Supplementary Figure S15 and Table S13. Chronotype and timing**

1382 **A**



1383

Primary near-eye chronotype and light-exposure timing
FDR-supported associations (left) and descriptive chronotype distributions by site (right)



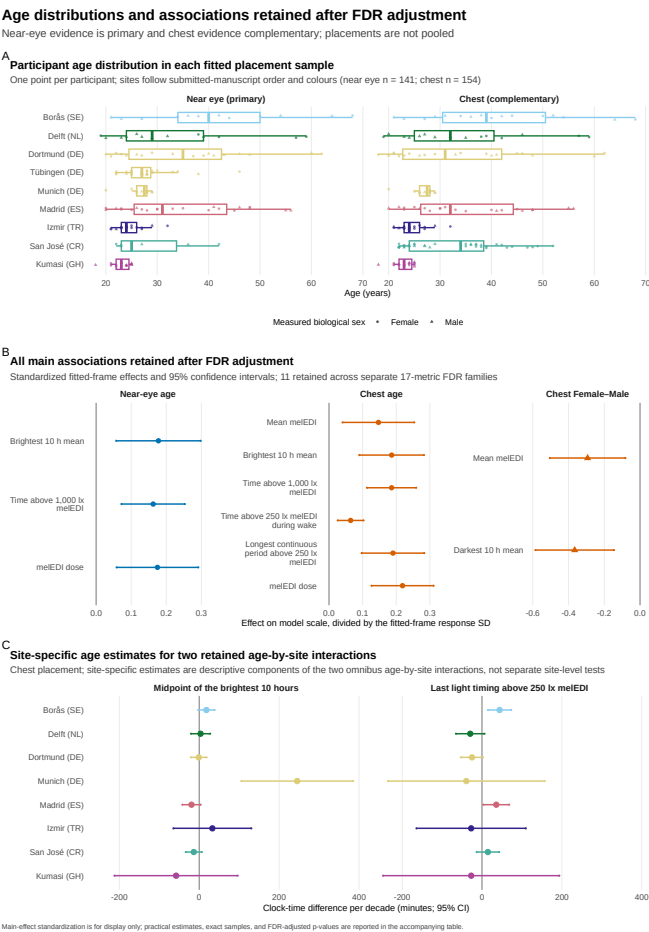
1385

1386 Supplementary Figure S15. Chronotype and timing of personal light exposure. A, Study-site-
1387 adjusted associations of sleep-timing-based corrected midsleep on Free days and
1388 questionnaire-based morningness-eveningness with the analysed timing metrics. Points are
1389 estimates and bars are 95% confidence intervals; the two chronotype instruments remain
1390 separate. B, Observed participant-day timing values across chronotype and country-coded
1391 study sites for the samples used in the corresponding models. Each point represents one
1392 participant-day in a fitted sample. The observed panel is descriptive and does not replace
1393 the adjusted models or estimate independent site effects. Near-eye estimates provide
1394 ocular-exposure evidence; chest estimates, where shown, are complementary non-ocular
1395 evidence.

Metric	Instrument	Signed association, h (95% CI)	FDR-adjusted p	Fitted sample: participants / days / observations / hours / sites
Midpoint of the brightest 10 hours	MCTQ MSFsc	+0.205 (+0.045 to +0.366)	0.017	140 / 810 / 810 / 18711.7 h / 9
Midpoint of the brightest 10 hours	MEQ	-0.278 (-0.451 to -0.105)	0.002	141 / 816 / 816 / 18851.0 h / 9
Midpoint of the darkest 10 hours	MCTQ MSFsc	+0.276 (+0.118 to +0.435)	0.003	140 / 810 / 810 / 18711.7 h / 9
Midpoint of the darkest 10 hours	MEQ	-0.314 (-0.487 to -0.142)	0.001	141 / 816 / 816 / 18851.0 h / 9
First light timing above 250 lx meLEDI	MCTQ MSFsc	+0.381 (+0.146 to +0.616)	0.003	139 / 722 / 722 / 16716.3 h / 9
First light timing above 250 lx meLEDI	MEQ	-0.444 (-0.699 to -0.189)	0.001	140 / 727 / 727 / 16832.5 h / 9
Last light timing above 250 lx meLEDI	MCTQ MSFsc	-0.025 (-0.264 to +0.214)	0.811	140 / 683 / 683 / 15900.8 h / 9
Last light timing above 250 lx meLEDI	MEQ	+0.029 (-0.236 to +0.294)	0.814	141 / 687 / 687 / 15995.0 h / 9
Midpoint of the longest continuous period above 250 lx meLEDI	MCTQ MSFsc	+0.161 (-0.113 to +0.434)	0.284	131 / 478 / 478 / 11325.5 h / 9
Midpoint of the longest continuous period above 250 lx meLEDI	MEQ	-0.228 (-0.521 to +0.064)	0.135	132 / 482 / 482 / 11419.8 h / 9
MCTQ associations are per one-hour later MSFsc; MEQ associations are per 10 points greater morning preference. Positive associations indicate later timing and negative associations earlier. Intervals are 95% Wald CIs. FDR-adjusted p-values are bold when FDR-adjusted p < 0.050 within the instrument-specific five-outcome family.				

1398 **Supplementary Figure S16 and Table S14. Age and biological sex**

1399



1400

1401 Supplementary Figure S16. Participant age distributions, the 11 main associations retained
1402 after FDR adjustment, and descriptive site-specific estimates for two retained age-by-site
1403 interactions in the complementary chest analysis. The site-specific interaction estimates are
1404 descriptive model contrasts, not independent site tests. Biological sex, coded Female or
1405 Male, was analysed; gender was recorded separately and was not analysed.

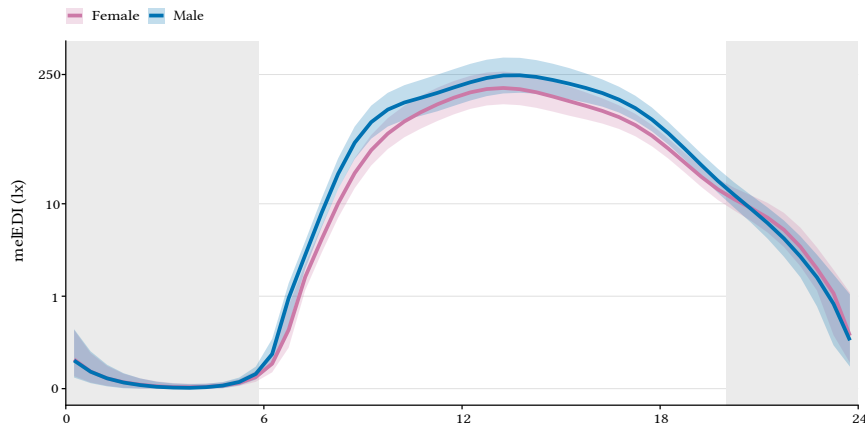
Placement and role	Predictor	Metric	Practical association (95% CI)	FDR-adjusted p	Sample
Near eye (primary)	Age, per 10 years	Brightest 10 h mean	1.31× (1.09–1.58)	0.018	141 participants; 816 participant-days/observations
Near eye (primary)	Age, per 10 years	Time above 1,000 lx melEDI	1.27× (1.11–1.45)	0.009	141 participants; 816 participant-days/observations
Near eye (primary)	Age, per 10 years	melEDI dose	1.31× (1.09–1.56)	0.018	141 participants; 761 participant-days/observations
Chest (complementary)	Age, per 10 years	Mean melEDI	1.16× (1.04–1.30)	0.017	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Brightest 10 h mean	1.35× (1.15–1.57)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Time above 1,000 lx melEDI	1.31× (1.18–1.46)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Time above 250 lx melEDI during wake	1.16× (1.06–1.26)	0.004	154 participants; 818 participant-days/observations
Chest (complementary)	Age, per 10 years	Longest continuous period above 250 lx melEDI	1.18× (1.09–1.27)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	melEDI dose	1.40× (1.21–1.61)	<0.001	154 participants; 851 participant-days/observations
Chest (complementary)	Measured biological sex, Female minus Male	Mean melEDI	0.74× (0.60–0.92)	0.048	154 participants; 902 participant-days/observations
Chest (complementary)	Measured biological sex, Female minus Male	Darkest 10 h mean	0.76× (0.65–0.90)	0.016	154 participants; 902 participant-days/observations
Associations are site-adjusted estimates per 10-year age increase or Female minus Male. Ratios are back-transformed to the practical scale. Each FDR-adjusted p-value belongs to its sensor-position- and predictor-specific complete 17-test family; bold values meet the FDR-adjusted p < 0.050 rule. Samples give the exact fitted participants and participant-day observations for these retained daily-response results.					

1408 **Supplementary Figure S17 and Table S15. Biological-sex-specific**
1409 **daily curves**

1410

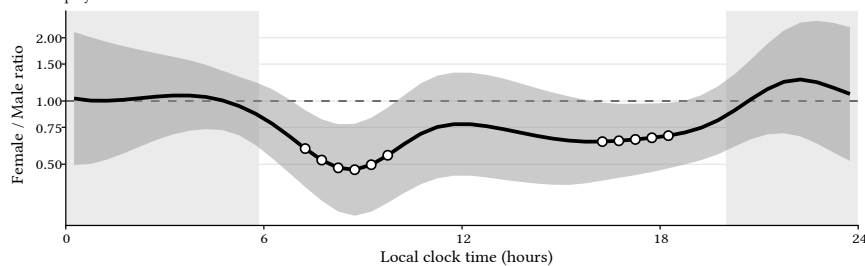
A Sex-specific near-eye curves

Equal-site conditional means; 141 participants, 816 participant-days, 37,756
30-minute observations



B Female-to-Male shifted-value ratio

Ribbons are participant-cluster-robust pointwise 95% intervals; open circles
mark displayed bins whose interval excludes 1.



Grey shading spans midnight to equal-site mean civil dawn and equal-site mean civil dusk to midnig
Intervals are pointwise, not simultaneous; the global participant-cluster-robust curve test is
inferential test and the display cannot establish a familywise-signi ficant time period. The melEDI
uses the approved symlog scale: linear from 0 to 1 lx and base-10 above 1 lx

1411

1412 Supplementary Figure S17. Near-eye biological-sex-specific fitted melanopic equivalent
1413 daylight illuminance curves and Female-to-Male shifted-value ratio. Ribbons and contrast
1414 intervals are participant-cluster-robust 95% confidence intervals at the displayed clock times.
1415 Grey shading gives site-average civil-night context with each site weighted equally and is not
1416 a model covariate. Biological sex, coded Female or Male, was analysed; gender was
1417 recorded separately and was not analysed.

1418 *Supplementary Table S15. Biological-sex-specific daily patterns and global tests*

	Role	Test	F statistic	Numerator df	Denominator df	Raw p	FDR-adjusted p	Support
Near eye	Primary	Complete Female-minus-Male 24-hour curve	20.946	9.692	131	0.028	0.028	Supported
Chest	Complementary	Complete Female-minus-Male 24-hour curve	4.876	1.003	152	0.029	0.029	Supported

Primary near-eye fitted sample: 141 participants, 816 participant-days, 37,756 30-minute observations, and 9 sites. Complementary chest fitted sample: 154 participants, 902 participant-days, 41,842 30-minute observations, and 8 sites.

Each row is one joint complete-curve test. Raw p is bold at raw p < 0.050; FDR-adjusted p is bold independently at adjusted p < 0.050. Each labelled global family contains one test, so the stored raw and adjusted values coincide.

1419

1420 **Measurement position, robustness, and reproducibility**

1421 Near-eye and complementary chest measurements remain separate sensor-position
1422 estimands. Common-sample analyses use the same participants and participant-days at
1423 both positions and fit the positions separately. They do not pool positions, test equivalence,
1424 or establish a universal correction. Detailed model checks, sensitivities, preregistration
1425 deviations, exact formulas, package versions, and source documentation are provided in the
1426 analysis reports and reproducibility package.